vanilla_gan_cifar10

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#

BÀI TẬP CUỐI KỲ

##

Môn: Thị giác máy tính chuyên sâu

1 First-level requirement:

1.0.1 Train and generate images for one class of the dataset using vanilla GAN.

[1]:
```
!pip install tensorflow_datasets

# Import libraries
from google.colab import files
import numpy as np
import tensorflow as tf
import tensorflow_datasets as tfds
import keras

from sklearn.model_selection import train_test_split
from matplotlib import pyplot as plt
from keras import Model
from keras import utils
from keras import callbacks
from keras import optimizers
from keras.datasets import cifar10
from keras.optimizers.legacy import Adam, SGD
from keras.models import Sequential
from keras.layers import Activation, Conv2D, Dense, MaxPooling2D, Flatten,␣
 ↪Conv2DTranspose, LeakyReLU, Reshape, Dropout, BatchNormalization,␣
 ↪GlobalMaxPooling2D, Flatten, Dense
```

```
Requirement already satisfied: tensorflow_datasets in
/usr/local/lib/python3.10/dist-packages (4.9.4)
Requirement already satisfied: absl-py in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (1.4.0)
Requirement already satisfied: click in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (8.1.7)
Requirement already satisfied: dm-tree in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (0.1.8)
Requirement already satisfied: etils[enp,epath,etree]>=0.9.0 in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (1.7.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (1.25.2)
Requirement already satisfied: promise in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (2.3)
Requirement already satisfied: protobuf>=3.20 in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (3.20.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (5.9.5)
Requirement already satisfied: requests>=2.19.0 in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (2.31.0)
Requirement already satisfied: tensorflow-metadata in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (1.14.0)
Requirement already satisfied: termcolor in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (2.4.0)
Requirement already satisfied: toml in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (0.10.2)
Requirement already satisfied: tqdm in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (4.66.2)
Requirement already satisfied: wrapt in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (1.14.1)
Requirement already satisfied: array-record>=0.5.0 in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (0.5.1)
Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages
(from etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (2023.6.0)
Requirement already satisfied: importlib_resources in
/usr/local/lib/python3.10/dist-packages (from
etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (6.4.0)
Requirement already satisfied: typing_extensions in
/usr/local/lib/python3.10/dist-packages (from
etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (4.11.0)
Requirement already satisfied: zipp in /usr/local/lib/python3.10/dist-packages
(from etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (3.18.1)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.10/dist-packages (from
requests>=2.19.0->tensorflow_datasets) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-
packages (from requests>=2.19.0->tensorflow_datasets) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in
```

```
/usr/local/lib/python3.10/dist-packages (from
requests>=2.19.0->tensorflow_datasets) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.10/dist-packages (from
requests>=2.19.0->tensorflow_datasets) (2024.2.2)
Requirement already satisfied: six in /usr/local/lib/python3.10/dist-packages
(from promise->tensorflow_datasets) (1.16.0)
Requirement already satisfied: googleapis-common-protos<2,>=1.52.0 in
/usr/local/lib/python3.10/dist-packages (from tensorflow-
metadata->tensorflow_datasets) (1.63.0)
```

[2]:

```python
# Get the full dataset (batch_size=-1) in NumPy arrays from the returned tf.
 ↪Tensor object
cifar10_train = tfds.load(name="cifar10", split=tfds.Split.TRAIN, batch_size=-1␣
 ↪)
cifar10_test = tfds.load(name="cifar10", split=tfds.Split.TEST, batch_size=-1)

# Convert tfds dataset to numpy array records
cifar10_train = tfds.as_numpy(cifar10_train)
cifar10_test = tfds.as_numpy(cifar10_test)

# Seperate feature X and label Y
X_train, y_train = cifar10_train["image"], cifar10_train["label"]
X_test, y_test = cifar10_test["image"], cifar10_test["label"]

# Print shapes of the entire training and test set of CIFAR 10
print("X_train shape: " + str(X_train.shape))
print("X_test shape: " + str(X_test.shape))
print("y_train shape: " + str(y_train.shape))
print("y_test shape: " + str(y_test.shape))
```

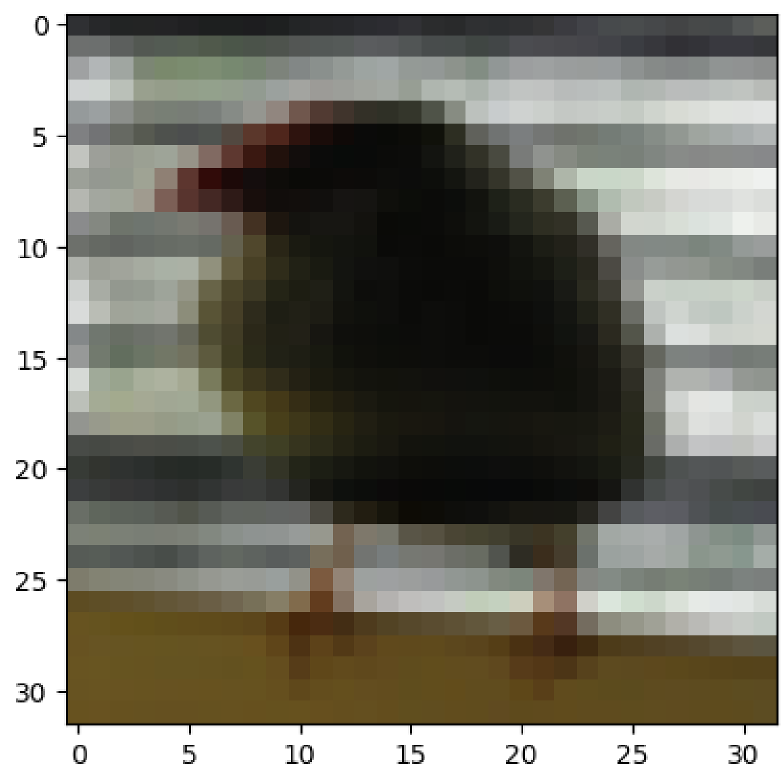
```
Downloading and preparing dataset 162.17 MiB (download: 162.17 MiB, generated:
132.40 MiB, total: 294.58 MiB) to /root/tensorflow_datasets/cifar10/3.0.2…

Dl Completed…: 0 url [00:00, ? url/s]

Dl Size…: 0 MiB [00:00, ? MiB/s]

Extraction completed…: 0 file [00:00, ? file/s]

Generating splits…:   0%|          | 0/2 [00:00<?, ? splits/s]

Generating train examples…:   0%|          | 0/50000 [00:00<?, ? examples/s]

Shuffling /root/tensorflow_datasets/cifar10/3.0.2.incomplete1OJB8E/cifar10-train.
 ↪tfrecord*…:   0%|          …

Generating test examples…:   0%|          | 0/10000 [00:00<?, ? examples/s]

Shuffling /root/tensorflow_datasets/cifar10/3.0.2.incomplete1OJB8E/cifar10-test.
 ↪tfrecord*…:   0%|          |…
```

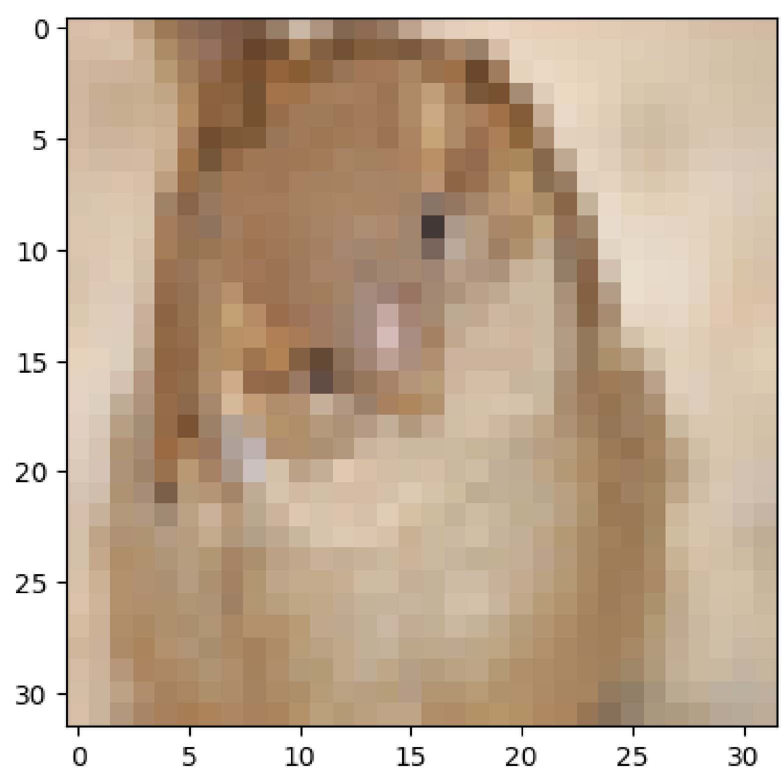
```
Dataset cifar10 downloaded and prepared to
/root/tensorflow_datasets/cifar10/3.0.2. Subsequent calls will reuse this data.
X_train shape: (50000, 32, 32, 3)
X_test shape: (10000, 32, 32, 3)
y_train shape: (50000,)
y_test shape: (10000,)
```

[3]:
```
# Get samples  of a certain label. You can choose from one of the classes:
# 0 - airplane; 1 - automobile; 2 - bird; 3 - cat; 4 - deer; 5 - dog; 6 - frog;␣
↪7 - horse; 8 - ship; 9 - truck

label = 2                                    # Class '2' is bird
train_indices = np.where(y_train == label)  # Get indices of the label
train_set = X_train[train_indices]          # Get train samples of the label
test_indices = np.where(y_test == label)    # Get indices of the label
test_set = X_test[test_indices]             # Get test samples of the label

# Print shapes of datasets of the selected class
print("Train_set shape of the selected class: " + str(train_set.shape))
print("Test_set shape of the selected class: " + str(test_set.shape))

# NOTE: Training a GAN doesn't require to have train/test split. You should␣
↪merge the train_set and test_set into one dataset, which can be used as real␣
↪data samples
data = np.vstack((train_set, test_set))
print("Dataset shape of the selected class: " + str(data.shape))
```

```
Train_set shape of the selected class: (5000, 32, 32, 3)
Test_set shape of the selected class: (1000, 32, 32, 3)
Dataset shape of the selected class: (6000, 32, 32, 3)
```

[4]:
```
# Visualize some data samples of class '2' (bird)
import matplotlib.pyplot as plt
imgplot = plt.imshow(data[50])
plt.show()
imgplot = plt.imshow(data[5500])
plt.show()
```


0
5
10
15
20
25
30
0
5
10
15
20
25
30


0
5
10
15
20
25
30
0
5
10
15
20
25
30

2 Build the Generator

[5]:
```
# Define the structure of Generator
def define_generator(latent_dim):
  # latent_dim: dimension of the random input vector
  model = Sequential()  # Initialize a sequential model for Generator

  # Foundation for 4x4 image
  n_nodes = 4*4*256    # Expected vector size for converting to a feature map
  model.add(Dense(input_dim=latent_dim, units=n_nodes))  # FC layer to convert␣
↪latent vector to a vector of size n_nodes
  model.add(Reshape((4, 4, 256)))
  model.add(LeakyReLU(alpha=0.2))

  # Upsample from 4x4x256 to 8x8x128
  model.add(Conv2DTranspose(128, (3,3), strides=(2,2), padding='same'))
  model.add(LeakyReLU(alpha=0.2))

  # Upsample from 8x8x128 to 16x16x64
  model.add(Conv2DTranspose(64, (3,3), strides=(2,2), padding='same'))
  model.add(LeakyReLU(alpha=0.2))

  # Upsample from 16x16x64 to 32x32x32
  model.add(Conv2DTranspose(32, (3,3), strides=(2,2), padding='same'))
  model.add(LeakyReLU(alpha=0.2))

  # Convert from 32x32x32 to 32x32x3 conforming to Discriminator's input
  model.add(Conv2D(3, (5,5), activation='tanh', padding='same'))

  return model          # Return the Generator

# Show a sample of Generator structure
generator = define_generator(256)
generator.summary()
```

```
Model: "sequential"

_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense (Dense)               (None, 4096)              1052672

 reshape (Reshape)           (None, 4, 4, 256)         0

 leaky_re_lu (LeakyReLU)     (None, 4, 4, 256)         0
```

```
 conv2d_transpose (Conv2DTr  (None, 8, 8, 128)         295040
 anspose)

 leaky_re_lu_1 (LeakyReLU)   (None, 8, 8, 128)         0

 conv2d_transpose_1 (Conv2D  (None, 16, 16, 64)        73792
 Transpose)

 leaky_re_lu_2 (LeakyReLU)   (None, 16, 16, 64)        0

 conv2d_transpose_2 (Conv2D  (None, 32, 32, 32)        18464
 Transpose)

 leaky_re_lu_3 (LeakyReLU)   (None, 32, 32, 32)        0

 conv2d (Conv2D)             (None, 32, 32, 3)         2403

=================================================================
Total params: 1442371 (5.50 MB)
Trainable params: 1442371 (5.50 MB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
```

3 Build the Discriminator

[6]:
```python
# Define the structure of Discriminator
def define_discriminator(in_shape=(32,32,3)):
  model = Sequential()  # Initialize a sequential model for Discriminator

  # Block 1 conv
  model.add(Conv2D(16, (3,3), padding='same', input_shape=in_shape))
  model.add(LeakyReLU(alpha=0.2))

  # Block 2 conv
  model.add(Conv2D(32, (3,3), strides=(2, 2), padding='same'))
  model.add(LeakyReLU(alpha=0.2))

  # Block 3 conv
  model.add(Conv2D(64, (3,3), strides=(2, 2), padding='same'))
  model.add(LeakyReLU(alpha=0.2))

  # Block 4 conv
  model.add(Conv2D(128, (3,3), strides=(2, 2), padding='same'))
  model.add(LeakyReLU(alpha=0.2))

  # Block 5 conv
```

```
    model.add(Conv2D(256, (3,3), strides=(2, 2), padding='same'))
    model.add(LeakyReLU(alpha=0.2))

    # Classifier
    model.add(Flatten())
    model.add(Dropout(0.5))
    model.add(Dense(1, activation='sigmoid'))

    return model          # Return the Discriminator

# Show a sample of Discriminator structure
sample_discriminator = define_discriminator()
sample_discriminator.summary()
```

```
Model: "sequential_1"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d_1 (Conv2D)           (None, 32, 32, 16)        448

 leaky_re_lu_4 (LeakyReLU)   (None, 32, 32, 16)        0

 conv2d_2 (Conv2D)           (None, 16, 16, 32)        4640

 leaky_re_lu_5 (LeakyReLU)   (None, 16, 16, 32)        0

 conv2d_3 (Conv2D)           (None, 8, 8, 64)          18496

 leaky_re_lu_6 (LeakyReLU)   (None, 8, 8, 64)          0

 conv2d_4 (Conv2D)           (None, 4, 4, 128)         73856

 leaky_re_lu_7 (LeakyReLU)   (None, 4, 4, 128)         0

 conv2d_5 (Conv2D)           (None, 2, 2, 256)         295168

 leaky_re_lu_8 (LeakyReLU)   (None, 2, 2, 256)         0

 flatten (Flatten)           (None, 1024)              0

 dropout (Dropout)           (None, 1024)              0

 dense_1 (Dense)             (None, 1)                 1025

=================================================================
Total params: 393633 (1.50 MB)
Trainable params: 393633 (1.50 MB)
Non-trainable params: 0 (0.00 Byte)
```

---

4 Build GAN with the Generator and the Discriminator

[7]:
```
# Define GAN network consisting of a Generator and a Discriminator
def define_gan(g_model, d_model):
    # g_model: Generator component; d_model: Discriminator component
    model = Sequential()  # Initialize a sequential model
    model.add(g_model)    # Add the Generator component
    model.add(d_model)    # Add the Discriminator component
    return model          # Return the GAN
```

5 Prepare real data samples from the CIFAR 10 dataset

[8]:
```
# Load and normalize CIFAR 10 images of the selected class
def load_real_samples():
    # Load dataset
    (X_train, y_train), (X_test, y_test) = cifar10.load_data()
    label = 2                                       # Class '2' is bird
    train_indices = np.where(y_train == label)[0]  # Get indices of the label
    train_set = X_train[train_indices]
    test_indices = np.where(y_test == label)[0]    # Get indices of the label
    test_set = X_test[test_indices]
    X = np.vstack((train_set, test_set))

    # Convert pixel values from unsigned int to float32 for normalization␣
↪calculation
    X = X.astype('float32')

    # Because the generator is using tanh activation, we need to normalize real␣
↪images from range [0,255] into range [-1,1]
    X = (X - 127.5) / 127.5

    return X  # Return the set of normalized images

# Select real samples from the real dataset for training purpose
def generate_real_samples(dataset, n_samples):
    # Generate random n_samples sample indices from the dataset
    ix = np.random.randint(0, dataset.shape[0], size = n_samples)

    # Get the images from the list of indices
    X = dataset[ix]

    # Generate labels for the real samples (label=1)
    y = np.ones((n_samples, 1))
```

```
    return X, y  # Return the real images and labels
```

6 Generate fake data samples by passing a random vector through the Generator

[9]:
```
# Generate random latent vectors as input for Generator
def generate_latent_points(latent_dim, n_samples):
    # Generate points in the latent space
    x_input = np.random.randn(latent_dim * n_samples)

    # Reshape to a batch of inputs for the Generator
    x_input = x_input.reshape(n_samples, latent_dim)

    return x_input

# Use Generator to generate n_samples fake samples
def generate_fake_samples(g_model, latent_dim, n_samples):
    # Generate random latent vectors as input for Generator
    x_input = generate_latent_points(latent_dim, n_samples)

    # Generate fake data samples by passing the random latent vectors through␣
 ↪the Generator
    X = g_model.predict(x_input)

    # Generate labels for the fake samples (label=0)
    y = np.zeros((n_samples, 1))

    return X, y  # Return the fake images and labels
```

7 Training process with the Generator and the Discriminator

[10]:
```
# Train the Generator and the Discriminator
def train(g_model, d_model, gan_model, dataset, latent_dim, n_epochs, n_batch):
    bat_per_epo = int(dataset.shape[0] / n_batch)  # Number of iterations per␣
 ↪epoch
    half_batch = int(n_batch / 2)                  # A batch to the␣
 ↪Discriminator contains real samples (half-batch) and fake samples␣
 ↪(half-batch)

    g_loss_values = []                             # Array containing Generator␣
 ↪loss over iterations
    d_loss_real_values = []                        # Array containing␣
 ↪Discriminator loss on real samples over iterations
    d_loss_fake_values = []                        # Array containing␣
 ↪Discriminator loss on fake samples over iterations
```

```
    # Number of epochs
    for i in range(n_epochs):
        # Number of iterations
        for j in range(bat_per_epo):

            # Randomly get half_batch real data samples from the real dataset
            X_real, y_real = generate_real_samples(dataset, half_batch)

            # Generate half_batch fake data samples
            X_fake, y_fake = generate_fake_samples(g_model, latent_dim,␣
↪half_batch)

            # Update discriminator model weights
            d_loss1, _ = d_model.train_on_batch(X_real, y_real)
            d_loss2, _ = d_model.train_on_batch(X_fake, y_fake)

            # Generate random latent vectors as input for Generator
            X_gan = generate_latent_points(latent_dim, n_batch)

            # Generate inverted labels for the fake samples to train the␣
↪Generator
            y_gan = np.ones((n_batch, 1))

            # Train the Generator
            g_loss = gan_model.train_on_batch(X_gan, y_gan)

            # Training log after every iteration
            print('Epoch %d, Iteration %d/%d, D_loss_real = %.3f, D_loss_fake =␣
↪%.3f, G_loss = %.3f' % (i+1, j+1, bat_per_epo, d_loss1, d_loss2, g_loss))

            # Add loss values to historical arrays
            g_loss_values.append(g_loss)
            d_loss_real_values.append(d_loss1)
            d_loss_fake_values.append(d_loss2)

        # Plot the generated images every certain number of epochs
        if (i+1) % 10 == 0:
            X_fake, _ = generate_fake_samples(g_model, latent_dim, 49)  #␣
↪Generate 49 fake data samples using the Generator
            save_plot(X_fake, i, n=7)                                   # Plot␣
↪a grid of 7x7 fake samples and save to file at epoch i

    return d_loss_real_values, d_loss_fake_values, g_loss_values        #␣
↪Return the loss arrays of Generator and Discriminator
```

8 Plot the Generator's outputs

[11]:
```
# Create and save a plot of generated images
def save_plot(samples, epoch, n):
    # Define a plot of (n rows, n columns)
    for i in range(n * n):
        # (i + 1) is the index position on a grid with nrows rows and ncols␣
↪columns. Index starts at 1 in the upper left corner and increases to the␣
↪right.
        plt.subplot(n, n, i+1)
        plt.axis('off')                                  # Disable axis
        # Plot the sample i, convert pixel values from range [-1,1] into range␣
↪[0,255]
        plt.imshow(((samples[i] * 127.5) + 127.5).astype(np.uint8))
    filename = 'Plot_epoch_%03d.png' % (epoch+1)
    plt.savefig(filename)                                # Save plot to file
    plt.close()                                          # Close the plot
```

9 Main code

[12]:
```
# Size of the latent space
latent_dim = 256
epochs = 500
batch_size = 128

# Initialize an optimizer
opt = Adam(learning_rate=0.0001, beta_1 = 0.5)

# Create the Generator
g_model = define_generator(latent_dim)

# Create the Discriminator
d_model = define_discriminator()
d_model.trainable = True
d_model.compile(loss='binary_crossentropy', optimizer=opt, metrics=['accuracy'])

# Create the GAN
d_model.trainable = False
gan_model = define_gan(g_model, d_model)
gan_model.compile(loss='binary_crossentropy', optimizer=opt)

# Load real image data from the CIFAR 10 dataset
dataset = load_real_samples()

# Train the GAN
```

```
d_losses_real, d_losses_fake, g_losses = train(g_model, d_model, gan_model,␣
↪dataset, latent_dim, epochs, batch_size)
```

```
Kết quả truyền trực tuyến bị cắt bớt đến 5000 dòng cuối.
Epoch 446, Iteration 34/46, D_loss_real = 0.601, D_loss_fake = 0.673, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 35/46, D_loss_real = 0.643, D_loss_fake = 0.616, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 36/46, D_loss_real = 0.630, D_loss_fake = 0.682, G_loss =
0.754
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 37/46, D_loss_real = 0.597, D_loss_fake = 0.673, G_loss =
0.776
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 38/46, D_loss_real = 0.589, D_loss_fake = 0.642, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 39/46, D_loss_real = 0.666, D_loss_fake = 0.658, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 40/46, D_loss_real = 0.664, D_loss_fake = 0.664, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 41/46, D_loss_real = 0.696, D_loss_fake = 0.615, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 446, Iteration 42/46, D_loss_real = 0.661, D_loss_fake = 0.600, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 43/46, D_loss_real = 0.648, D_loss_fake = 0.604, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 44/46, D_loss_real = 0.667, D_loss_fake = 0.617, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 45/46, D_loss_real = 0.625, D_loss_fake = 0.638, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 446, Iteration 46/46, D_loss_real = 0.674, D_loss_fake = 0.625, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 1/46, D_loss_real = 0.653, D_loss_fake = 0.627, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 2/46, D_loss_real = 0.657, D_loss_fake = 0.612, G_loss =
0.838
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 3/46, D_loss_real = 0.657, D_loss_fake = 0.649, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 4/46, D_loss_real = 0.688, D_loss_fake = 0.629, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 5/46, D_loss_real = 0.578, D_loss_fake = 0.691, G_loss =
0.790
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 6/46, D_loss_real = 0.624, D_loss_fake = 0.653, G_loss =
0.794
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 7/46, D_loss_real = 0.614, D_loss_fake = 0.611, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 8/46, D_loss_real = 0.642, D_loss_fake = 0.724, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 9/46, D_loss_real = 0.650, D_loss_fake = 0.623, G_loss =
0.802
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 10/46, D_loss_real = 0.639, D_loss_fake = 0.679, G_loss =
0.795
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 11/46, D_loss_real = 0.598, D_loss_fake = 0.679, G_loss =
0.797
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 12/46, D_loss_real = 0.614, D_loss_fake = 0.655, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 13/46, D_loss_real = 0.634, D_loss_fake = 0.632, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 14/46, D_loss_real = 0.625, D_loss_fake = 0.626, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 15/46, D_loss_real = 0.619, D_loss_fake = 0.662, G_loss =
0.739
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 16/46, D_loss_real = 0.639, D_loss_fake = 0.693, G_loss =
0.793
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 17/46, D_loss_real = 0.603, D_loss_fake = 0.726, G_loss =
0.797
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 18/46, D_loss_real = 0.659, D_loss_fake = 0.645, G_loss =
0.826
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 19/46, D_loss_real = 0.668, D_loss_fake = 0.608, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 20/46, D_loss_real = 0.714, D_loss_fake = 0.611, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 21/46, D_loss_real = 0.704, D_loss_fake = 0.627, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 22/46, D_loss_real = 0.637, D_loss_fake = 0.583, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 23/46, D_loss_real = 0.737, D_loss_fake = 0.631, G_loss =
0.867
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 24/46, D_loss_real = 0.725, D_loss_fake = 0.647, G_loss =
0.870
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 25/46, D_loss_real = 0.663, D_loss_fake = 0.592, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 26/46, D_loss_real = 0.685, D_loss_fake = 0.595, G_loss =
0.785
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 27/46, D_loss_real = 0.651, D_loss_fake = 0.683, G_loss =
0.789
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 28/46, D_loss_real = 0.660, D_loss_fake = 0.636, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 29/46, D_loss_real = 0.641, D_loss_fake = 0.626, G_loss =
0.775
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 30/46, D_loss_real = 0.632, D_loss_fake = 0.640, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 31/46, D_loss_real = 0.636, D_loss_fake = 0.697, G_loss =
0.782
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 32/46, D_loss_real = 0.620, D_loss_fake = 0.622, G_loss =
0.767
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 33/46, D_loss_real = 0.638, D_loss_fake = 0.642, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 34/46, D_loss_real = 0.624, D_loss_fake = 0.675, G_loss =
0.820
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 35/46, D_loss_real = 0.630, D_loss_fake = 0.614, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 36/46, D_loss_real = 0.647, D_loss_fake = 0.605, G_loss =
0.782
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 37/46, D_loss_real = 0.645, D_loss_fake = 0.631, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 38/46, D_loss_real = 0.646, D_loss_fake = 0.668, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 39/46, D_loss_real = 0.634, D_loss_fake = 0.658, G_loss =
0.780
2/2 [==============================] - 0s 4ms/step
Epoch 447, Iteration 40/46, D_loss_real = 0.643, D_loss_fake = 0.650, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 41/46, D_loss_real = 0.603, D_loss_fake = 0.623, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 42/46, D_loss_real = 0.662, D_loss_fake = 0.646, G_loss =
0.776
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.669, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 44/46, D_loss_real = 0.721, D_loss_fake = 0.595, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.624, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 447, Iteration 46/46, D_loss_real = 0.666, D_loss_fake = 0.624, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 1/46, D_loss_real = 0.659, D_loss_fake = 0.634, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 2/46, D_loss_real = 0.603, D_loss_fake = 0.593, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 3/46, D_loss_real = 0.666, D_loss_fake = 0.614, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 4/46, D_loss_real = 0.698, D_loss_fake = 0.630, G_loss =
0.876
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 5/46, D_loss_real = 0.689, D_loss_fake = 0.627, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 6/46, D_loss_real = 0.677, D_loss_fake = 0.584, G_loss =
0.842
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 7/46, D_loss_real = 0.675, D_loss_fake = 0.595, G_loss =
0.866
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 8/46, D_loss_real = 0.696, D_loss_fake = 0.599, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 9/46, D_loss_real = 0.689, D_loss_fake = 0.579, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 10/46, D_loss_real = 0.657, D_loss_fake = 0.631, G_loss =
0.786
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 11/46, D_loss_real = 0.693, D_loss_fake = 0.628, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 12/46, D_loss_real = 0.656, D_loss_fake = 0.639, G_loss =
0.796
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 13/46, D_loss_real = 0.651, D_loss_fake = 0.640, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 14/46, D_loss_real = 0.655, D_loss_fake = 0.616, G_loss =
0.775
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 15/46, D_loss_real = 0.637, D_loss_fake = 0.641, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 16/46, D_loss_real = 0.638, D_loss_fake = 0.637, G_loss =
0.797
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 17/46, D_loss_real = 0.622, D_loss_fake = 0.634, G_loss =
0.785
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 18/46, D_loss_real = 0.661, D_loss_fake = 0.678, G_loss =
0.812
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 19/46, D_loss_real = 0.647, D_loss_fake = 0.653, G_loss =
0.781
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 20/46, D_loss_real = 0.661, D_loss_fake = 0.675, G_loss =
0.778
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 21/46, D_loss_real = 0.605, D_loss_fake = 0.679, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 22/46, D_loss_real = 0.597, D_loss_fake = 0.648, G_loss =
0.783
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 23/46, D_loss_real = 0.585, D_loss_fake = 0.676, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 24/46, D_loss_real = 0.610, D_loss_fake = 0.653, G_loss =
0.783
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 25/46, D_loss_real = 0.631, D_loss_fake = 0.628, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 26/46, D_loss_real = 0.645, D_loss_fake = 0.638, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 27/46, D_loss_real = 0.665, D_loss_fake = 0.620, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 28/46, D_loss_real = 0.628, D_loss_fake = 0.648, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 29/46, D_loss_real = 0.674, D_loss_fake = 0.624, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 30/46, D_loss_real = 0.632, D_loss_fake = 0.576, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 31/46, D_loss_real = 0.637, D_loss_fake = 0.680, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 32/46, D_loss_real = 0.654, D_loss_fake = 0.590, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 33/46, D_loss_real = 0.590, D_loss_fake = 0.659, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.625, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 35/46, D_loss_real = 0.621, D_loss_fake = 0.618, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 36/46, D_loss_real = 0.606, D_loss_fake = 0.658, G_loss =
0.809
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 37/46, D_loss_real = 0.649, D_loss_fake = 0.624, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 38/46, D_loss_real = 0.619, D_loss_fake = 0.621, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 39/46, D_loss_real = 0.655, D_loss_fake = 0.605, G_loss =
0.816
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 40/46, D_loss_real = 0.641, D_loss_fake = 0.670, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 41/46, D_loss_real = 0.672, D_loss_fake = 0.606, G_loss =
0.766
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 42/46, D_loss_real = 0.642, D_loss_fake = 0.718, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 43/46, D_loss_real = 0.686, D_loss_fake = 0.624, G_loss =
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 448, Iteration 44/46, D_loss_real = 0.592, D_loss_fake = 0.598, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 45/46, D_loss_real = 0.666, D_loss_fake = 0.608, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 448, Iteration 46/46, D_loss_real = 0.695, D_loss_fake = 0.601, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 1/46, D_loss_real = 0.605, D_loss_fake = 0.642, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 2/46, D_loss_real = 0.663, D_loss_fake = 0.617, G_loss =
0.807
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 3/46, D_loss_real = 0.586, D_loss_fake = 0.632, G_loss =
0.773
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 4/46, D_loss_real = 0.602, D_loss_fake = 0.732, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 5/46, D_loss_real = 0.669, D_loss_fake = 0.667, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 6/46, D_loss_real = 0.605, D_loss_fake = 0.649, G_loss =
0.795
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 7/46, D_loss_real = 0.619, D_loss_fake = 0.640, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 8/46, D_loss_real = 0.637, D_loss_fake = 0.676, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 9/46, D_loss_real = 0.666, D_loss_fake = 0.617, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 10/46, D_loss_real = 0.670, D_loss_fake = 0.601, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.611, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 12/46, D_loss_real = 0.656, D_loss_fake = 0.656, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 13/46, D_loss_real = 0.628, D_loss_fake = 0.646, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 14/46, D_loss_real = 0.668, D_loss_fake = 0.581, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 15/46, D_loss_real = 0.656, D_loss_fake = 0.601, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 16/46, D_loss_real = 0.651, D_loss_fake = 0.647, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 17/46, D_loss_real = 0.677, D_loss_fake = 0.596, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 18/46, D_loss_real = 0.709, D_loss_fake = 0.652, G_loss =
0.881
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 19/46, D_loss_real = 0.700, D_loss_fake = 0.614, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 20/46, D_loss_real = 0.682, D_loss_fake = 0.594, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 21/46, D_loss_real = 0.664, D_loss_fake = 0.615, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 22/46, D_loss_real = 0.613, D_loss_fake = 0.633, G_loss =
0.839
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 23/46, D_loss_real = 0.623, D_loss_fake = 0.662, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 24/46, D_loss_real = 0.625, D_loss_fake = 0.626, G_loss =
0.790
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 25/46, D_loss_real = 0.606, D_loss_fake = 0.651, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 26/46, D_loss_real = 0.607, D_loss_fake = 0.635, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 27/46, D_loss_real = 0.639, D_loss_fake = 0.618, G_loss =
0.782
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 28/46, D_loss_real = 0.566, D_loss_fake = 0.643, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 29/46, D_loss_real = 0.614, D_loss_fake = 0.661, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 30/46, D_loss_real = 0.654, D_loss_fake = 0.638, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.645, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 32/46, D_loss_real = 0.716, D_loss_fake = 0.616, G_loss =
0.825
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 33/46, D_loss_real = 0.713, D_loss_fake = 0.611, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 34/46, D_loss_real = 0.690, D_loss_fake = 0.666, G_loss =
0.760
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 35/46, D_loss_real = 0.609, D_loss_fake = 0.621, G_loss =
0.785
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 36/46, D_loss_real = 0.657, D_loss_fake = 0.653, G_loss =
0.795
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 37/46, D_loss_real = 0.660, D_loss_fake = 0.626, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 38/46, D_loss_real = 0.628, D_loss_fake = 0.639, G_loss =
0.799
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 39/46, D_loss_real = 0.622, D_loss_fake = 0.652, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 40/46, D_loss_real = 0.630, D_loss_fake = 0.657, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 41/46, D_loss_real = 0.614, D_loss_fake = 0.687, G_loss =
0.803
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 42/46, D_loss_real = 0.632, D_loss_fake = 0.609, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 43/46, D_loss_real = 0.617, D_loss_fake = 0.636, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 44/46, D_loss_real = 0.648, D_loss_fake = 0.702, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 449, Iteration 45/46, D_loss_real = 0.681, D_loss_fake = 0.627, G_loss =
0.787
2/2 [==============================] - 0s 4ms/step
Epoch 449, Iteration 46/46, D_loss_real = 0.597, D_loss_fake = 0.635, G_loss =
0.760
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 1/46, D_loss_real = 0.599, D_loss_fake = 0.683, G_loss =
0.770
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 2/46, D_loss_real = 0.633, D_loss_fake = 0.673, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 3/46, D_loss_real = 0.622, D_loss_fake = 0.686, G_loss =
0.785
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 4/46, D_loss_real = 0.590, D_loss_fake = 0.638, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 5/46, D_loss_real = 0.644, D_loss_fake = 0.647, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 6/46, D_loss_real = 0.678, D_loss_fake = 0.628, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 7/46, D_loss_real = 0.683, D_loss_fake = 0.659, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 8/46, D_loss_real = 0.649, D_loss_fake = 0.651, G_loss =
0.811
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 9/46, D_loss_real = 0.678, D_loss_fake = 0.631, G_loss =
0.820
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 10/46, D_loss_real = 0.677, D_loss_fake = 0.588, G_loss =
0.783
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 11/46, D_loss_real = 0.621, D_loss_fake = 0.653, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 12/46, D_loss_real = 0.653, D_loss_fake = 0.634, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 13/46, D_loss_real = 0.730, D_loss_fake = 0.578, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 14/46, D_loss_real = 0.663, D_loss_fake = 0.645, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 15/46, D_loss_real = 0.595, D_loss_fake = 0.630, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 16/46, D_loss_real = 0.695, D_loss_fake = 0.618, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 17/46, D_loss_real = 0.649, D_loss_fake = 0.640, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 18/46, D_loss_real = 0.646, D_loss_fake = 0.630, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 19/46, D_loss_real = 0.604, D_loss_fake = 0.610, G_loss =
0.772
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 20/46, D_loss_real = 0.645, D_loss_fake = 0.619, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 21/46, D_loss_real = 0.609, D_loss_fake = 0.681, G_loss =
0.794
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 22/46, D_loss_real = 0.642, D_loss_fake = 0.695, G_loss =
0.781
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 23/46, D_loss_real = 0.593, D_loss_fake = 0.655, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 24/46, D_loss_real = 0.677, D_loss_fake = 0.630, G_loss =
0.845
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 25/46, D_loss_real = 0.659, D_loss_fake = 0.621, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.652, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.605, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 28/46, D_loss_real = 0.639, D_loss_fake = 0.620, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 29/46, D_loss_real = 0.635, D_loss_fake = 0.601, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 30/46, D_loss_real = 0.649, D_loss_fake = 0.673, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 31/46, D_loss_real = 0.660, D_loss_fake = 0.663, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 32/46, D_loss_real = 0.639, D_loss_fake = 0.627, G_loss =
0.778
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 33/46, D_loss_real = 0.637, D_loss_fake = 0.618, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 34/46, D_loss_real = 0.657, D_loss_fake = 0.670, G_loss =
0.786
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 35/46, D_loss_real = 0.645, D_loss_fake = 0.661, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 36/46, D_loss_real = 0.646, D_loss_fake = 0.646, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 37/46, D_loss_real = 0.621, D_loss_fake = 0.604, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 38/46, D_loss_real = 0.661, D_loss_fake = 0.608, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 39/46, D_loss_real = 0.624, D_loss_fake = 0.686, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 40/46, D_loss_real = 0.636, D_loss_fake = 0.641, G_loss =
0.823
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 41/46, D_loss_real = 0.684, D_loss_fake = 0.631, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.611, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 43/46, D_loss_real = 0.640, D_loss_fake = 0.625, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 44/46, D_loss_real = 0.602, D_loss_fake = 0.630, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 450, Iteration 45/46, D_loss_real = 0.604, D_loss_fake = 0.620, G_loss =
0.794
2/2 [==============================] - 0s 3ms/step
Epoch 450, Iteration 46/46, D_loss_real = 0.611, D_loss_fake = 0.615, G_loss =
0.780
2/2 [==============================] - 0s 4ms/step
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 1/46, D_loss_real = 0.645, D_loss_fake = 0.648, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 2/46, D_loss_real = 0.680, D_loss_fake = 0.681, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 3/46, D_loss_real = 0.641, D_loss_fake = 0.638, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 4/46, D_loss_real = 0.651, D_loss_fake = 0.648, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 5/46, D_loss_real = 0.704, D_loss_fake = 0.616, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 6/46, D_loss_real = 0.697, D_loss_fake = 0.595, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 7/46, D_loss_real = 0.680, D_loss_fake = 0.620, G_loss =
0.769
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 8/46, D_loss_real = 0.637, D_loss_fake = 0.657, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 9/46, D_loss_real = 0.663, D_loss_fake = 0.623, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 10/46, D_loss_real = 0.623, D_loss_fake = 0.615, G_loss =
```

```
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 11/46, D_loss_real = 0.663, D_loss_fake = 0.612, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 12/46, D_loss_real = 0.609, D_loss_fake = 0.615, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 13/46, D_loss_real = 0.677, D_loss_fake = 0.633, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 14/46, D_loss_real = 0.601, D_loss_fake = 0.694, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 15/46, D_loss_real = 0.638, D_loss_fake = 0.654, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 16/46, D_loss_real = 0.628, D_loss_fake = 0.626, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.631, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 18/46, D_loss_real = 0.675, D_loss_fake = 0.656, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 19/46, D_loss_real = 0.630, D_loss_fake = 0.650, G_loss =
0.856
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 20/46, D_loss_real = 0.635, D_loss_fake = 0.603, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.656, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 22/46, D_loss_real = 0.698, D_loss_fake = 0.594, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 23/46, D_loss_real = 0.630, D_loss_fake = 0.622, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 24/46, D_loss_real = 0.659, D_loss_fake = 0.659, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 25/46, D_loss_real = 0.606, D_loss_fake = 0.629, G_loss =
0.792
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 26/46, D_loss_real = 0.660, D_loss_fake = 0.616, G_loss =
```

```
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 27/46, D_loss_real = 0.653, D_loss_fake = 0.630, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 28/46, D_loss_real = 0.640, D_loss_fake = 0.631, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 29/46, D_loss_real = 0.661, D_loss_fake = 0.673, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 30/46, D_loss_real = 0.651, D_loss_fake = 0.646, G_loss =
0.792
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 31/46, D_loss_real = 0.580, D_loss_fake = 0.632, G_loss =
0.810
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 32/46, D_loss_real = 0.559, D_loss_fake = 0.610, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 33/46, D_loss_real = 0.688, D_loss_fake = 0.623, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 34/46, D_loss_real = 0.632, D_loss_fake = 0.621, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 35/46, D_loss_real = 0.577, D_loss_fake = 0.661, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 36/46, D_loss_real = 0.623, D_loss_fake = 0.636, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 37/46, D_loss_real = 0.732, D_loss_fake = 0.606, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 38/46, D_loss_real = 0.733, D_loss_fake = 0.580, G_loss =
0.870
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 39/46, D_loss_real = 0.721, D_loss_fake = 0.584, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.581, G_loss =
0.899
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 41/46, D_loss_real = 0.617, D_loss_fake = 0.619, G_loss =
0.803
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 42/46, D_loss_real = 0.613, D_loss_fake = 0.635, G_loss =
```

```
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 451, Iteration 43/46, D_loss_real = 0.635, D_loss_fake = 0.673, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 44/46, D_loss_real = 0.709, D_loss_fake = 0.672, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 45/46, D_loss_real = 0.642, D_loss_fake = 0.628, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 451, Iteration 46/46, D_loss_real = 0.636, D_loss_fake = 0.629, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 1/46, D_loss_real = 0.623, D_loss_fake = 0.632, G_loss =
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 2/46, D_loss_real = 0.650, D_loss_fake = 0.660, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 3/46, D_loss_real = 0.633, D_loss_fake = 0.600, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 4/46, D_loss_real = 0.645, D_loss_fake = 0.718, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 5/46, D_loss_real = 0.641, D_loss_fake = 0.622, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 6/46, D_loss_real = 0.604, D_loss_fake = 0.717, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 7/46, D_loss_real = 0.588, D_loss_fake = 0.608, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 8/46, D_loss_real = 0.651, D_loss_fake = 0.629, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 9/46, D_loss_real = 0.691, D_loss_fake = 0.632, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 10/46, D_loss_real = 0.668, D_loss_fake = 0.651, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 11/46, D_loss_real = 0.641, D_loss_fake = 0.644, G_loss =
0.767
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 12/46, D_loss_real = 0.658, D_loss_fake = 0.620, G_loss =
```

```
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 13/46, D_loss_real = 0.577, D_loss_fake = 0.670, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 14/46, D_loss_real = 0.665, D_loss_fake = 0.677, G_loss =
0.859
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.603, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 16/46, D_loss_real = 0.656, D_loss_fake = 0.637, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 17/46, D_loss_real = 0.657, D_loss_fake = 0.652, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 18/46, D_loss_real = 0.683, D_loss_fake = 0.637, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.629, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 20/46, D_loss_real = 0.633, D_loss_fake = 0.637, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 21/46, D_loss_real = 0.668, D_loss_fake = 0.650, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 22/46, D_loss_real = 0.613, D_loss_fake = 0.643, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 23/46, D_loss_real = 0.662, D_loss_fake = 0.674, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 24/46, D_loss_real = 0.686, D_loss_fake = 0.639, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 25/46, D_loss_real = 0.673, D_loss_fake = 0.607, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 26/46, D_loss_real = 0.644, D_loss_fake = 0.587, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 27/46, D_loss_real = 0.651, D_loss_fake = 0.598, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 28/46, D_loss_real = 0.648, D_loss_fake = 0.636, G_loss =
```

```
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 29/46, D_loss_real = 0.624, D_loss_fake = 0.654, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 30/46, D_loss_real = 0.629, D_loss_fake = 0.611, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 31/46, D_loss_real = 0.685, D_loss_fake = 0.644, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 32/46, D_loss_real = 0.615, D_loss_fake = 0.619, G_loss =
0.783
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 33/46, D_loss_real = 0.659, D_loss_fake = 0.676, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 34/46, D_loss_real = 0.650, D_loss_fake = 0.602, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 35/46, D_loss_real = 0.630, D_loss_fake = 0.624, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 36/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 37/46, D_loss_real = 0.652, D_loss_fake = 0.568, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 38/46, D_loss_real = 0.699, D_loss_fake = 0.651, G_loss =
0.804
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 39/46, D_loss_real = 0.711, D_loss_fake = 0.632, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 40/46, D_loss_real = 0.621, D_loss_fake = 0.620, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 41/46, D_loss_real = 0.711, D_loss_fake = 0.653, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 42/46, D_loss_real = 0.624, D_loss_fake = 0.701, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 43/46, D_loss_real = 0.631, D_loss_fake = 0.572, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 44/46, D_loss_real = 0.652, D_loss_fake = 0.570, G_loss =
```

```
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 452, Iteration 45/46, D_loss_real = 0.652, D_loss_fake = 0.613, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 452, Iteration 46/46, D_loss_real = 0.626, D_loss_fake = 0.680, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 1/46, D_loss_real = 0.641, D_loss_fake = 0.631, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 2/46, D_loss_real = 0.602, D_loss_fake = 0.679, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 3/46, D_loss_real = 0.674, D_loss_fake = 0.645, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 4/46, D_loss_real = 0.628, D_loss_fake = 0.654, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 5/46, D_loss_real = 0.654, D_loss_fake = 0.602, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 6/46, D_loss_real = 0.647, D_loss_fake = 0.635, G_loss =
0.813
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.653, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 8/46, D_loss_real = 0.590, D_loss_fake = 0.613, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 9/46, D_loss_real = 0.650, D_loss_fake = 0.644, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 10/46, D_loss_real = 0.587, D_loss_fake = 0.717, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 11/46, D_loss_real = 0.620, D_loss_fake = 0.639, G_loss =
0.776
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 12/46, D_loss_real = 0.584, D_loss_fake = 0.658, G_loss =
0.743
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 13/46, D_loss_real = 0.571, D_loss_fake = 0.654, G_loss =
0.782
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 14/46, D_loss_real = 0.644, D_loss_fake = 0.681, G_loss =
```

```
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 15/46, D_loss_real = 0.669, D_loss_fake = 0.630, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 17/46, D_loss_real = 0.655, D_loss_fake = 0.671, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 18/46, D_loss_real = 0.667, D_loss_fake = 0.600, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 19/46, D_loss_real = 0.639, D_loss_fake = 0.645, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 20/46, D_loss_real = 0.666, D_loss_fake = 0.638, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 21/46, D_loss_real = 0.620, D_loss_fake = 0.620, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 22/46, D_loss_real = 0.683, D_loss_fake = 0.659, G_loss =
0.785
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 23/46, D_loss_real = 0.661, D_loss_fake = 0.604, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 24/46, D_loss_real = 0.714, D_loss_fake = 0.634, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 25/46, D_loss_real = 0.645, D_loss_fake = 0.619, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.576, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 27/46, D_loss_real = 0.595, D_loss_fake = 0.639, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 28/46, D_loss_real = 0.669, D_loss_fake = 0.628, G_loss =
0.802
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 29/46, D_loss_real = 0.595, D_loss_fake = 0.698, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 30/46, D_loss_real = 0.580, D_loss_fake = 0.629, G_loss =
```

```
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 31/46, D_loss_real = 0.685, D_loss_fake = 0.617, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 453, Iteration 32/46, D_loss_real = 0.647, D_loss_fake = 0.607, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 33/46, D_loss_real = 0.664, D_loss_fake = 0.646, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 34/46, D_loss_real = 0.633, D_loss_fake = 0.645, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.664, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 36/46, D_loss_real = 0.650, D_loss_fake = 0.626, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 37/46, D_loss_real = 0.662, D_loss_fake = 0.584, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.591, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 39/46, D_loss_real = 0.628, D_loss_fake = 0.602, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 40/46, D_loss_real = 0.674, D_loss_fake = 0.589, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 41/46, D_loss_real = 0.600, D_loss_fake = 0.595, G_loss =
0.868
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 42/46, D_loss_real = 0.718, D_loss_fake = 0.655, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 43/46, D_loss_real = 0.661, D_loss_fake = 0.645, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 44/46, D_loss_real = 0.630, D_loss_fake = 0.603, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 45/46, D_loss_real = 0.593, D_loss_fake = 0.631, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 453, Iteration 46/46, D_loss_real = 0.667, D_loss_fake = 0.577, G_loss =
```

```
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 1/46, D_loss_real = 0.602, D_loss_fake = 0.590, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 2/46, D_loss_real = 0.643, D_loss_fake = 0.639, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 3/46, D_loss_real = 0.668, D_loss_fake = 0.621, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 4/46, D_loss_real = 0.611, D_loss_fake = 0.646, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 5/46, D_loss_real = 0.620, D_loss_fake = 0.630, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 6/46, D_loss_real = 0.640, D_loss_fake = 0.629, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 7/46, D_loss_real = 0.616, D_loss_fake = 0.652, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 8/46, D_loss_real = 0.647, D_loss_fake = 0.685, G_loss =
0.873
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 9/46, D_loss_real = 0.618, D_loss_fake = 0.630, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 10/46, D_loss_real = 0.656, D_loss_fake = 0.652, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.594, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 12/46, D_loss_real = 0.669, D_loss_fake = 0.615, G_loss =
0.795
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 13/46, D_loss_real = 0.620, D_loss_fake = 0.665, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 14/46, D_loss_real = 0.707, D_loss_fake = 0.610, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 15/46, D_loss_real = 0.627, D_loss_fake = 0.686, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 16/46, D_loss_real = 0.700, D_loss_fake = 0.653, G_loss =
```

0.867
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 17/46, D_loss_real = 0.660, D_loss_fake = 0.677, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 18/46, D_loss_real = 0.662, D_loss_fake = 0.589, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 19/46, D_loss_real = 0.653, D_loss_fake = 0.644, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 20/46, D_loss_real = 0.639, D_loss_fake = 0.634, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 21/46, D_loss_real = 0.552, D_loss_fake = 0.642, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 22/46, D_loss_real = 0.601, D_loss_fake = 0.617, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 23/46, D_loss_real = 0.650, D_loss_fake = 0.614, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 24/46, D_loss_real = 0.626, D_loss_fake = 0.646, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 25/46, D_loss_real = 0.638, D_loss_fake = 0.652, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 26/46, D_loss_real = 0.631, D_loss_fake = 0.612, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.621, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 28/46, D_loss_real = 0.717, D_loss_fake = 0.617, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 29/46, D_loss_real = 0.622, D_loss_fake = 0.664, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 30/46, D_loss_real = 0.558, D_loss_fake = 0.641, G_loss =
0.778
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 31/46, D_loss_real = 0.611, D_loss_fake = 0.642, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 32/46, D_loss_real = 0.643, D_loss_fake = 0.667, G_loss =

```
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 33/46, D_loss_real = 0.662, D_loss_fake = 0.629, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 34/46, D_loss_real = 0.726, D_loss_fake = 0.625, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 35/46, D_loss_real = 0.730, D_loss_fake = 0.583, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 36/46, D_loss_real = 0.623, D_loss_fake = 0.573, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 37/46, D_loss_real = 0.634, D_loss_fake = 0.629, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 454, Iteration 38/46, D_loss_real = 0.636, D_loss_fake = 0.648, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.645, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.645, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 41/46, D_loss_real = 0.596, D_loss_fake = 0.670, G_loss =
0.798
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 42/46, D_loss_real = 0.645, D_loss_fake = 0.667, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 43/46, D_loss_real = 0.612, D_loss_fake = 0.637, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 44/46, D_loss_real = 0.604, D_loss_fake = 0.659, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 45/46, D_loss_real = 0.670, D_loss_fake = 0.604, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 454, Iteration 46/46, D_loss_real = 0.622, D_loss_fake = 0.678, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 1/46, D_loss_real = 0.623, D_loss_fake = 0.612, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 2/46, D_loss_real = 0.648, D_loss_fake = 0.592, G_loss =
```

```
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 3/46, D_loss_real = 0.561, D_loss_fake = 0.669, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 4/46, D_loss_real = 0.611, D_loss_fake = 0.687, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 5/46, D_loss_real = 0.639, D_loss_fake = 0.665, G_loss =
0.764
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 6/46, D_loss_real = 0.636, D_loss_fake = 0.636, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 7/46, D_loss_real = 0.672, D_loss_fake = 0.611, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 8/46, D_loss_real = 0.692, D_loss_fake = 0.594, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 9/46, D_loss_real = 0.673, D_loss_fake = 0.611, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.638, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.607, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 12/46, D_loss_real = 0.703, D_loss_fake = 0.598, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 13/46, D_loss_real = 0.628, D_loss_fake = 0.684, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 14/46, D_loss_real = 0.616, D_loss_fake = 0.644, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 15/46, D_loss_real = 0.620, D_loss_fake = 0.676, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 16/46, D_loss_real = 0.707, D_loss_fake = 0.641, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 17/46, D_loss_real = 0.725, D_loss_fake = 0.622, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 18/46, D_loss_real = 0.638, D_loss_fake = 0.621, G_loss =
```

```
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 19/46, D_loss_real = 0.619, D_loss_fake = 0.592, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 20/46, D_loss_real = 0.644, D_loss_fake = 0.636, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 21/46, D_loss_real = 0.610, D_loss_fake = 0.648, G_loss =
0.753
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 22/46, D_loss_real = 0.636, D_loss_fake = 0.651, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 23/46, D_loss_real = 0.677, D_loss_fake = 0.634, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 24/46, D_loss_real = 0.668, D_loss_fake = 0.677, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 25/46, D_loss_real = 0.601, D_loss_fake = 0.676, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 26/46, D_loss_real = 0.641, D_loss_fake = 0.651, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 27/46, D_loss_real = 0.664, D_loss_fake = 0.590, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.607, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 29/46, D_loss_real = 0.719, D_loss_fake = 0.636, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 30/46, D_loss_real = 0.652, D_loss_fake = 0.589, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 31/46, D_loss_real = 0.663, D_loss_fake = 0.604, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 32/46, D_loss_real = 0.637, D_loss_fake = 0.644, G_loss =
0.825
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 33/46, D_loss_real = 0.668, D_loss_fake = 0.646, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 34/46, D_loss_real = 0.701, D_loss_fake = 0.633, G_loss =
```

```
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 35/46, D_loss_real = 0.651, D_loss_fake = 0.645, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 36/46, D_loss_real = 0.639, D_loss_fake = 0.658, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 37/46, D_loss_real = 0.651, D_loss_fake = 0.617, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 38/46, D_loss_real = 0.618, D_loss_fake = 0.652, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 39/46, D_loss_real = 0.654, D_loss_fake = 0.646, G_loss =
0.794
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 40/46, D_loss_real = 0.654, D_loss_fake = 0.706, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 41/46, D_loss_real = 0.623, D_loss_fake = 0.627, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 42/46, D_loss_real = 0.625, D_loss_fake = 0.669, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 43/46, D_loss_real = 0.714, D_loss_fake = 0.654, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 44/46, D_loss_real = 0.563, D_loss_fake = 0.655, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 455, Iteration 45/46, D_loss_real = 0.579, D_loss_fake = 0.670, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 455, Iteration 46/46, D_loss_real = 0.596, D_loss_fake = 0.632, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 1/46, D_loss_real = 0.657, D_loss_fake = 0.634, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 2/46, D_loss_real = 0.691, D_loss_fake = 0.611, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 3/46, D_loss_real = 0.550, D_loss_fake = 0.576, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 4/46, D_loss_real = 0.646, D_loss_fake = 0.646, G_loss =
```

```
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 5/46, D_loss_real = 0.632, D_loss_fake = 0.576, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.585, G_loss =
0.878
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 7/46, D_loss_real = 0.720, D_loss_fake = 0.614, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 8/46, D_loss_real = 0.613, D_loss_fake = 0.584, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 9/46, D_loss_real = 0.671, D_loss_fake = 0.674, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 10/46, D_loss_real = 0.659, D_loss_fake = 0.582, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 11/46, D_loss_real = 0.631, D_loss_fake = 0.657, G_loss =
0.866
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 12/46, D_loss_real = 0.613, D_loss_fake = 0.606, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 13/46, D_loss_real = 0.652, D_loss_fake = 0.651, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 14/46, D_loss_real = 0.664, D_loss_fake = 0.643, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 15/46, D_loss_real = 0.668, D_loss_fake = 0.630, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 16/46, D_loss_real = 0.562, D_loss_fake = 0.599, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 17/46, D_loss_real = 0.606, D_loss_fake = 0.670, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 18/46, D_loss_real = 0.621, D_loss_fake = 0.677, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 19/46, D_loss_real = 0.644, D_loss_fake = 0.654, G_loss =
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 20/46, D_loss_real = 0.599, D_loss_fake = 0.655, G_loss =
```

```
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 21/46, D_loss_real = 0.723, D_loss_fake = 0.591, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 22/46, D_loss_real = 0.648, D_loss_fake = 0.578, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 23/46, D_loss_real = 0.642, D_loss_fake = 0.635, G_loss =
0.847
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 24/46, D_loss_real = 0.660, D_loss_fake = 0.609, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 25/46, D_loss_real = 0.661, D_loss_fake = 0.689, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 26/46, D_loss_real = 0.638, D_loss_fake = 0.659, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 27/46, D_loss_real = 0.615, D_loss_fake = 0.679, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 28/46, D_loss_real = 0.688, D_loss_fake = 0.611, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.619, G_loss =
0.822
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 30/46, D_loss_real = 0.680, D_loss_fake = 0.574, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 31/46, D_loss_real = 0.618, D_loss_fake = 0.662, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 32/46, D_loss_real = 0.665, D_loss_fake = 0.657, G_loss =
0.821
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 33/46, D_loss_real = 0.668, D_loss_fake = 0.604, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 34/46, D_loss_real = 0.645, D_loss_fake = 0.589, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 35/46, D_loss_real = 0.672, D_loss_fake = 0.613, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 36/46, D_loss_real = 0.700, D_loss_fake = 0.610, G_loss =
```

```
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 37/46, D_loss_real = 0.688, D_loss_fake = 0.684, G_loss =
0.829
2/2 [==============================] - 0s 5ms/step
Epoch 456, Iteration 38/46, D_loss_real = 0.646, D_loss_fake = 0.640, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 39/46, D_loss_real = 0.643, D_loss_fake = 0.616, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.650, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 41/46, D_loss_real = 0.701, D_loss_fake = 0.652, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 42/46, D_loss_real = 0.620, D_loss_fake = 0.588, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 456, Iteration 43/46, D_loss_real = 0.641, D_loss_fake = 0.639, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 44/46, D_loss_real = 0.639, D_loss_fake = 0.682, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 45/46, D_loss_real = 0.625, D_loss_fake = 0.645, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 456, Iteration 46/46, D_loss_real = 0.658, D_loss_fake = 0.652, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 1/46, D_loss_real = 0.656, D_loss_fake = 0.622, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 2/46, D_loss_real = 0.643, D_loss_fake = 0.617, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 3/46, D_loss_real = 0.632, D_loss_fake = 0.632, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 4/46, D_loss_real = 0.657, D_loss_fake = 0.585, G_loss =
0.788
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 5/46, D_loss_real = 0.595, D_loss_fake = 0.659, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 6/46, D_loss_real = 0.644, D_loss_fake = 0.661, G_loss =
```

```
0.778
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 7/46, D_loss_real = 0.649, D_loss_fake = 0.629, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 8/46, D_loss_real = 0.698, D_loss_fake = 0.648, G_loss =
0.858
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 9/46, D_loss_real = 0.642, D_loss_fake = 0.620, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 10/46, D_loss_real = 0.689, D_loss_fake = 0.625, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 11/46, D_loss_real = 0.605, D_loss_fake = 0.677, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 12/46, D_loss_real = 0.654, D_loss_fake = 0.615, G_loss =
0.800
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 13/46, D_loss_real = 0.627, D_loss_fake = 0.664, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 14/46, D_loss_real = 0.618, D_loss_fake = 0.636, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 15/46, D_loss_real = 0.621, D_loss_fake = 0.679, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 16/46, D_loss_real = 0.677, D_loss_fake = 0.636, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 17/46, D_loss_real = 0.626, D_loss_fake = 0.640, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 18/46, D_loss_real = 0.653, D_loss_fake = 0.647, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 19/46, D_loss_real = 0.622, D_loss_fake = 0.618, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 20/46, D_loss_real = 0.654, D_loss_fake = 0.640, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 21/46, D_loss_real = 0.592, D_loss_fake = 0.624, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.581, G_loss =
```

```
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 23/46, D_loss_real = 0.651, D_loss_fake = 0.624, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 24/46, D_loss_real = 0.585, D_loss_fake = 0.577, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.633, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 26/46, D_loss_real = 0.603, D_loss_fake = 0.650, G_loss =
0.767
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 27/46, D_loss_real = 0.663, D_loss_fake = 0.689, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 28/46, D_loss_real = 0.658, D_loss_fake = 0.669, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 29/46, D_loss_real = 0.655, D_loss_fake = 0.630, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 30/46, D_loss_real = 0.590, D_loss_fake = 0.644, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 31/46, D_loss_real = 0.662, D_loss_fake = 0.620, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 32/46, D_loss_real = 0.604, D_loss_fake = 0.642, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 33/46, D_loss_real = 0.640, D_loss_fake = 0.691, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 34/46, D_loss_real = 0.649, D_loss_fake = 0.665, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 35/46, D_loss_real = 0.652, D_loss_fake = 0.645, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 36/46, D_loss_real = 0.652, D_loss_fake = 0.630, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 37/46, D_loss_real = 0.680, D_loss_fake = 0.640, G_loss =
0.863
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 38/46, D_loss_real = 0.612, D_loss_fake = 0.635, G_loss =
```

```
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 39/46, D_loss_real = 0.651, D_loss_fake = 0.666, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 40/46, D_loss_real = 0.617, D_loss_fake = 0.674, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 41/46, D_loss_real = 0.637, D_loss_fake = 0.557, G_loss =
0.878
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 42/46, D_loss_real = 0.709, D_loss_fake = 0.563, G_loss =
0.910
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 43/46, D_loss_real = 0.695, D_loss_fake = 0.584, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 457, Iteration 45/46, D_loss_real = 0.676, D_loss_fake = 0.689, G_loss =
0.859
2/2 [==============================] - 0s 4ms/step
Epoch 457, Iteration 46/46, D_loss_real = 0.714, D_loss_fake = 0.614, G_loss =
0.896
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 1/46, D_loss_real = 0.617, D_loss_fake = 0.632, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 2/46, D_loss_real = 0.672, D_loss_fake = 0.685, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 3/46, D_loss_real = 0.729, D_loss_fake = 0.584, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 4/46, D_loss_real = 0.668, D_loss_fake = 0.620, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 5/46, D_loss_real = 0.620, D_loss_fake = 0.589, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 6/46, D_loss_real = 0.643, D_loss_fake = 0.669, G_loss =
0.786
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 7/46, D_loss_real = 0.636, D_loss_fake = 0.629, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 8/46, D_loss_real = 0.678, D_loss_fake = 0.612, G_loss =
```

```
0.785
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 9/46, D_loss_real = 0.615, D_loss_fake = 0.677, G_loss =
0.784
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 10/46, D_loss_real = 0.667, D_loss_fake = 0.663, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 11/46, D_loss_real = 0.682, D_loss_fake = 0.639, G_loss =
0.781
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 12/46, D_loss_real = 0.628, D_loss_fake = 0.672, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 13/46, D_loss_real = 0.640, D_loss_fake = 0.671, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 14/46, D_loss_real = 0.682, D_loss_fake = 0.616, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 15/46, D_loss_real = 0.612, D_loss_fake = 0.653, G_loss =
0.895
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 16/46, D_loss_real = 0.637, D_loss_fake = 0.587, G_loss =
0.899
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 17/46, D_loss_real = 0.670, D_loss_fake = 0.583, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 18/46, D_loss_real = 0.653, D_loss_fake = 0.662, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 19/46, D_loss_real = 0.652, D_loss_fake = 0.670, G_loss =
0.810
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 20/46, D_loss_real = 0.648, D_loss_fake = 0.628, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 21/46, D_loss_real = 0.605, D_loss_fake = 0.631, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 22/46, D_loss_real = 0.629, D_loss_fake = 0.690, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 23/46, D_loss_real = 0.614, D_loss_fake = 0.582, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 24/46, D_loss_real = 0.590, D_loss_fake = 0.662, G_loss =
```

```
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 25/46, D_loss_real = 0.689, D_loss_fake = 0.596, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.625, G_loss =
0.858
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 27/46, D_loss_real = 0.633, D_loss_fake = 0.599, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 28/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 29/46, D_loss_real = 0.662, D_loss_fake = 0.641, G_loss =
0.807
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 30/46, D_loss_real = 0.709, D_loss_fake = 0.649, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 31/46, D_loss_real = 0.613, D_loss_fake = 0.654, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 32/46, D_loss_real = 0.654, D_loss_fake = 0.666, G_loss =
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 33/46, D_loss_real = 0.665, D_loss_fake = 0.675, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 34/46, D_loss_real = 0.605, D_loss_fake = 0.651, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 35/46, D_loss_real = 0.622, D_loss_fake = 0.665, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 36/46, D_loss_real = 0.735, D_loss_fake = 0.643, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 37/46, D_loss_real = 0.670, D_loss_fake = 0.631, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 38/46, D_loss_real = 0.639, D_loss_fake = 0.582, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 39/46, D_loss_real = 0.675, D_loss_fake = 0.561, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 40/46, D_loss_real = 0.668, D_loss_fake = 0.611, G_loss =
```

```
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 41/46, D_loss_real = 0.606, D_loss_fake = 0.594, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 42/46, D_loss_real = 0.682, D_loss_fake = 0.629, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 43/46, D_loss_real = 0.631, D_loss_fake = 0.626, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 44/46, D_loss_real = 0.630, D_loss_fake = 0.670, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 458, Iteration 45/46, D_loss_real = 0.623, D_loss_fake = 0.656, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 458, Iteration 46/46, D_loss_real = 0.615, D_loss_fake = 0.612, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 1/46, D_loss_real = 0.655, D_loss_fake = 0.613, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 2/46, D_loss_real = 0.672, D_loss_fake = 0.684, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 3/46, D_loss_real = 0.635, D_loss_fake = 0.632, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 4/46, D_loss_real = 0.637, D_loss_fake = 0.652, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 5/46, D_loss_real = 0.653, D_loss_fake = 0.634, G_loss =
0.796
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 6/46, D_loss_real = 0.594, D_loss_fake = 0.695, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 8/46, D_loss_real = 0.634, D_loss_fake = 0.604, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.629, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 10/46, D_loss_real = 0.601, D_loss_fake = 0.634, G_loss =
```

```
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 11/46, D_loss_real = 0.609, D_loss_fake = 0.615, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 12/46, D_loss_real = 0.633, D_loss_fake = 0.644, G_loss =
0.758
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 13/46, D_loss_real = 0.687, D_loss_fake = 0.687, G_loss =
0.798
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 14/46, D_loss_real = 0.609, D_loss_fake = 0.644, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 15/46, D_loss_real = 0.623, D_loss_fake = 0.622, G_loss =
0.865
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 16/46, D_loss_real = 0.657, D_loss_fake = 0.639, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 17/46, D_loss_real = 0.732, D_loss_fake = 0.612, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 18/46, D_loss_real = 0.591, D_loss_fake = 0.565, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 19/46, D_loss_real = 0.665, D_loss_fake = 0.636, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 20/46, D_loss_real = 0.677, D_loss_fake = 0.667, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 21/46, D_loss_real = 0.689, D_loss_fake = 0.645, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 22/46, D_loss_real = 0.698, D_loss_fake = 0.620, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 23/46, D_loss_real = 0.668, D_loss_fake = 0.682, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 24/46, D_loss_real = 0.663, D_loss_fake = 0.603, G_loss =
0.858
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 25/46, D_loss_real = 0.665, D_loss_fake = 0.602, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.611, G_loss =
```

```
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 27/46, D_loss_real = 0.633, D_loss_fake = 0.632, G_loss =
0.793
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 28/46, D_loss_real = 0.609, D_loss_fake = 0.600, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 29/46, D_loss_real = 0.660, D_loss_fake = 0.637, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 30/46, D_loss_real = 0.631, D_loss_fake = 0.688, G_loss =
0.796
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 31/46, D_loss_real = 0.616, D_loss_fake = 0.655, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 32/46, D_loss_real = 0.635, D_loss_fake = 0.628, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 33/46, D_loss_real = 0.561, D_loss_fake = 0.681, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 34/46, D_loss_real = 0.590, D_loss_fake = 0.606, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 35/46, D_loss_real = 0.685, D_loss_fake = 0.584, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 36/46, D_loss_real = 0.683, D_loss_fake = 0.603, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 37/46, D_loss_real = 0.582, D_loss_fake = 0.647, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 38/46, D_loss_real = 0.665, D_loss_fake = 0.672, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.644, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 40/46, D_loss_real = 0.637, D_loss_fake = 0.635, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 41/46, D_loss_real = 0.635, D_loss_fake = 0.619, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 42/46, D_loss_real = 0.646, D_loss_fake = 0.629, G_loss =
```

```
0.813
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 43/46, D_loss_real = 0.592, D_loss_fake = 0.679, G_loss =
0.873
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 44/46, D_loss_real = 0.675, D_loss_fake = 0.627, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 459, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.631, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 459, Iteration 46/46, D_loss_real = 0.612, D_loss_fake = 0.632, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.654, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 2/46, D_loss_real = 0.612, D_loss_fake = 0.620, G_loss =
0.847
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 3/46, D_loss_real = 0.659, D_loss_fake = 0.619, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.614, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 5/46, D_loss_real = 0.625, D_loss_fake = 0.588, G_loss =
0.825
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 6/46, D_loss_real = 0.634, D_loss_fake = 0.653, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 7/46, D_loss_real = 0.634, D_loss_fake = 0.600, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 8/46, D_loss_real = 0.663, D_loss_fake = 0.632, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 9/46, D_loss_real = 0.649, D_loss_fake = 0.587, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 10/46, D_loss_real = 0.632, D_loss_fake = 0.664, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 11/46, D_loss_real = 0.633, D_loss_fake = 0.603, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 12/46, D_loss_real = 0.686, D_loss_fake = 0.587, G_loss =
```

```
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 13/46, D_loss_real = 0.613, D_loss_fake = 0.621, G_loss =
0.822
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 14/46, D_loss_real = 0.636, D_loss_fake = 0.635, G_loss =
0.820
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 15/46, D_loss_real = 0.622, D_loss_fake = 0.613, G_loss =
0.872
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 16/46, D_loss_real = 0.651, D_loss_fake = 0.645, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 17/46, D_loss_real = 0.656, D_loss_fake = 0.624, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 18/46, D_loss_real = 0.716, D_loss_fake = 0.666, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 19/46, D_loss_real = 0.608, D_loss_fake = 0.647, G_loss =
0.790
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.644, G_loss =
0.773
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 21/46, D_loss_real = 0.637, D_loss_fake = 0.627, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 22/46, D_loss_real = 0.656, D_loss_fake = 0.678, G_loss =
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 23/46, D_loss_real = 0.628, D_loss_fake = 0.658, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 24/46, D_loss_real = 0.648, D_loss_fake = 0.625, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 25/46, D_loss_real = 0.667, D_loss_fake = 0.585, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 26/46, D_loss_real = 0.634, D_loss_fake = 0.611, G_loss =
0.790
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 27/46, D_loss_real = 0.654, D_loss_fake = 0.615, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 28/46, D_loss_real = 0.621, D_loss_fake = 0.663, G_loss =
```

```
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 29/46, D_loss_real = 0.656, D_loss_fake = 0.631, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.622, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 31/46, D_loss_real = 0.638, D_loss_fake = 0.649, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 32/46, D_loss_real = 0.591, D_loss_fake = 0.631, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 33/46, D_loss_real = 0.643, D_loss_fake = 0.649, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 34/46, D_loss_real = 0.623, D_loss_fake = 0.609, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.643, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 36/46, D_loss_real = 0.645, D_loss_fake = 0.610, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.692, G_loss =
0.872
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 38/46, D_loss_real = 0.678, D_loss_fake = 0.568, G_loss =
0.810
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 39/46, D_loss_real = 0.695, D_loss_fake = 0.594, G_loss =
0.884
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 40/46, D_loss_real = 0.622, D_loss_fake = 0.663, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 41/46, D_loss_real = 0.643, D_loss_fake = 0.608, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.647, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 43/46, D_loss_real = 0.684, D_loss_fake = 0.745, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 44/46, D_loss_real = 0.632, D_loss_fake = 0.663, G_loss =
```

```
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 460, Iteration 45/46, D_loss_real = 0.641, D_loss_fake = 0.631, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 460, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.632, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 1/46, D_loss_real = 0.655, D_loss_fake = 0.669, G_loss =
0.781
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 2/46, D_loss_real = 0.627, D_loss_fake = 0.708, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.630, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 4/46, D_loss_real = 0.666, D_loss_fake = 0.593, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 5/46, D_loss_real = 0.662, D_loss_fake = 0.608, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 6/46, D_loss_real = 0.649, D_loss_fake = 0.624, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 7/46, D_loss_real = 0.637, D_loss_fake = 0.617, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 8/46, D_loss_real = 0.624, D_loss_fake = 0.596, G_loss =
0.798
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.623, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 10/46, D_loss_real = 0.649, D_loss_fake = 0.666, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 11/46, D_loss_real = 0.657, D_loss_fake = 0.613, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 12/46, D_loss_real = 0.565, D_loss_fake = 0.641, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 13/46, D_loss_real = 0.650, D_loss_fake = 0.668, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 461, Iteration 14/46, D_loss_real = 0.661, D_loss_fake = 0.622, G_loss =
0.794
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 15/46, D_loss_real = 0.646, D_loss_fake = 0.646, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 16/46, D_loss_real = 0.627, D_loss_fake = 0.674, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 17/46, D_loss_real = 0.585, D_loss_fake = 0.695, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 18/46, D_loss_real = 0.597, D_loss_fake = 0.627, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 19/46, D_loss_real = 0.615, D_loss_fake = 0.621, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 20/46, D_loss_real = 0.669, D_loss_fake = 0.638, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 21/46, D_loss_real = 0.567, D_loss_fake = 0.622, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 22/46, D_loss_real = 0.652, D_loss_fake = 0.663, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 23/46, D_loss_real = 0.679, D_loss_fake = 0.609, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 24/46, D_loss_real = 0.629, D_loss_fake = 0.636, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 25/46, D_loss_real = 0.677, D_loss_fake = 0.580, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 26/46, D_loss_real = 0.676, D_loss_fake = 0.607, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 27/46, D_loss_real = 0.663, D_loss_fake = 0.697, G_loss =
0.804
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 28/46, D_loss_real = 0.692, D_loss_fake = 0.627, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 29/46, D_loss_real = 0.660, D_loss_fake = 0.668, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 461, Iteration 30/46, D_loss_real = 0.612, D_loss_fake = 0.627, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 31/46, D_loss_real = 0.623, D_loss_fake = 0.646, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 32/46, D_loss_real = 0.651, D_loss_fake = 0.605, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 33/46, D_loss_real = 0.658, D_loss_fake = 0.580, G_loss =
0.842
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 34/46, D_loss_real = 0.623, D_loss_fake = 0.613, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 35/46, D_loss_real = 0.665, D_loss_fake = 0.660, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 36/46, D_loss_real = 0.617, D_loss_fake = 0.653, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 37/46, D_loss_real = 0.669, D_loss_fake = 0.629, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 38/46, D_loss_real = 0.640, D_loss_fake = 0.617, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 39/46, D_loss_real = 0.709, D_loss_fake = 0.631, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 40/46, D_loss_real = 0.624, D_loss_fake = 0.655, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 41/46, D_loss_real = 0.632, D_loss_fake = 0.619, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 42/46, D_loss_real = 0.607, D_loss_fake = 0.641, G_loss =
0.800
2/2 [==============================] - 0s 4ms/step
Epoch 461, Iteration 43/46, D_loss_real = 0.608, D_loss_fake = 0.619, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 44/46, D_loss_real = 0.619, D_loss_fake = 0.685, G_loss =
0.795
2/2 [==============================] - 0s 3ms/step
Epoch 461, Iteration 45/46, D_loss_real = 0.653, D_loss_fake = 0.603, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 461, Iteration 46/46, D_loss_real = 0.545, D_loss_fake = 0.623, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 1/46, D_loss_real = 0.623, D_loss_fake = 0.675, G_loss =
0.775
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 2/46, D_loss_real = 0.670, D_loss_fake = 0.674, G_loss =
0.821
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 3/46, D_loss_real = 0.685, D_loss_fake = 0.644, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 4/46, D_loss_real = 0.623, D_loss_fake = 0.741, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 5/46, D_loss_real = 0.651, D_loss_fake = 0.668, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 6/46, D_loss_real = 0.658, D_loss_fake = 0.595, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 7/46, D_loss_real = 0.736, D_loss_fake = 0.632, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 8/46, D_loss_real = 0.683, D_loss_fake = 0.617, G_loss =
0.825
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 9/46, D_loss_real = 0.640, D_loss_fake = 0.688, G_loss =
0.878
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 10/46, D_loss_real = 0.628, D_loss_fake = 0.657, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.614, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 12/46, D_loss_real = 0.664, D_loss_fake = 0.625, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 13/46, D_loss_real = 0.667, D_loss_fake = 0.619, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 14/46, D_loss_real = 0.646, D_loss_fake = 0.591, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 15/46, D_loss_real = 0.656, D_loss_fake = 0.653, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 462, Iteration 16/46, D_loss_real = 0.571, D_loss_fake = 0.625, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 17/46, D_loss_real = 0.652, D_loss_fake = 0.656, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 18/46, D_loss_real = 0.675, D_loss_fake = 0.605, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 19/46, D_loss_real = 0.627, D_loss_fake = 0.608, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 20/46, D_loss_real = 0.658, D_loss_fake = 0.596, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 21/46, D_loss_real = 0.646, D_loss_fake = 0.622, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 22/46, D_loss_real = 0.610, D_loss_fake = 0.698, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.616, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 24/46, D_loss_real = 0.638, D_loss_fake = 0.634, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 25/46, D_loss_real = 0.678, D_loss_fake = 0.614, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 26/46, D_loss_real = 0.623, D_loss_fake = 0.606, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 27/46, D_loss_real = 0.617, D_loss_fake = 0.609, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 28/46, D_loss_real = 0.638, D_loss_fake = 0.594, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 29/46, D_loss_real = 0.638, D_loss_fake = 0.605, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 30/46, D_loss_real = 0.671, D_loss_fake = 0.635, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 31/46, D_loss_real = 0.612, D_loss_fake = 0.667, G_loss =
0.891
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 462, Iteration 32/46, D_loss_real = 0.694, D_loss_fake = 0.610, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 33/46, D_loss_real = 0.656, D_loss_fake = 0.620, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 34/46, D_loss_real = 0.655, D_loss_fake = 0.606, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 35/46, D_loss_real = 0.658, D_loss_fake = 0.601, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 36/46, D_loss_real = 0.623, D_loss_fake = 0.688, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 37/46, D_loss_real = 0.615, D_loss_fake = 0.660, G_loss =
0.794
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 38/46, D_loss_real = 0.655, D_loss_fake = 0.604, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 39/46, D_loss_real = 0.682, D_loss_fake = 0.646, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 40/46, D_loss_real = 0.593, D_loss_fake = 0.623, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 41/46, D_loss_real = 0.683, D_loss_fake = 0.646, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 42/46, D_loss_real = 0.679, D_loss_fake = 0.647, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 43/46, D_loss_real = 0.590, D_loss_fake = 0.629, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 44/46, D_loss_real = 0.577, D_loss_fake = 0.604, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 462, Iteration 45/46, D_loss_real = 0.590, D_loss_fake = 0.616, G_loss =
0.837
2/2 [==============================] - 0s 4ms/step
Epoch 462, Iteration 46/46, D_loss_real = 0.654, D_loss_fake = 0.627, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 1/46, D_loss_real = 0.657, D_loss_fake = 0.661, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 463, Iteration 2/46, D_loss_real = 0.637, D_loss_fake = 0.611, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 3/46, D_loss_real = 0.750, D_loss_fake = 0.664, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 4/46, D_loss_real = 0.725, D_loss_fake = 0.597, G_loss =
0.876
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.537, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.614, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 7/46, D_loss_real = 0.642, D_loss_fake = 0.621, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 8/46, D_loss_real = 0.648, D_loss_fake = 0.677, G_loss =
0.782
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 9/46, D_loss_real = 0.611, D_loss_fake = 0.635, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 10/46, D_loss_real = 0.633, D_loss_fake = 0.675, G_loss =
0.796
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 11/46, D_loss_real = 0.610, D_loss_fake = 0.701, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 12/46, D_loss_real = 0.634, D_loss_fake = 0.651, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 13/46, D_loss_real = 0.622, D_loss_fake = 0.677, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 14/46, D_loss_real = 0.662, D_loss_fake = 0.587, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 15/46, D_loss_real = 0.689, D_loss_fake = 0.606, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 16/46, D_loss_real = 0.695, D_loss_fake = 0.657, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 17/46, D_loss_real = 0.614, D_loss_fake = 0.627, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 463, Iteration 18/46, D_loss_real = 0.647, D_loss_fake = 0.601, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 19/46, D_loss_real = 0.645, D_loss_fake = 0.664, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.649, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 21/46, D_loss_real = 0.613, D_loss_fake = 0.673, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 22/46, D_loss_real = 0.636, D_loss_fake = 0.609, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.595, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 24/46, D_loss_real = 0.686, D_loss_fake = 0.668, G_loss =
0.820
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 25/46, D_loss_real = 0.670, D_loss_fake = 0.646, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 26/46, D_loss_real = 0.632, D_loss_fake = 0.602, G_loss =
0.800
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 27/46, D_loss_real = 0.669, D_loss_fake = 0.737, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 28/46, D_loss_real = 0.667, D_loss_fake = 0.642, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 29/46, D_loss_real = 0.651, D_loss_fake = 0.710, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 30/46, D_loss_real = 0.634, D_loss_fake = 0.666, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 31/46, D_loss_real = 0.643, D_loss_fake = 0.707, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 32/46, D_loss_real = 0.659, D_loss_fake = 0.627, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 33/46, D_loss_real = 0.692, D_loss_fake = 0.628, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 463, Iteration 34/46, D_loss_real = 0.731, D_loss_fake = 0.612, G_loss =
0.789
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 35/46, D_loss_real = 0.657, D_loss_fake = 0.648, G_loss =
0.856
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 36/46, D_loss_real = 0.686, D_loss_fake = 0.652, G_loss =
0.792
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 37/46, D_loss_real = 0.615, D_loss_fake = 0.636, G_loss =
0.789
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 38/46, D_loss_real = 0.643, D_loss_fake = 0.717, G_loss =
0.812
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 39/46, D_loss_real = 0.643, D_loss_fake = 0.621, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 40/46, D_loss_real = 0.647, D_loss_fake = 0.674, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 463, Iteration 41/46, D_loss_real = 0.671, D_loss_fake = 0.650, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 42/46, D_loss_real = 0.673, D_loss_fake = 0.583, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 43/46, D_loss_real = 0.655, D_loss_fake = 0.676, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 44/46, D_loss_real = 0.661, D_loss_fake = 0.628, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 45/46, D_loss_real = 0.634, D_loss_fake = 0.625, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 463, Iteration 46/46, D_loss_real = 0.658, D_loss_fake = 0.636, G_loss =
0.783
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 1/46, D_loss_real = 0.668, D_loss_fake = 0.636, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 2/46, D_loss_real = 0.640, D_loss_fake = 0.698, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 3/46, D_loss_real = 0.668, D_loss_fake = 0.592, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 464, Iteration 4/46, D_loss_real = 0.652, D_loss_fake = 0.606, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 5/46, D_loss_real = 0.609, D_loss_fake = 0.621, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 6/46, D_loss_real = 0.683, D_loss_fake = 0.643, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 7/46, D_loss_real = 0.608, D_loss_fake = 0.617, G_loss =
0.800
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 8/46, D_loss_real = 0.712, D_loss_fake = 0.669, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 9/46, D_loss_real = 0.597, D_loss_fake = 0.676, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 10/46, D_loss_real = 0.570, D_loss_fake = 0.637, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 11/46, D_loss_real = 0.560, D_loss_fake = 0.627, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 12/46, D_loss_real = 0.678, D_loss_fake = 0.644, G_loss =
0.800
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 13/46, D_loss_real = 0.611, D_loss_fake = 0.660, G_loss =
0.779
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 14/46, D_loss_real = 0.671, D_loss_fake = 0.668, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 15/46, D_loss_real = 0.606, D_loss_fake = 0.660, G_loss =
0.781
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 16/46, D_loss_real = 0.655, D_loss_fake = 0.641, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 17/46, D_loss_real = 0.624, D_loss_fake = 0.692, G_loss =
0.812
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 18/46, D_loss_real = 0.612, D_loss_fake = 0.642, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 19/46, D_loss_real = 0.629, D_loss_fake = 0.586, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 464, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.647, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 21/46, D_loss_real = 0.632, D_loss_fake = 0.625, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 22/46, D_loss_real = 0.667, D_loss_fake = 0.637, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 23/46, D_loss_real = 0.640, D_loss_fake = 0.637, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 24/46, D_loss_real = 0.629, D_loss_fake = 0.673, G_loss =
0.881
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.603, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 26/46, D_loss_real = 0.666, D_loss_fake = 0.608, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 27/46, D_loss_real = 0.719, D_loss_fake = 0.606, G_loss =
0.897
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 28/46, D_loss_real = 0.678, D_loss_fake = 0.621, G_loss =
0.889
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 29/46, D_loss_real = 0.702, D_loss_fake = 0.620, G_loss =
0.891
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 30/46, D_loss_real = 0.703, D_loss_fake = 0.590, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 31/46, D_loss_real = 0.648, D_loss_fake = 0.662, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 32/46, D_loss_real = 0.649, D_loss_fake = 0.615, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 33/46, D_loss_real = 0.728, D_loss_fake = 0.606, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.605, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 35/46, D_loss_real = 0.627, D_loss_fake = 0.617, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 464, Iteration 36/46, D_loss_real = 0.652, D_loss_fake = 0.632, G_loss =
0.781
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 37/46, D_loss_real = 0.594, D_loss_fake = 0.694, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 38/46, D_loss_real = 0.663, D_loss_fake = 0.638, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 39/46, D_loss_real = 0.636, D_loss_fake = 0.688, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 40/46, D_loss_real = 0.624, D_loss_fake = 0.643, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 41/46, D_loss_real = 0.649, D_loss_fake = 0.677, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 42/46, D_loss_real = 0.655, D_loss_fake = 0.624, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 43/46, D_loss_real = 0.661, D_loss_fake = 0.658, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 44/46, D_loss_real = 0.643, D_loss_fake = 0.621, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 464, Iteration 45/46, D_loss_real = 0.614, D_loss_fake = 0.618, G_loss =
0.816
2/2 [==============================] - 0s 4ms/step
Epoch 464, Iteration 46/46, D_loss_real = 0.611, D_loss_fake = 0.721, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 1/46, D_loss_real = 0.638, D_loss_fake = 0.620, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 2/46, D_loss_real = 0.630, D_loss_fake = 0.680, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 3/46, D_loss_real = 0.642, D_loss_fake = 0.647, G_loss =
0.810
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 4/46, D_loss_real = 0.587, D_loss_fake = 0.654, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 5/46, D_loss_real = 0.621, D_loss_fake = 0.677, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 465, Iteration 6/46, D_loss_real = 0.651, D_loss_fake = 0.613, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 7/46, D_loss_real = 0.618, D_loss_fake = 0.665, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 8/46, D_loss_real = 0.655, D_loss_fake = 0.739, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 9/46, D_loss_real = 0.634, D_loss_fake = 0.667, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 10/46, D_loss_real = 0.658, D_loss_fake = 0.647, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 11/46, D_loss_real = 0.638, D_loss_fake = 0.608, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 12/46, D_loss_real = 0.708, D_loss_fake = 0.570, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 13/46, D_loss_real = 0.656, D_loss_fake = 0.643, G_loss =
0.794
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 14/46, D_loss_real = 0.622, D_loss_fake = 0.669, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 15/46, D_loss_real = 0.592, D_loss_fake = 0.643, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 16/46, D_loss_real = 0.640, D_loss_fake = 0.628, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 17/46, D_loss_real = 0.623, D_loss_fake = 0.659, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 18/46, D_loss_real = 0.682, D_loss_fake = 0.647, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.637, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 20/46, D_loss_real = 0.616, D_loss_fake = 0.628, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 21/46, D_loss_real = 0.661, D_loss_fake = 0.591, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 465, Iteration 22/46, D_loss_real = 0.672, D_loss_fake = 0.615, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 23/46, D_loss_real = 0.643, D_loss_fake = 0.622, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 24/46, D_loss_real = 0.625, D_loss_fake = 0.610, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 25/46, D_loss_real = 0.724, D_loss_fake = 0.626, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 26/46, D_loss_real = 0.638, D_loss_fake = 0.651, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 27/46, D_loss_real = 0.675, D_loss_fake = 0.594, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 28/46, D_loss_real = 0.497, D_loss_fake = 0.718, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 29/46, D_loss_real = 0.640, D_loss_fake = 0.653, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 30/46, D_loss_real = 0.626, D_loss_fake = 0.623, G_loss =
0.787
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 31/46, D_loss_real = 0.631, D_loss_fake = 0.694, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 32/46, D_loss_real = 0.619, D_loss_fake = 0.613, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 33/46, D_loss_real = 0.641, D_loss_fake = 0.645, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 34/46, D_loss_real = 0.574, D_loss_fake = 0.614, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 35/46, D_loss_real = 0.622, D_loss_fake = 0.673, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 36/46, D_loss_real = 0.640, D_loss_fake = 0.650, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 37/46, D_loss_real = 0.619, D_loss_fake = 0.623, G_loss =
0.907
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 465, Iteration 38/46, D_loss_real = 0.624, D_loss_fake = 0.603, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 39/46, D_loss_real = 0.636, D_loss_fake = 0.585, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.627, G_loss =
0.886
2/2 [==============================] - 0s 4ms/step
Epoch 465, Iteration 41/46, D_loss_real = 0.723, D_loss_fake = 0.605, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 42/46, D_loss_real = 0.653, D_loss_fake = 0.563, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 43/46, D_loss_real = 0.635, D_loss_fake = 0.614, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 44/46, D_loss_real = 0.664, D_loss_fake = 0.702, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 45/46, D_loss_real = 0.663, D_loss_fake = 0.609, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 465, Iteration 46/46, D_loss_real = 0.608, D_loss_fake = 0.626, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 1/46, D_loss_real = 0.626, D_loss_fake = 0.671, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 2/46, D_loss_real = 0.578, D_loss_fake = 0.647, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 3/46, D_loss_real = 0.629, D_loss_fake = 0.665, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 4/46, D_loss_real = 0.666, D_loss_fake = 0.607, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 5/46, D_loss_real = 0.647, D_loss_fake = 0.644, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 6/46, D_loss_real = 0.593, D_loss_fake = 0.670, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 7/46, D_loss_real = 0.662, D_loss_fake = 0.705, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 466, Iteration 8/46, D_loss_real = 0.642, D_loss_fake = 0.673, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 9/46, D_loss_real = 0.648, D_loss_fake = 0.649, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 10/46, D_loss_real = 0.671, D_loss_fake = 0.665, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 11/46, D_loss_real = 0.665, D_loss_fake = 0.667, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 12/46, D_loss_real = 0.663, D_loss_fake = 0.645, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 13/46, D_loss_real = 0.651, D_loss_fake = 0.617, G_loss =
0.920
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 14/46, D_loss_real = 0.772, D_loss_fake = 0.623, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 15/46, D_loss_real = 0.719, D_loss_fake = 0.583, G_loss =
0.888
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 16/46, D_loss_real = 0.728, D_loss_fake = 0.564, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 17/46, D_loss_real = 0.629, D_loss_fake = 0.596, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 18/46, D_loss_real = 0.658, D_loss_fake = 0.614, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 19/46, D_loss_real = 0.668, D_loss_fake = 0.631, G_loss =
0.880
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 20/46, D_loss_real = 0.644, D_loss_fake = 0.643, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 21/46, D_loss_real = 0.665, D_loss_fake = 0.597, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 22/46, D_loss_real = 0.655, D_loss_fake = 0.662, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 23/46, D_loss_real = 0.588, D_loss_fake = 0.661, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 466, Iteration 24/46, D_loss_real = 0.591, D_loss_fake = 0.658, G_loss =
0.816
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 25/46, D_loss_real = 0.668, D_loss_fake = 0.625, G_loss =
0.793
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 26/46, D_loss_real = 0.610, D_loss_fake = 0.629, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 27/46, D_loss_real = 0.616, D_loss_fake = 0.627, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 28/46, D_loss_real = 0.649, D_loss_fake = 0.625, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 29/46, D_loss_real = 0.645, D_loss_fake = 0.625, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 30/46, D_loss_real = 0.590, D_loss_fake = 0.614, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.607, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 32/46, D_loss_real = 0.666, D_loss_fake = 0.613, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 33/46, D_loss_real = 0.612, D_loss_fake = 0.637, G_loss =
0.770
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 34/46, D_loss_real = 0.614, D_loss_fake = 0.658, G_loss =
0.777
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 35/46, D_loss_real = 0.714, D_loss_fake = 0.656, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 36/46, D_loss_real = 0.596, D_loss_fake = 0.656, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 37/46, D_loss_real = 0.586, D_loss_fake = 0.624, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 38/46, D_loss_real = 0.641, D_loss_fake = 0.566, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 39/46, D_loss_real = 0.652, D_loss_fake = 0.651, G_loss =
0.796
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 466, Iteration 40/46, D_loss_real = 0.580, D_loss_fake = 0.650, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 41/46, D_loss_real = 0.635, D_loss_fake = 0.624, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 42/46, D_loss_real = 0.616, D_loss_fake = 0.627, G_loss =
0.766
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 43/46, D_loss_real = 0.620, D_loss_fake = 0.683, G_loss =
0.779
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 44/46, D_loss_real = 0.649, D_loss_fake = 0.619, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 466, Iteration 45/46, D_loss_real = 0.579, D_loss_fake = 0.677, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 466, Iteration 46/46, D_loss_real = 0.645, D_loss_fake = 0.709, G_loss =
0.850
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 1/46, D_loss_real = 0.632, D_loss_fake = 0.590, G_loss =
0.803
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 2/46, D_loss_real = 0.635, D_loss_fake = 0.685, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 3/46, D_loss_real = 0.631, D_loss_fake = 0.617, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 4/46, D_loss_real = 0.670, D_loss_fake = 0.644, G_loss =
0.886
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 5/46, D_loss_real = 0.698, D_loss_fake = 0.574, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 6/46, D_loss_real = 0.639, D_loss_fake = 0.597, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 7/46, D_loss_real = 0.683, D_loss_fake = 0.683, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 8/46, D_loss_real = 0.664, D_loss_fake = 0.710, G_loss =
0.881
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 9/46, D_loss_real = 0.670, D_loss_fake = 0.584, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 467, Iteration 10/46, D_loss_real = 0.676, D_loss_fake = 0.641, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 11/46, D_loss_real = 0.688, D_loss_fake = 0.617, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 12/46, D_loss_real = 0.582, D_loss_fake = 0.618, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 13/46, D_loss_real = 0.639, D_loss_fake = 0.633, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 14/46, D_loss_real = 0.668, D_loss_fake = 0.669, G_loss =
0.802
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 15/46, D_loss_real = 0.614, D_loss_fake = 0.623, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 16/46, D_loss_real = 0.642, D_loss_fake = 0.665, G_loss =
0.775
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 17/46, D_loss_real = 0.575, D_loss_fake = 0.649, G_loss =
0.796
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 18/46, D_loss_real = 0.589, D_loss_fake = 0.707, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 19/46, D_loss_real = 0.621, D_loss_fake = 0.725, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 20/46, D_loss_real = 0.632, D_loss_fake = 0.617, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.580, G_loss =
0.822
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 22/46, D_loss_real = 0.601, D_loss_fake = 0.679, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 23/46, D_loss_real = 0.622, D_loss_fake = 0.590, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 24/46, D_loss_real = 0.637, D_loss_fake = 0.677, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 25/46, D_loss_real = 0.661, D_loss_fake = 0.695, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 467, Iteration 26/46, D_loss_real = 0.629, D_loss_fake = 0.652, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 27/46, D_loss_real = 0.613, D_loss_fake = 0.630, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.615, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 29/46, D_loss_real = 0.660, D_loss_fake = 0.618, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 30/46, D_loss_real = 0.629, D_loss_fake = 0.601, G_loss =
0.842
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 31/46, D_loss_real = 0.627, D_loss_fake = 0.641, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 32/46, D_loss_real = 0.621, D_loss_fake = 0.702, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 33/46, D_loss_real = 0.619, D_loss_fake = 0.643, G_loss =
0.782
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.654, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 35/46, D_loss_real = 0.601, D_loss_fake = 0.662, G_loss =
0.786
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 36/46, D_loss_real = 0.661, D_loss_fake = 0.658, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 37/46, D_loss_real = 0.656, D_loss_fake = 0.645, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 38/46, D_loss_real = 0.647, D_loss_fake = 0.664, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 39/46, D_loss_real = 0.641, D_loss_fake = 0.621, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.655, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 41/46, D_loss_real = 0.650, D_loss_fake = 0.639, G_loss =
0.783
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 467, Iteration 42/46, D_loss_real = 0.666, D_loss_fake = 0.593, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 43/46, D_loss_real = 0.614, D_loss_fake = 0.608, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 467, Iteration 44/46, D_loss_real = 0.568, D_loss_fake = 0.728, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 45/46, D_loss_real = 0.704, D_loss_fake = 0.677, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 467, Iteration 46/46, D_loss_real = 0.647, D_loss_fake = 0.605, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 1/46, D_loss_real = 0.625, D_loss_fake = 0.592, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 2/46, D_loss_real = 0.610, D_loss_fake = 0.611, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 3/46, D_loss_real = 0.606, D_loss_fake = 0.628, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 4/46, D_loss_real = 0.679, D_loss_fake = 0.600, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 5/46, D_loss_real = 0.625, D_loss_fake = 0.613, G_loss =
0.793
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 6/46, D_loss_real = 0.664, D_loss_fake = 0.633, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 7/46, D_loss_real = 0.654, D_loss_fake = 0.627, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 8/46, D_loss_real = 0.625, D_loss_fake = 0.658, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 9/46, D_loss_real = 0.677, D_loss_fake = 0.594, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 10/46, D_loss_real = 0.679, D_loss_fake = 0.592, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 11/46, D_loss_real = 0.732, D_loss_fake = 0.622, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 468, Iteration 12/46, D_loss_real = 0.709, D_loss_fake = 0.635, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 13/46, D_loss_real = 0.635, D_loss_fake = 0.598, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 14/46, D_loss_real = 0.652, D_loss_fake = 0.614, G_loss =
0.825
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 15/46, D_loss_real = 0.621, D_loss_fake = 0.616, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 16/46, D_loss_real = 0.658, D_loss_fake = 0.615, G_loss =
0.803
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 17/46, D_loss_real = 0.656, D_loss_fake = 0.674, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 18/46, D_loss_real = 0.666, D_loss_fake = 0.642, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 19/46, D_loss_real = 0.721, D_loss_fake = 0.611, G_loss =
0.842
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 20/46, D_loss_real = 0.681, D_loss_fake = 0.617, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 21/46, D_loss_real = 0.648, D_loss_fake = 0.579, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 22/46, D_loss_real = 0.644, D_loss_fake = 0.667, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 23/46, D_loss_real = 0.575, D_loss_fake = 0.624, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 24/46, D_loss_real = 0.602, D_loss_fake = 0.649, G_loss =
0.765
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 25/46, D_loss_real = 0.648, D_loss_fake = 0.698, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.619, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 27/46, D_loss_real = 0.647, D_loss_fake = 0.641, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 468, Iteration 28/46, D_loss_real = 0.634, D_loss_fake = 0.632, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 29/46, D_loss_real = 0.676, D_loss_fake = 0.599, G_loss =
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 30/46, D_loss_real = 0.606, D_loss_fake = 0.610, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 31/46, D_loss_real = 0.651, D_loss_fake = 0.698, G_loss =
0.783
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 32/46, D_loss_real = 0.631, D_loss_fake = 0.646, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 33/46, D_loss_real = 0.608, D_loss_fake = 0.819, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 34/46, D_loss_real = 0.679, D_loss_fake = 0.565, G_loss =
0.897
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 35/46, D_loss_real = 0.669, D_loss_fake = 0.666, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 36/46, D_loss_real = 0.674, D_loss_fake = 0.563, G_loss =
0.877
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 37/46, D_loss_real = 0.653, D_loss_fake = 0.624, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 38/46, D_loss_real = 0.656, D_loss_fake = 0.639, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 39/46, D_loss_real = 0.641, D_loss_fake = 0.594, G_loss =
0.865
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 40/46, D_loss_real = 0.653, D_loss_fake = 0.634, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 41/46, D_loss_real = 0.649, D_loss_fake = 0.638, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 468, Iteration 42/46, D_loss_real = 0.685, D_loss_fake = 0.609, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 43/46, D_loss_real = 0.646, D_loss_fake = 0.599, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 468, Iteration 44/46, D_loss_real = 0.639, D_loss_fake = 0.669, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 45/46, D_loss_real = 0.645, D_loss_fake = 0.641, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 468, Iteration 46/46, D_loss_real = 0.596, D_loss_fake = 0.606, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 1/46, D_loss_real = 0.662, D_loss_fake = 0.623, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 2/46, D_loss_real = 0.633, D_loss_fake = 0.599, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 3/46, D_loss_real = 0.616, D_loss_fake = 0.603, G_loss =
0.905
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 4/46, D_loss_real = 0.644, D_loss_fake = 0.646, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 5/46, D_loss_real = 0.638, D_loss_fake = 0.650, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 6/46, D_loss_real = 0.639, D_loss_fake = 0.601, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 7/46, D_loss_real = 0.600, D_loss_fake = 0.646, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 8/46, D_loss_real = 0.651, D_loss_fake = 0.689, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 9/46, D_loss_real = 0.607, D_loss_fake = 0.645, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 10/46, D_loss_real = 0.675, D_loss_fake = 0.616, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 11/46, D_loss_real = 0.640, D_loss_fake = 0.660, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 12/46, D_loss_real = 0.640, D_loss_fake = 0.628, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 13/46, D_loss_real = 0.645, D_loss_fake = 0.606, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 469, Iteration 14/46, D_loss_real = 0.648, D_loss_fake = 0.628, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 15/46, D_loss_real = 0.575, D_loss_fake = 0.612, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 16/46, D_loss_real = 0.658, D_loss_fake = 0.606, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 17/46, D_loss_real = 0.632, D_loss_fake = 0.668, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 18/46, D_loss_real = 0.659, D_loss_fake = 0.588, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 19/46, D_loss_real = 0.630, D_loss_fake = 0.655, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 20/46, D_loss_real = 0.621, D_loss_fake = 0.645, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 21/46, D_loss_real = 0.602, D_loss_fake = 0.680, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 22/46, D_loss_real = 0.631, D_loss_fake = 0.614, G_loss =
0.864
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 23/46, D_loss_real = 0.635, D_loss_fake = 0.629, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 24/46, D_loss_real = 0.568, D_loss_fake = 0.628, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 25/46, D_loss_real = 0.665, D_loss_fake = 0.623, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 26/46, D_loss_real = 0.630, D_loss_fake = 0.642, G_loss =
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 27/46, D_loss_real = 0.639, D_loss_fake = 0.654, G_loss =
0.791
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 28/46, D_loss_real = 0.603, D_loss_fake = 0.657, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 29/46, D_loss_real = 0.664, D_loss_fake = 0.657, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 469, Iteration 30/46, D_loss_real = 0.628, D_loss_fake = 0.649, G_loss =
0.812
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 31/46, D_loss_real = 0.601, D_loss_fake = 0.622, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 32/46, D_loss_real = 0.660, D_loss_fake = 0.637, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 33/46, D_loss_real = 0.610, D_loss_fake = 0.671, G_loss =
0.864
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 34/46, D_loss_real = 0.633, D_loss_fake = 0.682, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 35/46, D_loss_real = 0.687, D_loss_fake = 0.607, G_loss =
0.872
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 36/46, D_loss_real = 0.657, D_loss_fake = 0.603, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 37/46, D_loss_real = 0.671, D_loss_fake = 0.618, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 38/46, D_loss_real = 0.651, D_loss_fake = 0.596, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 39/46, D_loss_real = 0.676, D_loss_fake = 0.607, G_loss =
0.875
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.599, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 41/46, D_loss_real = 0.689, D_loss_fake = 0.657, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 42/46, D_loss_real = 0.595, D_loss_fake = 0.635, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 469, Iteration 43/46, D_loss_real = 0.627, D_loss_fake = 0.704, G_loss =
0.890
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 44/46, D_loss_real = 0.673, D_loss_fake = 0.603, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 469, Iteration 45/46, D_loss_real = 0.728, D_loss_fake = 0.589, G_loss =
0.807
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 469, Iteration 46/46, D_loss_real = 0.662, D_loss_fake = 0.628, G_loss =
0.810
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 1/46, D_loss_real = 0.605, D_loss_fake = 0.635, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 2/46, D_loss_real = 0.654, D_loss_fake = 0.665, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 3/46, D_loss_real = 0.651, D_loss_fake = 0.641, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 4/46, D_loss_real = 0.554, D_loss_fake = 0.648, G_loss =
0.785
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 5/46, D_loss_real = 0.600, D_loss_fake = 0.700, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 6/46, D_loss_real = 0.642, D_loss_fake = 0.675, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 7/46, D_loss_real = 0.669, D_loss_fake = 0.603, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 8/46, D_loss_real = 0.651, D_loss_fake = 0.619, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 9/46, D_loss_real = 0.646, D_loss_fake = 0.613, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 10/46, D_loss_real = 0.678, D_loss_fake = 0.601, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 11/46, D_loss_real = 0.593, D_loss_fake = 0.612, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 12/46, D_loss_real = 0.673, D_loss_fake = 0.669, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 13/46, D_loss_real = 0.669, D_loss_fake = 0.646, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 14/46, D_loss_real = 0.691, D_loss_fake = 0.669, G_loss =
0.803
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 15/46, D_loss_real = 0.731, D_loss_fake = 0.672, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 470, Iteration 16/46, D_loss_real = 0.734, D_loss_fake = 0.632, G_loss =
0.873
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 17/46, D_loss_real = 0.646, D_loss_fake = 0.600, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 18/46, D_loss_real = 0.656, D_loss_fake = 0.619, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 19/46, D_loss_real = 0.653, D_loss_fake = 0.644, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 20/46, D_loss_real = 0.671, D_loss_fake = 0.638, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 21/46, D_loss_real = 0.615, D_loss_fake = 0.718, G_loss =
0.786
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 22/46, D_loss_real = 0.590, D_loss_fake = 0.628, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 23/46, D_loss_real = 0.673, D_loss_fake = 0.633, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 24/46, D_loss_real = 0.645, D_loss_fake = 0.689, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 25/46, D_loss_real = 0.638, D_loss_fake = 0.624, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.624, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 27/46, D_loss_real = 0.644, D_loss_fake = 0.640, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 28/46, D_loss_real = 0.616, D_loss_fake = 0.666, G_loss =
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 29/46, D_loss_real = 0.669, D_loss_fake = 0.671, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 30/46, D_loss_real = 0.640, D_loss_fake = 0.642, G_loss =
0.862
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 31/46, D_loss_real = 0.624, D_loss_fake = 0.642, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 470, Iteration 32/46, D_loss_real = 0.549, D_loss_fake = 0.654, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 33/46, D_loss_real = 0.616, D_loss_fake = 0.672, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 34/46, D_loss_real = 0.604, D_loss_fake = 0.652, G_loss =
0.778
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 35/46, D_loss_real = 0.631, D_loss_fake = 0.684, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 36/46, D_loss_real = 0.712, D_loss_fake = 0.727, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.606, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 38/46, D_loss_real = 0.623, D_loss_fake = 0.647, G_loss =
0.842
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 39/46, D_loss_real = 0.649, D_loss_fake = 0.600, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 40/46, D_loss_real = 0.637, D_loss_fake = 0.629, G_loss =
0.886
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 41/46, D_loss_real = 0.620, D_loss_fake = 0.608, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.603, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 43/46, D_loss_real = 0.677, D_loss_fake = 0.597, G_loss =
0.871
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 44/46, D_loss_real = 0.670, D_loss_fake = 0.633, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 470, Iteration 45/46, D_loss_real = 0.633, D_loss_fake = 0.589, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 470, Iteration 46/46, D_loss_real = 0.591, D_loss_fake = 0.624, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 1/46, D_loss_real = 0.663, D_loss_fake = 0.631, G_loss =
0.814
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 2/46, D_loss_real = 0.642, D_loss_fake = 0.658, G_loss =
0.872
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 3/46, D_loss_real = 0.687, D_loss_fake = 0.670, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 4/46, D_loss_real = 0.601, D_loss_fake = 0.639, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 5/46, D_loss_real = 0.723, D_loss_fake = 0.569, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 6/46, D_loss_real = 0.677, D_loss_fake = 0.604, G_loss =
0.897
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.651, G_loss =
0.892
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 8/46, D_loss_real = 0.652, D_loss_fake = 0.631, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 9/46, D_loss_real = 0.739, D_loss_fake = 0.658, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 10/46, D_loss_real = 0.684, D_loss_fake = 0.610, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 11/46, D_loss_real = 0.668, D_loss_fake = 0.592, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 12/46, D_loss_real = 0.632, D_loss_fake = 0.621, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 13/46, D_loss_real = 0.645, D_loss_fake = 0.619, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 14/46, D_loss_real = 0.639, D_loss_fake = 0.639, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 15/46, D_loss_real = 0.640, D_loss_fake = 0.651, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 16/46, D_loss_real = 0.573, D_loss_fake = 0.679, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 17/46, D_loss_real = 0.636, D_loss_fake = 0.636, G_loss =
0.806
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 18/46, D_loss_real = 0.671, D_loss_fake = 0.655, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 19/46, D_loss_real = 0.581, D_loss_fake = 0.637, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 20/46, D_loss_real = 0.622, D_loss_fake = 0.716, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 21/46, D_loss_real = 0.671, D_loss_fake = 0.734, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 22/46, D_loss_real = 0.690, D_loss_fake = 0.678, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 23/46, D_loss_real = 0.606, D_loss_fake = 0.610, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 24/46, D_loss_real = 0.668, D_loss_fake = 0.603, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 25/46, D_loss_real = 0.598, D_loss_fake = 0.650, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 26/46, D_loss_real = 0.678, D_loss_fake = 0.591, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 27/46, D_loss_real = 0.612, D_loss_fake = 0.694, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 28/46, D_loss_real = 0.646, D_loss_fake = 0.614, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 29/46, D_loss_real = 0.594, D_loss_fake = 0.624, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 30/46, D_loss_real = 0.664, D_loss_fake = 0.627, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 31/46, D_loss_real = 0.606, D_loss_fake = 0.626, G_loss =
0.796
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 32/46, D_loss_real = 0.668, D_loss_fake = 0.649, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 33/46, D_loss_real = 0.652, D_loss_fake = 0.660, G_loss =
0.842
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 34/46, D_loss_real = 0.676, D_loss_fake = 0.620, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 35/46, D_loss_real = 0.591, D_loss_fake = 0.668, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 36/46, D_loss_real = 0.626, D_loss_fake = 0.625, G_loss =
0.813
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 37/46, D_loss_real = 0.629, D_loss_fake = 0.670, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 38/46, D_loss_real = 0.552, D_loss_fake = 0.651, G_loss =
0.791
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 39/46, D_loss_real = 0.629, D_loss_fake = 0.637, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 40/46, D_loss_real = 0.657, D_loss_fake = 0.704, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 41/46, D_loss_real = 0.666, D_loss_fake = 0.633, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 42/46, D_loss_real = 0.707, D_loss_fake = 0.625, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 43/46, D_loss_real = 0.682, D_loss_fake = 0.592, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 44/46, D_loss_real = 0.690, D_loss_fake = 0.637, G_loss =
0.856
2/2 [==============================] - 0s 4ms/step
Epoch 471, Iteration 45/46, D_loss_real = 0.614, D_loss_fake = 0.584, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 471, Iteration 46/46, D_loss_real = 0.702, D_loss_fake = 0.626, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 1/46, D_loss_real = 0.631, D_loss_fake = 0.649, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 2/46, D_loss_real = 0.591, D_loss_fake = 0.585, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 3/46, D_loss_real = 0.645, D_loss_fake = 0.619, G_loss =
0.798
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 4/46, D_loss_real = 0.670, D_loss_fake = 0.621, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 5/46, D_loss_real = 0.616, D_loss_fake = 0.663, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 6/46, D_loss_real = 0.614, D_loss_fake = 0.635, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 7/46, D_loss_real = 0.628, D_loss_fake = 0.584, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 8/46, D_loss_real = 0.629, D_loss_fake = 0.714, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 9/46, D_loss_real = 0.597, D_loss_fake = 0.625, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 10/46, D_loss_real = 0.623, D_loss_fake = 0.660, G_loss =
0.772
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 11/46, D_loss_real = 0.610, D_loss_fake = 0.638, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 12/46, D_loss_real = 0.569, D_loss_fake = 0.672, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 13/46, D_loss_real = 0.622, D_loss_fake = 0.633, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 14/46, D_loss_real = 0.626, D_loss_fake = 0.637, G_loss =
0.784
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 15/46, D_loss_real = 0.637, D_loss_fake = 0.646, G_loss =
0.775
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 16/46, D_loss_real = 0.597, D_loss_fake = 0.669, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 17/46, D_loss_real = 0.664, D_loss_fake = 0.595, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 18/46, D_loss_real = 0.645, D_loss_fake = 0.653, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 19/46, D_loss_real = 0.675, D_loss_fake = 0.638, G_loss =
0.842
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 20/46, D_loss_real = 0.675, D_loss_fake = 0.613, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 21/46, D_loss_real = 0.670, D_loss_fake = 0.617, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 22/46, D_loss_real = 0.654, D_loss_fake = 0.618, G_loss =
0.939
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.567, G_loss =
0.904
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 24/46, D_loss_real = 0.773, D_loss_fake = 0.593, G_loss =
0.886
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 25/46, D_loss_real = 0.732, D_loss_fake = 0.639, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 26/46, D_loss_real = 0.682, D_loss_fake = 0.612, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.636, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 28/46, D_loss_real = 0.630, D_loss_fake = 0.635, G_loss =
0.840
2/2 [==============================] - 0s 11ms/step
Epoch 472, Iteration 29/46, D_loss_real = 0.682, D_loss_fake = 0.601, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 30/46, D_loss_real = 0.632, D_loss_fake = 0.614, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 31/46, D_loss_real = 0.642, D_loss_fake = 0.641, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 32/46, D_loss_real = 0.610, D_loss_fake = 0.588, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 33/46, D_loss_real = 0.603, D_loss_fake = 0.652, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 34/46, D_loss_real = 0.661, D_loss_fake = 0.691, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 35/46, D_loss_real = 0.667, D_loss_fake = 0.614, G_loss =
0.827
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 36/46, D_loss_real = 0.664, D_loss_fake = 0.659, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 37/46, D_loss_real = 0.608, D_loss_fake = 0.628, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 38/46, D_loss_real = 0.625, D_loss_fake = 0.603, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 39/46, D_loss_real = 0.651, D_loss_fake = 0.638, G_loss =
0.798
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 40/46, D_loss_real = 0.650, D_loss_fake = 0.683, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 41/46, D_loss_real = 0.589, D_loss_fake = 0.671, G_loss =
0.794
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 42/46, D_loss_real = 0.660, D_loss_fake = 0.605, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 43/46, D_loss_real = 0.646, D_loss_fake = 0.648, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 44/46, D_loss_real = 0.581, D_loss_fake = 0.623, G_loss =
0.788
2/2 [==============================] - 0s 3ms/step
Epoch 472, Iteration 45/46, D_loss_real = 0.629, D_loss_fake = 0.671, G_loss =
0.804
2/2 [==============================] - 0s 4ms/step
Epoch 472, Iteration 46/46, D_loss_real = 0.610, D_loss_fake = 0.692, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 1/46, D_loss_real = 0.635, D_loss_fake = 0.681, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 2/46, D_loss_real = 0.641, D_loss_fake = 0.643, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 3/46, D_loss_real = 0.626, D_loss_fake = 0.607, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 4/46, D_loss_real = 0.640, D_loss_fake = 0.642, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.624, G_loss =
0.865
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 6/46, D_loss_real = 0.731, D_loss_fake = 0.640, G_loss =
0.897
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 7/46, D_loss_real = 0.700, D_loss_fake = 0.570, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 8/46, D_loss_real = 0.678, D_loss_fake = 0.556, G_loss =
0.901
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 9/46, D_loss_real = 0.747, D_loss_fake = 0.586, G_loss =
0.879
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 10/46, D_loss_real = 0.637, D_loss_fake = 0.657, G_loss =
0.861
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 11/46, D_loss_real = 0.687, D_loss_fake = 0.582, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 12/46, D_loss_real = 0.763, D_loss_fake = 0.598, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 13/46, D_loss_real = 0.683, D_loss_fake = 0.634, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 14/46, D_loss_real = 0.650, D_loss_fake = 0.612, G_loss =
0.759
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 15/46, D_loss_real = 0.646, D_loss_fake = 0.780, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.637, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 17/46, D_loss_real = 0.644, D_loss_fake = 0.608, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 18/46, D_loss_real = 0.712, D_loss_fake = 0.622, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 19/46, D_loss_real = 0.632, D_loss_fake = 0.607, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 20/46, D_loss_real = 0.649, D_loss_fake = 0.590, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 21/46, D_loss_real = 0.581, D_loss_fake = 0.629, G_loss =
0.812
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 22/46, D_loss_real = 0.555, D_loss_fake = 0.695, G_loss =
0.786
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 23/46, D_loss_real = 0.606, D_loss_fake = 0.642, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 24/46, D_loss_real = 0.574, D_loss_fake = 0.614, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 25/46, D_loss_real = 0.628, D_loss_fake = 0.635, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 26/46, D_loss_real = 0.658, D_loss_fake = 0.660, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 27/46, D_loss_real = 0.588, D_loss_fake = 0.621, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 28/46, D_loss_real = 0.548, D_loss_fake = 0.618, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 29/46, D_loss_real = 0.623, D_loss_fake = 0.623, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 30/46, D_loss_real = 0.630, D_loss_fake = 0.646, G_loss =
0.785
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 31/46, D_loss_real = 0.634, D_loss_fake = 0.694, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 32/46, D_loss_real = 0.631, D_loss_fake = 0.647, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 33/46, D_loss_real = 0.662, D_loss_fake = 0.639, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 34/46, D_loss_real = 0.671, D_loss_fake = 0.666, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 35/46, D_loss_real = 0.673, D_loss_fake = 0.553, G_loss =
0.868
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 36/46, D_loss_real = 0.713, D_loss_fake = 0.605, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.588, G_loss =
0.867
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 38/46, D_loss_real = 0.675, D_loss_fake = 0.603, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 39/46, D_loss_real = 0.643, D_loss_fake = 0.606, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 40/46, D_loss_real = 0.739, D_loss_fake = 0.606, G_loss =
0.915
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 41/46, D_loss_real = 0.677, D_loss_fake = 0.635, G_loss =
0.842
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 42/46, D_loss_real = 0.647, D_loss_fake = 0.627, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 473, Iteration 43/46, D_loss_real = 0.620, D_loss_fake = 0.583, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 44/46, D_loss_real = 0.613, D_loss_fake = 0.619, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 45/46, D_loss_real = 0.669, D_loss_fake = 0.621, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 473, Iteration 46/46, D_loss_real = 0.598, D_loss_fake = 0.624, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 1/46, D_loss_real = 0.631, D_loss_fake = 0.614, G_loss =
0.866
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 2/46, D_loss_real = 0.670, D_loss_fake = 0.591, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 3/46, D_loss_real = 0.594, D_loss_fake = 0.702, G_loss =
0.837
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 4/46, D_loss_real = 0.593, D_loss_fake = 0.611, G_loss =
0.775
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 5/46, D_loss_real = 0.656, D_loss_fake = 0.626, G_loss =
0.803
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 6/46, D_loss_real = 0.597, D_loss_fake = 0.610, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 7/46, D_loss_real = 0.630, D_loss_fake = 0.616, G_loss =
0.827
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 8/46, D_loss_real = 0.651, D_loss_fake = 0.614, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 9/46, D_loss_real = 0.642, D_loss_fake = 0.645, G_loss =
0.821
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 10/46, D_loss_real = 0.625, D_loss_fake = 0.639, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 11/46, D_loss_real = 0.618, D_loss_fake = 0.671, G_loss =
0.788
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 12/46, D_loss_real = 0.570, D_loss_fake = 0.628, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 13/46, D_loss_real = 0.600, D_loss_fake = 0.632, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 14/46, D_loss_real = 0.584, D_loss_fake = 0.626, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 15/46, D_loss_real = 0.644, D_loss_fake = 0.631, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 16/46, D_loss_real = 0.649, D_loss_fake = 0.605, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 17/46, D_loss_real = 0.609, D_loss_fake = 0.713, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 18/46, D_loss_real = 0.667, D_loss_fake = 0.742, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 19/46, D_loss_real = 0.607, D_loss_fake = 0.656, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 20/46, D_loss_real = 0.615, D_loss_fake = 0.642, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 21/46, D_loss_real = 0.652, D_loss_fake = 0.592, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 22/46, D_loss_real = 0.608, D_loss_fake = 0.617, G_loss =
0.863
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 23/46, D_loss_real = 0.702, D_loss_fake = 0.600, G_loss =
0.854
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 24/46, D_loss_real = 0.630, D_loss_fake = 0.625, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 25/46, D_loss_real = 0.624, D_loss_fake = 0.651, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 26/46, D_loss_real = 0.666, D_loss_fake = 0.588, G_loss =
0.822
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 27/46, D_loss_real = 0.638, D_loss_fake = 0.650, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 28/46, D_loss_real = 0.671, D_loss_fake = 0.603, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 29/46, D_loss_real = 0.664, D_loss_fake = 0.645, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 30/46, D_loss_real = 0.631, D_loss_fake = 0.622, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 31/46, D_loss_real = 0.639, D_loss_fake = 0.680, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.632, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.631, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 34/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 35/46, D_loss_real = 0.617, D_loss_fake = 0.703, G_loss =
0.810
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 36/46, D_loss_real = 0.621, D_loss_fake = 0.642, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 37/46, D_loss_real = 0.668, D_loss_fake = 0.628, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 38/46, D_loss_real = 0.706, D_loss_fake = 0.627, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 39/46, D_loss_real = 0.648, D_loss_fake = 0.691, G_loss =
0.821
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 40/46, D_loss_real = 0.671, D_loss_fake = 0.688, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 474, Iteration 41/46, D_loss_real = 0.645, D_loss_fake = 0.623, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 42/46, D_loss_real = 0.638, D_loss_fake = 0.628, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 43/46, D_loss_real = 0.685, D_loss_fake = 0.624, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 44/46, D_loss_real = 0.673, D_loss_fake = 0.619, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 45/46, D_loss_real = 0.673, D_loss_fake = 0.610, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 474, Iteration 46/46, D_loss_real = 0.611, D_loss_fake = 0.616, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 1/46, D_loss_real = 0.601, D_loss_fake = 0.684, G_loss =
0.816
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 2/46, D_loss_real = 0.645, D_loss_fake = 0.584, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 3/46, D_loss_real = 0.663, D_loss_fake = 0.639, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 4/46, D_loss_real = 0.660, D_loss_fake = 0.648, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 5/46, D_loss_real = 0.666, D_loss_fake = 0.629, G_loss =
0.796
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 6/46, D_loss_real = 0.642, D_loss_fake = 0.655, G_loss =
0.858
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 7/46, D_loss_real = 0.629, D_loss_fake = 0.625, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 8/46, D_loss_real = 0.599, D_loss_fake = 0.661, G_loss =
0.787
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 9/46, D_loss_real = 0.584, D_loss_fake = 0.701, G_loss =
0.790
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 10/46, D_loss_real = 0.685, D_loss_fake = 0.671, G_loss =
0.863
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 11/46, D_loss_real = 0.639, D_loss_fake = 0.618, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 12/46, D_loss_real = 0.669, D_loss_fake = 0.581, G_loss =
0.870
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 13/46, D_loss_real = 0.687, D_loss_fake = 0.592, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 14/46, D_loss_real = 0.587, D_loss_fake = 0.624, G_loss =
0.883
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 15/46, D_loss_real = 0.654, D_loss_fake = 0.643, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 16/46, D_loss_real = 0.648, D_loss_fake = 0.585, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 17/46, D_loss_real = 0.649, D_loss_fake = 0.600, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 18/46, D_loss_real = 0.652, D_loss_fake = 0.667, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 19/46, D_loss_real = 0.662, D_loss_fake = 0.587, G_loss =
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.621, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.638, G_loss =
0.884
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 22/46, D_loss_real = 0.690, D_loss_fake = 0.616, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 23/46, D_loss_real = 0.624, D_loss_fake = 0.626, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 24/46, D_loss_real = 0.605, D_loss_fake = 0.661, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 25/46, D_loss_real = 0.595, D_loss_fake = 0.678, G_loss =
0.797
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 26/46, D_loss_real = 0.653, D_loss_fake = 0.685, G_loss =
0.803
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 27/46, D_loss_real = 0.637, D_loss_fake = 0.701, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 28/46, D_loss_real = 0.616, D_loss_fake = 0.601, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 29/46, D_loss_real = 0.652, D_loss_fake = 0.662, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 30/46, D_loss_real = 0.606, D_loss_fake = 0.655, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 31/46, D_loss_real = 0.606, D_loss_fake = 0.608, G_loss =
0.780
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 32/46, D_loss_real = 0.561, D_loss_fake = 0.682, G_loss =
0.781
2/2 [==============================] - 0s 13ms/step
Epoch 475, Iteration 33/46, D_loss_real = 0.614, D_loss_fake = 0.704, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 34/46, D_loss_real = 0.657, D_loss_fake = 0.657, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 35/46, D_loss_real = 0.657, D_loss_fake = 0.663, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 36/46, D_loss_real = 0.621, D_loss_fake = 0.619, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.737, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 38/46, D_loss_real = 0.682, D_loss_fake = 0.582, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.645, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 40/46, D_loss_real = 0.710, D_loss_fake = 0.596, G_loss =
0.876
2/2 [==============================] - 0s 4ms/step
Epoch 475, Iteration 41/46, D_loss_real = 0.691, D_loss_fake = 0.583, G_loss =
0.858
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 42/46, D_loss_real = 0.662, D_loss_fake = 0.594, G_loss =
0.898
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 43/46, D_loss_real = 0.656, D_loss_fake = 0.590, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 44/46, D_loss_real = 0.661, D_loss_fake = 0.628, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 45/46, D_loss_real = 0.658, D_loss_fake = 0.683, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 475, Iteration 46/46, D_loss_real = 0.671, D_loss_fake = 0.648, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 1/46, D_loss_real = 0.643, D_loss_fake = 0.615, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 2/46, D_loss_real = 0.636, D_loss_fake = 0.650, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 3/46, D_loss_real = 0.641, D_loss_fake = 0.736, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 4/46, D_loss_real = 0.627, D_loss_fake = 0.667, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 5/46, D_loss_real = 0.634, D_loss_fake = 0.641, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 476, Iteration 6/46, D_loss_real = 0.657, D_loss_fake = 0.618, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 7/46, D_loss_real = 0.604, D_loss_fake = 0.629, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 8/46, D_loss_real = 0.610, D_loss_fake = 0.643, G_loss =
0.787
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 9/46, D_loss_real = 0.583, D_loss_fake = 0.650, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 10/46, D_loss_real = 0.633, D_loss_fake = 0.662, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 11/46, D_loss_real = 0.600, D_loss_fake = 0.692, G_loss =
0.854
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 12/46, D_loss_real = 0.612, D_loss_fake = 0.663, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 13/46, D_loss_real = 0.588, D_loss_fake = 0.641, G_loss =
0.788
2/2 [==============================] - 0s 4ms/step
Epoch 476, Iteration 14/46, D_loss_real = 0.687, D_loss_fake = 0.684, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 15/46, D_loss_real = 0.578, D_loss_fake = 0.668, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.578, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 17/46, D_loss_real = 0.707, D_loss_fake = 0.582, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 18/46, D_loss_real = 0.602, D_loss_fake = 0.626, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 19/46, D_loss_real = 0.661, D_loss_fake = 0.622, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 476, Iteration 20/46, D_loss_real = 0.586, D_loss_fake = 0.656, G_loss =
0.868
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 21/46, D_loss_real = 0.620, D_loss_fake = 0.638, G_loss =
0.886
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 22/46, D_loss_real = 0.671, D_loss_fake = 0.602, G_loss =
0.890
2/2 [==============================] - 0s 4ms/step
Epoch 476, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.617, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 24/46, D_loss_real = 0.632, D_loss_fake = 0.635, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 25/46, D_loss_real = 0.667, D_loss_fake = 0.622, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 26/46, D_loss_real = 0.683, D_loss_fake = 0.595, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 27/46, D_loss_real = 0.601, D_loss_fake = 0.615, G_loss =
0.827
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 28/46, D_loss_real = 0.660, D_loss_fake = 0.606, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 29/46, D_loss_real = 0.715, D_loss_fake = 0.651, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 476, Iteration 30/46, D_loss_real = 0.701, D_loss_fake = 0.613, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 31/46, D_loss_real = 0.653, D_loss_fake = 0.610, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 32/46, D_loss_real = 0.629, D_loss_fake = 0.645, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 33/46, D_loss_real = 0.635, D_loss_fake = 0.681, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 34/46, D_loss_real = 0.558, D_loss_fake = 0.615, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 35/46, D_loss_real = 0.640, D_loss_fake = 0.604, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 36/46, D_loss_real = 0.611, D_loss_fake = 0.632, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 37/46, D_loss_real = 0.648, D_loss_fake = 0.589, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 38/46, D_loss_real = 0.594, D_loss_fake = 0.638, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 39/46, D_loss_real = 0.612, D_loss_fake = 0.686, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 40/46, D_loss_real = 0.584, D_loss_fake = 0.639, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 41/46, D_loss_real = 0.697, D_loss_fake = 0.638, G_loss =
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 42/46, D_loss_real = 0.656, D_loss_fake = 0.662, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 43/46, D_loss_real = 0.657, D_loss_fake = 0.604, G_loss =
0.833
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 476, Iteration 44/46, D_loss_real = 0.600, D_loss_fake = 0.584, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 45/46, D_loss_real = 0.663, D_loss_fake = 0.639, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 476, Iteration 46/46, D_loss_real = 0.670, D_loss_fake = 0.621, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 1/46, D_loss_real = 0.648, D_loss_fake = 0.619, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 2/46, D_loss_real = 0.664, D_loss_fake = 0.618, G_loss =
0.865
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 3/46, D_loss_real = 0.658, D_loss_fake = 0.598, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 4/46, D_loss_real = 0.671, D_loss_fake = 0.588, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 5/46, D_loss_real = 0.721, D_loss_fake = 0.579, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 6/46, D_loss_real = 0.700, D_loss_fake = 0.599, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.595, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 8/46, D_loss_real = 0.678, D_loss_fake = 0.580, G_loss =
0.850
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 9/46, D_loss_real = 0.642, D_loss_fake = 0.643, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 10/46, D_loss_real = 0.724, D_loss_fake = 0.631, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 11/46, D_loss_real = 0.655, D_loss_fake = 0.582, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 12/46, D_loss_real = 0.641, D_loss_fake = 0.592, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 13/46, D_loss_real = 0.650, D_loss_fake = 0.611, G_loss =
0.849
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 14/46, D_loss_real = 0.586, D_loss_fake = 0.603, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.659, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 16/46, D_loss_real = 0.623, D_loss_fake = 0.659, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 17/46, D_loss_real = 0.642, D_loss_fake = 0.674, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 18/46, D_loss_real = 0.677, D_loss_fake = 0.614, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 19/46, D_loss_real = 0.663, D_loss_fake = 0.621, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 20/46, D_loss_real = 0.600, D_loss_fake = 0.653, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 21/46, D_loss_real = 0.650, D_loss_fake = 0.672, G_loss =
0.785
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 22/46, D_loss_real = 0.642, D_loss_fake = 0.680, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 23/46, D_loss_real = 0.638, D_loss_fake = 0.675, G_loss =
0.879
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 24/46, D_loss_real = 0.666, D_loss_fake = 0.643, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 25/46, D_loss_real = 0.731, D_loss_fake = 0.625, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 26/46, D_loss_real = 0.673, D_loss_fake = 0.635, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 27/46, D_loss_real = 0.653, D_loss_fake = 0.618, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 28/46, D_loss_real = 0.670, D_loss_fake = 0.635, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.622, G_loss =
0.804
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 30/46, D_loss_real = 0.628, D_loss_fake = 0.620, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 31/46, D_loss_real = 0.651, D_loss_fake = 0.607, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 32/46, D_loss_real = 0.668, D_loss_fake = 0.632, G_loss =
0.766
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 33/46, D_loss_real = 0.604, D_loss_fake = 0.701, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 34/46, D_loss_real = 0.638, D_loss_fake = 0.606, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 35/46, D_loss_real = 0.657, D_loss_fake = 0.628, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 36/46, D_loss_real = 0.652, D_loss_fake = 0.638, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 37/46, D_loss_real = 0.646, D_loss_fake = 0.648, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 38/46, D_loss_real = 0.670, D_loss_fake = 0.644, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 39/46, D_loss_real = 0.638, D_loss_fake = 0.682, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.640, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 477, Iteration 41/46, D_loss_real = 0.663, D_loss_fake = 0.614, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 42/46, D_loss_real = 0.638, D_loss_fake = 0.669, G_loss =
0.876
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.645, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 44/46, D_loss_real = 0.693, D_loss_fake = 0.598, G_loss =
0.889
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 45/46, D_loss_real = 0.663, D_loss_fake = 0.638, G_loss =
0.867
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 477, Iteration 46/46, D_loss_real = 0.647, D_loss_fake = 0.635, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 1/46, D_loss_real = 0.645, D_loss_fake = 0.566, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 2/46, D_loss_real = 0.649, D_loss_fake = 0.634, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 3/46, D_loss_real = 0.642, D_loss_fake = 0.648, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 4/46, D_loss_real = 0.637, D_loss_fake = 0.636, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 5/46, D_loss_real = 0.605, D_loss_fake = 0.683, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 6/46, D_loss_real = 0.576, D_loss_fake = 0.632, G_loss =
0.795
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 7/46, D_loss_real = 0.615, D_loss_fake = 0.591, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 8/46, D_loss_real = 0.593, D_loss_fake = 0.583, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 9/46, D_loss_real = 0.592, D_loss_fake = 0.697, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 10/46, D_loss_real = 0.699, D_loss_fake = 0.652, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 11/46, D_loss_real = 0.635, D_loss_fake = 0.664, G_loss =
0.761
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 12/46, D_loss_real = 0.652, D_loss_fake = 0.643, G_loss =
0.796
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 13/46, D_loss_real = 0.578, D_loss_fake = 0.647, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 14/46, D_loss_real = 0.547, D_loss_fake = 0.682, G_loss =
0.775
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 15/46, D_loss_real = 0.672, D_loss_fake = 0.662, G_loss =
0.866
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 16/46, D_loss_real = 0.590, D_loss_fake = 0.578, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 17/46, D_loss_real = 0.627, D_loss_fake = 0.629, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 18/46, D_loss_real = 0.661, D_loss_fake = 0.613, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 19/46, D_loss_real = 0.633, D_loss_fake = 0.655, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.613, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 21/46, D_loss_real = 0.678, D_loss_fake = 0.598, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 22/46, D_loss_real = 0.700, D_loss_fake = 0.596, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 23/46, D_loss_real = 0.672, D_loss_fake = 0.606, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 24/46, D_loss_real = 0.629, D_loss_fake = 0.654, G_loss =
0.800
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 25/46, D_loss_real = 0.650, D_loss_fake = 0.598, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 26/46, D_loss_real = 0.626, D_loss_fake = 0.619, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 27/46, D_loss_real = 0.671, D_loss_fake = 0.620, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 28/46, D_loss_real = 0.608, D_loss_fake = 0.700, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 29/46, D_loss_real = 0.683, D_loss_fake = 0.592, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 30/46, D_loss_real = 0.645, D_loss_fake = 0.723, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.616, G_loss =
0.807
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 32/46, D_loss_real = 0.673, D_loss_fake = 0.623, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 33/46, D_loss_real = 0.622, D_loss_fake = 0.639, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 34/46, D_loss_real = 0.641, D_loss_fake = 0.638, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 35/46, D_loss_real = 0.645, D_loss_fake = 0.668, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 36/46, D_loss_real = 0.569, D_loss_fake = 0.792, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 37/46, D_loss_real = 0.669, D_loss_fake = 0.622, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.614, G_loss =
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 39/46, D_loss_real = 0.675, D_loss_fake = 0.643, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 40/46, D_loss_real = 0.618, D_loss_fake = 0.616, G_loss =
0.798
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 41/46, D_loss_real = 0.700, D_loss_fake = 0.629, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 42/46, D_loss_real = 0.668, D_loss_fake = 0.643, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.601, G_loss =
0.872
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 44/46, D_loss_real = 0.713, D_loss_fake = 0.597, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 478, Iteration 45/46, D_loss_real = 0.657, D_loss_fake = 0.595, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 478, Iteration 46/46, D_loss_real = 0.727, D_loss_fake = 0.624, G_loss =
0.899
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 1/46, D_loss_real = 0.651, D_loss_fake = 0.583, G_loss =
0.893
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.586, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 3/46, D_loss_real = 0.608, D_loss_fake = 0.597, G_loss =
0.876
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.644, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.660, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.610, G_loss =
0.856
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 7/46, D_loss_real = 0.707, D_loss_fake = 0.610, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 8/46, D_loss_real = 0.710, D_loss_fake = 0.626, G_loss =
0.872
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 9/46, D_loss_real = 0.673, D_loss_fake = 0.588, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 10/46, D_loss_real = 0.640, D_loss_fake = 0.632, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 11/46, D_loss_real = 0.582, D_loss_fake = 0.586, G_loss =
0.879
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 12/46, D_loss_real = 0.637, D_loss_fake = 0.586, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 13/46, D_loss_real = 0.644, D_loss_fake = 0.616, G_loss =
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 14/46, D_loss_real = 0.597, D_loss_fake = 0.719, G_loss =
0.810
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 15/46, D_loss_real = 0.627, D_loss_fake = 0.649, G_loss =
0.813
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 16/46, D_loss_real = 0.623, D_loss_fake = 0.629, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 17/46, D_loss_real = 0.601, D_loss_fake = 0.636, G_loss =
0.782
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 18/46, D_loss_real = 0.603, D_loss_fake = 0.671, G_loss =
0.801
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 19/46, D_loss_real = 0.596, D_loss_fake = 0.671, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 20/46, D_loss_real = 0.622, D_loss_fake = 0.625, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 21/46, D_loss_real = 0.666, D_loss_fake = 0.660, G_loss =
0.875
2/2 [==============================] - 0s 5ms/step
Epoch 479, Iteration 22/46, D_loss_real = 0.634, D_loss_fake = 0.593, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 23/46, D_loss_real = 0.689, D_loss_fake = 0.613, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 24/46, D_loss_real = 0.639, D_loss_fake = 0.626, G_loss =
0.796
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 25/46, D_loss_real = 0.598, D_loss_fake = 0.632, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 26/46, D_loss_real = 0.605, D_loss_fake = 0.655, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 27/46, D_loss_real = 0.641, D_loss_fake = 0.640, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 28/46, D_loss_real = 0.576, D_loss_fake = 0.728, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 29/46, D_loss_real = 0.608, D_loss_fake = 0.650, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 30/46, D_loss_real = 0.625, D_loss_fake = 0.603, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 31/46, D_loss_real = 0.635, D_loss_fake = 0.732, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 32/46, D_loss_real = 0.588, D_loss_fake = 0.624, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 33/46, D_loss_real = 0.629, D_loss_fake = 0.620, G_loss =
0.843
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 34/46, D_loss_real = 0.585, D_loss_fake = 0.624, G_loss =
0.821
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 35/46, D_loss_real = 0.625, D_loss_fake = 0.628, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 36/46, D_loss_real = 0.627, D_loss_fake = 0.624, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 37/46, D_loss_real = 0.548, D_loss_fake = 0.609, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 38/46, D_loss_real = 0.663, D_loss_fake = 0.647, G_loss =
0.800
2/2 [==============================] - 0s 3ms/step
Epoch 479, Iteration 39/46, D_loss_real = 0.571, D_loss_fake = 0.576, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 40/46, D_loss_real = 0.679, D_loss_fake = 0.652, G_loss =
0.837
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 41/46, D_loss_real = 0.636, D_loss_fake = 0.636, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 42/46, D_loss_real = 0.724, D_loss_fake = 0.664, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 43/46, D_loss_real = 0.638, D_loss_fake = 0.626, G_loss =
0.872
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 44/46, D_loss_real = 0.678, D_loss_fake = 0.612, G_loss =
0.861
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 45/46, D_loss_real = 0.685, D_loss_fake = 0.591, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
Epoch 479, Iteration 46/46, D_loss_real = 0.637, D_loss_fake = 0.630, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 1/46, D_loss_real = 0.637, D_loss_fake = 0.618, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 2/46, D_loss_real = 0.651, D_loss_fake = 0.624, G_loss =
0.797
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 3/46, D_loss_real = 0.563, D_loss_fake = 0.647, G_loss =
0.822
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 4/46, D_loss_real = 0.591, D_loss_fake = 0.686, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 5/46, D_loss_real = 0.715, D_loss_fake = 0.678, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 6/46, D_loss_real = 0.717, D_loss_fake = 0.638, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.585, G_loss =
0.798
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 8/46, D_loss_real = 0.674, D_loss_fake = 0.610, G_loss =
0.782
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.637, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 10/46, D_loss_real = 0.673, D_loss_fake = 0.657, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 11/46, D_loss_real = 0.603, D_loss_fake = 0.629, G_loss =
0.800
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 12/46, D_loss_real = 0.633, D_loss_fake = 0.647, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 13/46, D_loss_real = 0.546, D_loss_fake = 0.672, G_loss =
0.804
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 14/46, D_loss_real = 0.526, D_loss_fake = 0.652, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 15/46, D_loss_real = 0.603, D_loss_fake = 0.648, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 16/46, D_loss_real = 0.657, D_loss_fake = 0.641, G_loss =
0.847
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 17/46, D_loss_real = 0.584, D_loss_fake = 0.679, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 18/46, D_loss_real = 0.715, D_loss_fake = 0.573, G_loss =
0.870
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 19/46, D_loss_real = 0.649, D_loss_fake = 0.679, G_loss =
0.805
```

```
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 20/46, D_loss_real = 0.581, D_loss_fake = 0.635, G_loss =
0.908
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.641, G_loss =
0.895
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 22/46, D_loss_real = 0.677, D_loss_fake = 0.607, G_loss =
0.903
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 23/46, D_loss_real = 0.633, D_loss_fake = 0.606, G_loss =
0.901
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 24/46, D_loss_real = 0.677, D_loss_fake = 0.647, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 25/46, D_loss_real = 0.645, D_loss_fake = 0.630, G_loss =
0.894
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 26/46, D_loss_real = 0.625, D_loss_fake = 0.663, G_loss =
0.868
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 27/46, D_loss_real = 0.668, D_loss_fake = 0.592, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 28/46, D_loss_real = 0.657, D_loss_fake = 0.653, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 29/46, D_loss_real = 0.647, D_loss_fake = 0.613, G_loss =
0.888
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.598, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 31/46, D_loss_real = 0.681, D_loss_fake = 0.597, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 32/46, D_loss_real = 0.665, D_loss_fake = 0.630, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 33/46, D_loss_real = 0.645, D_loss_fake = 0.592, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 34/46, D_loss_real = 0.642, D_loss_fake = 0.645, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 35/46, D_loss_real = 0.664, D_loss_fake = 0.624, G_loss =
0.863
```

```
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 36/46, D_loss_real = 0.654, D_loss_fake = 0.649, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 37/46, D_loss_real = 0.631, D_loss_fake = 0.631, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 38/46, D_loss_real = 0.633, D_loss_fake = 0.624, G_loss =
0.841
2/2 [==============================] - 0s 12ms/step
Epoch 480, Iteration 39/46, D_loss_real = 0.629, D_loss_fake = 0.622, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 40/46, D_loss_real = 0.642, D_loss_fake = 0.606, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 41/46, D_loss_real = 0.624, D_loss_fake = 0.656, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 42/46, D_loss_real = 0.576, D_loss_fake = 0.632, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 43/46, D_loss_real = 0.618, D_loss_fake = 0.657, G_loss =
0.788
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 44/46, D_loss_real = 0.598, D_loss_fake = 0.719, G_loss =
0.801
2/2 [==============================] - 0s 4ms/step
Epoch 480, Iteration 45/46, D_loss_real = 0.605, D_loss_fake = 0.650, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 480, Iteration 46/46, D_loss_real = 0.656, D_loss_fake = 0.684, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 1/46, D_loss_real = 0.625, D_loss_fake = 0.666, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 2/46, D_loss_real = 0.649, D_loss_fake = 0.632, G_loss =
0.888
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 3/46, D_loss_real = 0.624, D_loss_fake = 0.617, G_loss =
0.882
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 4/46, D_loss_real = 0.596, D_loss_fake = 0.668, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.618, G_loss =
```

```
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 6/46, D_loss_real = 0.682, D_loss_fake = 0.619, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 7/46, D_loss_real = 0.639, D_loss_fake = 0.684, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 8/46, D_loss_real = 0.666, D_loss_fake = 0.671, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 9/46, D_loss_real = 0.675, D_loss_fake = 0.592, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 10/46, D_loss_real = 0.691, D_loss_fake = 0.662, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 11/46, D_loss_real = 0.653, D_loss_fake = 0.605, G_loss =
0.825
2/2 [==============================] - 0s 13ms/step
Epoch 481, Iteration 12/46, D_loss_real = 0.613, D_loss_fake = 0.623, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 13/46, D_loss_real = 0.711, D_loss_fake = 0.650, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.599, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 15/46, D_loss_real = 0.652, D_loss_fake = 0.617, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 16/46, D_loss_real = 0.699, D_loss_fake = 0.606, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 17/46, D_loss_real = 0.563, D_loss_fake = 0.604, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 18/46, D_loss_real = 0.609, D_loss_fake = 0.618, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 19/46, D_loss_real = 0.620, D_loss_fake = 0.592, G_loss =
0.772
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 20/46, D_loss_real = 0.579, D_loss_fake = 0.730, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.641, G_loss =
```

```
0.792
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.721, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 23/46, D_loss_real = 0.684, D_loss_fake = 0.597, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 24/46, D_loss_real = 0.645, D_loss_fake = 0.570, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 25/46, D_loss_real = 0.672, D_loss_fake = 0.682, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 26/46, D_loss_real = 0.661, D_loss_fake = 0.625, G_loss =
0.873
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 27/46, D_loss_real = 0.655, D_loss_fake = 0.634, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 28/46, D_loss_real = 0.584, D_loss_fake = 0.628, G_loss =
0.847
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 29/46, D_loss_real = 0.704, D_loss_fake = 0.615, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 30/46, D_loss_real = 0.680, D_loss_fake = 0.580, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 31/46, D_loss_real = 0.631, D_loss_fake = 0.680, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 32/46, D_loss_real = 0.600, D_loss_fake = 0.657, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 33/46, D_loss_real = 0.640, D_loss_fake = 0.630, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 34/46, D_loss_real = 0.658, D_loss_fake = 0.606, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 35/46, D_loss_real = 0.652, D_loss_fake = 0.682, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 36/46, D_loss_real = 0.652, D_loss_fake = 0.610, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 37/46, D_loss_real = 0.680, D_loss_fake = 0.626, G_loss =
```

```
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 38/46, D_loss_real = 0.592, D_loss_fake = 0.661, G_loss =
0.881
2/2 [==============================] - 0s 4ms/step
Epoch 481, Iteration 39/46, D_loss_real = 0.576, D_loss_fake = 0.603, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 40/46, D_loss_real = 0.660, D_loss_fake = 0.574, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 41/46, D_loss_real = 0.625, D_loss_fake = 0.654, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 42/46, D_loss_real = 0.650, D_loss_fake = 0.625, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 43/46, D_loss_real = 0.670, D_loss_fake = 0.687, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 44/46, D_loss_real = 0.619, D_loss_fake = 0.605, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.598, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 481, Iteration 46/46, D_loss_real = 0.640, D_loss_fake = 0.657, G_loss =
0.886
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 1/46, D_loss_real = 0.615, D_loss_fake = 0.634, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 2/46, D_loss_real = 0.597, D_loss_fake = 0.596, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 3/46, D_loss_real = 0.614, D_loss_fake = 0.668, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 4/46, D_loss_real = 0.633, D_loss_fake = 0.662, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.614, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 6/46, D_loss_real = 0.613, D_loss_fake = 0.674, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 7/46, D_loss_real = 0.633, D_loss_fake = 0.664, G_loss =
```

```
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 8/46, D_loss_real = 0.639, D_loss_fake = 0.642, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 9/46, D_loss_real = 0.612, D_loss_fake = 0.622, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 10/46, D_loss_real = 0.668, D_loss_fake = 0.619, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 11/46, D_loss_real = 0.643, D_loss_fake = 0.589, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 12/46, D_loss_real = 0.625, D_loss_fake = 0.586, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 482, Iteration 13/46, D_loss_real = 0.612, D_loss_fake = 0.617, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 14/46, D_loss_real = 0.615, D_loss_fake = 0.689, G_loss =
0.873
2/2 [==============================] - 0s 4ms/step
Epoch 482, Iteration 15/46, D_loss_real = 0.598, D_loss_fake = 0.597, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 16/46, D_loss_real = 0.621, D_loss_fake = 0.629, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 17/46, D_loss_real = 0.631, D_loss_fake = 0.673, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 18/46, D_loss_real = 0.612, D_loss_fake = 0.629, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 19/46, D_loss_real = 0.698, D_loss_fake = 0.608, G_loss =
0.915
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 20/46, D_loss_real = 0.705, D_loss_fake = 0.622, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 21/46, D_loss_real = 0.655, D_loss_fake = 0.604, G_loss =
0.885
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 22/46, D_loss_real = 0.680, D_loss_fake = 0.704, G_loss =
0.891
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 23/46, D_loss_real = 0.662, D_loss_fake = 0.623, G_loss =
```

```
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 24/46, D_loss_real = 0.697, D_loss_fake = 0.616, G_loss =
0.865
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 25/46, D_loss_real = 0.683, D_loss_fake = 0.589, G_loss =
0.893
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 26/46, D_loss_real = 0.668, D_loss_fake = 0.585, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 27/46, D_loss_real = 0.666, D_loss_fake = 0.596, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 28/46, D_loss_real = 0.589, D_loss_fake = 0.601, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 29/46, D_loss_real = 0.553, D_loss_fake = 0.655, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 30/46, D_loss_real = 0.608, D_loss_fake = 0.656, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 482, Iteration 31/46, D_loss_real = 0.607, D_loss_fake = 0.627, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 32/46, D_loss_real = 0.552, D_loss_fake = 0.658, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 33/46, D_loss_real = 0.634, D_loss_fake = 0.639, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 34/46, D_loss_real = 0.634, D_loss_fake = 0.672, G_loss =
0.781
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 35/46, D_loss_real = 0.654, D_loss_fake = 0.617, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 36/46, D_loss_real = 0.609, D_loss_fake = 0.675, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 37/46, D_loss_real = 0.587, D_loss_fake = 0.673, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 38/46, D_loss_real = 0.615, D_loss_fake = 0.598, G_loss =
0.904
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 39/46, D_loss_real = 0.642, D_loss_fake = 0.598, G_loss =
```

```
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 40/46, D_loss_real = 0.735, D_loss_fake = 0.580, G_loss =
0.888
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 41/46, D_loss_real = 0.675, D_loss_fake = 0.632, G_loss =
0.888
2/2 [==============================] - 0s 4ms/step
Epoch 482, Iteration 42/46, D_loss_real = 0.705, D_loss_fake = 0.618, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 43/46, D_loss_real = 0.679, D_loss_fake = 0.605, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 44/46, D_loss_real = 0.644, D_loss_fake = 0.603, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.593, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 482, Iteration 46/46, D_loss_real = 0.588, D_loss_fake = 0.626, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 1/46, D_loss_real = 0.669, D_loss_fake = 0.559, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 2/46, D_loss_real = 0.610, D_loss_fake = 0.667, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 3/46, D_loss_real = 0.659, D_loss_fake = 0.591, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 4/46, D_loss_real = 0.650, D_loss_fake = 0.619, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 5/46, D_loss_real = 0.700, D_loss_fake = 0.622, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.638, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 7/46, D_loss_real = 0.607, D_loss_fake = 0.619, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 8/46, D_loss_real = 0.619, D_loss_fake = 0.658, G_loss =
0.865
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 9/46, D_loss_real = 0.594, D_loss_fake = 0.648, G_loss =
```

```
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.659, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 483, Iteration 11/46, D_loss_real = 0.573, D_loss_fake = 0.596, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 12/46, D_loss_real = 0.644, D_loss_fake = 0.658, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 13/46, D_loss_real = 0.663, D_loss_fake = 0.622, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 14/46, D_loss_real = 0.583, D_loss_fake = 0.653, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 15/46, D_loss_real = 0.626, D_loss_fake = 0.653, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 16/46, D_loss_real = 0.637, D_loss_fake = 0.653, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 17/46, D_loss_real = 0.621, D_loss_fake = 0.677, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 18/46, D_loss_real = 0.614, D_loss_fake = 0.636, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 19/46, D_loss_real = 0.678, D_loss_fake = 0.625, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 20/46, D_loss_real = 0.675, D_loss_fake = 0.642, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 21/46, D_loss_real = 0.621, D_loss_fake = 0.588, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 22/46, D_loss_real = 0.658, D_loss_fake = 0.613, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 23/46, D_loss_real = 0.595, D_loss_fake = 0.629, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 24/46, D_loss_real = 0.612, D_loss_fake = 0.615, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 483, Iteration 25/46, D_loss_real = 0.551, D_loss_fake = 0.661, G_loss =
```

```
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 26/46, D_loss_real = 0.592, D_loss_fake = 0.667, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 27/46, D_loss_real = 0.605, D_loss_fake = 0.639, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 28/46, D_loss_real = 0.651, D_loss_fake = 0.609, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 29/46, D_loss_real = 0.654, D_loss_fake = 0.649, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 30/46, D_loss_real = 0.632, D_loss_fake = 0.680, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 31/46, D_loss_real = 0.683, D_loss_fake = 0.627, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 32/46, D_loss_real = 0.590, D_loss_fake = 0.673, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 33/46, D_loss_real = 0.643, D_loss_fake = 0.651, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 34/46, D_loss_real = 0.664, D_loss_fake = 0.627, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 35/46, D_loss_real = 0.584, D_loss_fake = 0.683, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 36/46, D_loss_real = 0.660, D_loss_fake = 0.629, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 37/46, D_loss_real = 0.649, D_loss_fake = 0.653, G_loss =
0.809
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 38/46, D_loss_real = 0.622, D_loss_fake = 0.622, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 483, Iteration 39/46, D_loss_real = 0.664, D_loss_fake = 0.624, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 40/46, D_loss_real = 0.626, D_loss_fake = 0.607, G_loss =
0.850
2/2 [==============================] - 0s 4ms/step
Epoch 483, Iteration 41/46, D_loss_real = 0.652, D_loss_fake = 0.596, G_loss =
```

```
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 42/46, D_loss_real = 0.661, D_loss_fake = 0.608, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 483, Iteration 43/46, D_loss_real = 0.658, D_loss_fake = 0.606, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 483, Iteration 44/46, D_loss_real = 0.673, D_loss_fake = 0.633, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 483, Iteration 45/46, D_loss_real = 0.680, D_loss_fake = 0.645, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 483, Iteration 46/46, D_loss_real = 0.654, D_loss_fake = 0.677, G_loss =
0.868
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 1/46, D_loss_real = 0.596, D_loss_fake = 0.588, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 2/46, D_loss_real = 0.723, D_loss_fake = 0.596, G_loss =
0.864
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 3/46, D_loss_real = 0.673, D_loss_fake = 0.608, G_loss =
0.875
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.611, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 5/46, D_loss_real = 0.604, D_loss_fake = 0.636, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 6/46, D_loss_real = 0.603, D_loss_fake = 0.668, G_loss =
0.858
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 7/46, D_loss_real = 0.627, D_loss_fake = 0.638, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 8/46, D_loss_real = 0.686, D_loss_fake = 0.603, G_loss =
0.802
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 9/46, D_loss_real = 0.605, D_loss_fake = 0.620, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 10/46, D_loss_real = 0.677, D_loss_fake = 0.626, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 11/46, D_loss_real = 0.668, D_loss_fake = 0.650, G_loss =
```

```
0.821
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 12/46, D_loss_real = 0.666, D_loss_fake = 0.645, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 13/46, D_loss_real = 0.624, D_loss_fake = 0.637, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 14/46, D_loss_real = 0.675, D_loss_fake = 0.617, G_loss =
0.781
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 15/46, D_loss_real = 0.658, D_loss_fake = 0.611, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 16/46, D_loss_real = 0.590, D_loss_fake = 0.686, G_loss =
0.867
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 17/46, D_loss_real = 0.625, D_loss_fake = 0.633, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 18/46, D_loss_real = 0.665, D_loss_fake = 0.631, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 19/46, D_loss_real = 0.658, D_loss_fake = 0.704, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 20/46, D_loss_real = 0.616, D_loss_fake = 0.627, G_loss =
0.837
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 21/46, D_loss_real = 0.620, D_loss_fake = 0.657, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 22/46, D_loss_real = 0.675, D_loss_fake = 0.672, G_loss =
0.845
2/2 [==============================] - 0s 5ms/step
Epoch 484, Iteration 23/46, D_loss_real = 0.663, D_loss_fake = 0.664, G_loss =
0.875
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 24/46, D_loss_real = 0.728, D_loss_fake = 0.611, G_loss =
0.911
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 25/46, D_loss_real = 0.709, D_loss_fake = 0.560, G_loss =
0.868
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 26/46, D_loss_real = 0.666, D_loss_fake = 0.585, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 27/46, D_loss_real = 0.672, D_loss_fake = 0.624, G_loss =
```

```
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 28/46, D_loss_real = 0.660, D_loss_fake = 0.634, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 29/46, D_loss_real = 0.607, D_loss_fake = 0.649, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 30/46, D_loss_real = 0.618, D_loss_fake = 0.647, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 31/46, D_loss_real = 0.667, D_loss_fake = 0.663, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 32/46, D_loss_real = 0.614, D_loss_fake = 0.599, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.586, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 34/46, D_loss_real = 0.620, D_loss_fake = 0.683, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 35/46, D_loss_real = 0.579, D_loss_fake = 0.629, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 36/46, D_loss_real = 0.656, D_loss_fake = 0.593, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 37/46, D_loss_real = 0.622, D_loss_fake = 0.635, G_loss =
0.796
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 38/46, D_loss_real = 0.626, D_loss_fake = 0.645, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 39/46, D_loss_real = 0.682, D_loss_fake = 0.633, G_loss =
0.797
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 40/46, D_loss_real = 0.671, D_loss_fake = 0.692, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 41/46, D_loss_real = 0.656, D_loss_fake = 0.605, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 484, Iteration 42/46, D_loss_real = 0.671, D_loss_fake = 0.603, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 43/46, D_loss_real = 0.669, D_loss_fake = 0.661, G_loss =
```

```
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 44/46, D_loss_real = 0.643, D_loss_fake = 0.619, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 45/46, D_loss_real = 0.682, D_loss_fake = 0.620, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 484, Iteration 46/46, D_loss_real = 0.690, D_loss_fake = 0.599, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 1/46, D_loss_real = 0.633, D_loss_fake = 0.602, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 2/46, D_loss_real = 0.649, D_loss_fake = 0.643, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 3/46, D_loss_real = 0.644, D_loss_fake = 0.600, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 4/46, D_loss_real = 0.638, D_loss_fake = 0.680, G_loss =
0.858
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 5/46, D_loss_real = 0.623, D_loss_fake = 0.590, G_loss =
0.863
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 6/46, D_loss_real = 0.636, D_loss_fake = 0.645, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 7/46, D_loss_real = 0.634, D_loss_fake = 0.614, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 8/46, D_loss_real = 0.615, D_loss_fake = 0.621, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 9/46, D_loss_real = 0.603, D_loss_fake = 0.627, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 10/46, D_loss_real = 0.678, D_loss_fake = 0.619, G_loss =
0.818
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 11/46, D_loss_real = 0.627, D_loss_fake = 0.612, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 12/46, D_loss_real = 0.569, D_loss_fake = 0.652, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 13/46, D_loss_real = 0.636, D_loss_fake = 0.610, G_loss =
```

```
0.799
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 14/46, D_loss_real = 0.531, D_loss_fake = 0.671, G_loss =
0.816
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 15/46, D_loss_real = 0.559, D_loss_fake = 0.643, G_loss =
0.798
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 16/46, D_loss_real = 0.658, D_loss_fake = 0.662, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 17/46, D_loss_real = 0.571, D_loss_fake = 0.668, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 18/46, D_loss_real = 0.606, D_loss_fake = 0.680, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 19/46, D_loss_real = 0.619, D_loss_fake = 0.625, G_loss =
0.866
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 20/46, D_loss_real = 0.707, D_loss_fake = 0.614, G_loss =
0.862
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 21/46, D_loss_real = 0.665, D_loss_fake = 0.573, G_loss =
0.869
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 22/46, D_loss_real = 0.728, D_loss_fake = 0.632, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 23/46, D_loss_real = 0.627, D_loss_fake = 0.579, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 24/46, D_loss_real = 0.631, D_loss_fake = 0.670, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 25/46, D_loss_real = 0.628, D_loss_fake = 0.641, G_loss =
0.869
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 26/46, D_loss_real = 0.625, D_loss_fake = 0.614, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 27/46, D_loss_real = 0.601, D_loss_fake = 0.646, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 28/46, D_loss_real = 0.563, D_loss_fake = 0.601, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 29/46, D_loss_real = 0.607, D_loss_fake = 0.613, G_loss =
```

```
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 30/46, D_loss_real = 0.603, D_loss_fake = 0.624, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 31/46, D_loss_real = 0.638, D_loss_fake = 0.565, G_loss =
0.889
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 32/46, D_loss_real = 0.649, D_loss_fake = 0.619, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 33/46, D_loss_real = 0.638, D_loss_fake = 0.604, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 34/46, D_loss_real = 0.634, D_loss_fake = 0.656, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 35/46, D_loss_real = 0.647, D_loss_fake = 0.631, G_loss =
0.888
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 36/46, D_loss_real = 0.595, D_loss_fake = 0.617, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 37/46, D_loss_real = 0.660, D_loss_fake = 0.641, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 38/46, D_loss_real = 0.731, D_loss_fake = 0.618, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 39/46, D_loss_real = 0.636, D_loss_fake = 0.637, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 40/46, D_loss_real = 0.657, D_loss_fake = 0.626, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 41/46, D_loss_real = 0.649, D_loss_fake = 0.625, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 42/46, D_loss_real = 0.629, D_loss_fake = 0.620, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 43/46, D_loss_real = 0.624, D_loss_fake = 0.600, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 44/46, D_loss_real = 0.681, D_loss_fake = 0.612, G_loss =
0.801
2/2 [==============================] - 0s 3ms/step
Epoch 485, Iteration 45/46, D_loss_real = 0.613, D_loss_fake = 0.567, G_loss =
```

```
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 485, Iteration 46/46, D_loss_real = 0.635, D_loss_fake = 0.764, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 1/46, D_loss_real = 0.557, D_loss_fake = 0.582, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 2/46, D_loss_real = 0.585, D_loss_fake = 0.634, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 3/46, D_loss_real = 0.602, D_loss_fake = 0.641, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 4/46, D_loss_real = 0.614, D_loss_fake = 0.680, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 5/46, D_loss_real = 0.642, D_loss_fake = 0.628, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.617, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 7/46, D_loss_real = 0.706, D_loss_fake = 0.732, G_loss =
0.889
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 8/46, D_loss_real = 0.727, D_loss_fake = 0.574, G_loss =
0.890
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 9/46, D_loss_real = 0.677, D_loss_fake = 0.579, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 10/46, D_loss_real = 0.602, D_loss_fake = 0.677, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 11/46, D_loss_real = 0.659, D_loss_fake = 0.561, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 12/46, D_loss_real = 0.622, D_loss_fake = 0.632, G_loss =
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 13/46, D_loss_real = 0.683, D_loss_fake = 0.619, G_loss =
0.878
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 14/46, D_loss_real = 0.642, D_loss_fake = 0.620, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 15/46, D_loss_real = 0.619, D_loss_fake = 0.641, G_loss =
```

```
0.820
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 16/46, D_loss_real = 0.611, D_loss_fake = 0.644, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 17/46, D_loss_real = 0.648, D_loss_fake = 0.610, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 18/46, D_loss_real = 0.569, D_loss_fake = 0.574, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 19/46, D_loss_real = 0.613, D_loss_fake = 0.601, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 20/46, D_loss_real = 0.651, D_loss_fake = 0.646, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 21/46, D_loss_real = 0.623, D_loss_fake = 0.567, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 22/46, D_loss_real = 0.641, D_loss_fake = 0.652, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 23/46, D_loss_real = 0.624, D_loss_fake = 0.669, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 24/46, D_loss_real = 0.669, D_loss_fake = 0.677, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 25/46, D_loss_real = 0.633, D_loss_fake = 0.646, G_loss =
0.822
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 26/46, D_loss_real = 0.659, D_loss_fake = 0.663, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.692, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 28/46, D_loss_real = 0.653, D_loss_fake = 0.635, G_loss =
0.889
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 29/46, D_loss_real = 0.718, D_loss_fake = 0.610, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 30/46, D_loss_real = 0.685, D_loss_fake = 0.668, G_loss =
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.624, G_loss =
```

```
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 32/46, D_loss_real = 0.651, D_loss_fake = 0.591, G_loss =
0.795
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 33/46, D_loss_real = 0.685, D_loss_fake = 0.640, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 34/46, D_loss_real = 0.641, D_loss_fake = 0.632, G_loss =
0.803
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 35/46, D_loss_real = 0.673, D_loss_fake = 0.645, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 36/46, D_loss_real = 0.595, D_loss_fake = 0.634, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 37/46, D_loss_real = 0.583, D_loss_fake = 0.627, G_loss =
0.870
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 38/46, D_loss_real = 0.658, D_loss_fake = 0.608, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 39/46, D_loss_real = 0.618, D_loss_fake = 0.563, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 40/46, D_loss_real = 0.660, D_loss_fake = 0.646, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 41/46, D_loss_real = 0.633, D_loss_fake = 0.672, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 42/46, D_loss_real = 0.678, D_loss_fake = 0.638, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 43/46, D_loss_real = 0.613, D_loss_fake = 0.709, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 486, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.661, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 45/46, D_loss_real = 0.598, D_loss_fake = 0.632, G_loss =
0.784
2/2 [==============================] - 0s 3ms/step
Epoch 486, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.652, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 1/46, D_loss_real = 0.624, D_loss_fake = 0.684, G_loss =
```

```
0.894
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 2/46, D_loss_real = 0.695, D_loss_fake = 0.574, G_loss =
0.903
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 3/46, D_loss_real = 0.661, D_loss_fake = 0.598, G_loss =
0.951
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 4/46, D_loss_real = 0.649, D_loss_fake = 0.609, G_loss =
0.942
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.550, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 6/46, D_loss_real = 0.674, D_loss_fake = 0.573, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 7/46, D_loss_real = 0.571, D_loss_fake = 0.567, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 8/46, D_loss_real = 0.649, D_loss_fake = 0.684, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 9/46, D_loss_real = 0.579, D_loss_fake = 0.658, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 10/46, D_loss_real = 0.642, D_loss_fake = 0.632, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 11/46, D_loss_real = 0.670, D_loss_fake = 0.630, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 12/46, D_loss_real = 0.635, D_loss_fake = 0.599, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 13/46, D_loss_real = 0.665, D_loss_fake = 0.578, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 14/46, D_loss_real = 0.599, D_loss_fake = 0.607, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 15/46, D_loss_real = 0.570, D_loss_fake = 0.650, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 16/46, D_loss_real = 0.629, D_loss_fake = 0.622, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
Epoch 487, Iteration 17/46, D_loss_real = 0.649, D_loss_fake = 0.625, G_loss =
```

```
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 18/46, D_loss_real = 0.660, D_loss_fake = 0.623, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 487, Iteration 19/46, D_loss_real = 0.560, D_loss_fake = 0.682, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 20/46, D_loss_real = 0.627, D_loss_fake = 0.673, G_loss =
0.805
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 21/46, D_loss_real = 0.596, D_loss_fake = 0.612, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 22/46, D_loss_real = 0.708, D_loss_fake = 0.589, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 23/46, D_loss_real = 0.647, D_loss_fake = 0.588, G_loss =
0.921
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.637, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 25/46, D_loss_real = 0.688, D_loss_fake = 0.629, G_loss =
0.865
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 26/46, D_loss_real = 0.622, D_loss_fake = 0.624, G_loss =
0.885
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 27/46, D_loss_real = 0.663, D_loss_fake = 0.593, G_loss =
0.883
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 28/46, D_loss_real = 0.706, D_loss_fake = 0.598, G_loss =
0.878
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 29/46, D_loss_real = 0.641, D_loss_fake = 0.575, G_loss =
0.889
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 30/46, D_loss_real = 0.633, D_loss_fake = 0.600, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 31/46, D_loss_real = 0.622, D_loss_fake = 0.677, G_loss =
0.895
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 32/46, D_loss_real = 0.608, D_loss_fake = 0.722, G_loss =
0.881
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 33/46, D_loss_real = 0.721, D_loss_fake = 0.571, G_loss =
```

```
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 34/46, D_loss_real = 0.621, D_loss_fake = 0.662, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 35/46, D_loss_real = 0.613, D_loss_fake = 0.600, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 36/46, D_loss_real = 0.642, D_loss_fake = 0.589, G_loss =
0.862
2/2 [==============================] - 0s 4ms/step
Epoch 487, Iteration 37/46, D_loss_real = 0.606, D_loss_fake = 0.606, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 38/46, D_loss_real = 0.595, D_loss_fake = 0.604, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 39/46, D_loss_real = 0.605, D_loss_fake = 0.609, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
Epoch 487, Iteration 40/46, D_loss_real = 0.615, D_loss_fake = 0.620, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 41/46, D_loss_real = 0.637, D_loss_fake = 0.716, G_loss =
0.797
2/2 [==============================] - 0s 4ms/step
Epoch 487, Iteration 42/46, D_loss_real = 0.622, D_loss_fake = 0.670, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 43/46, D_loss_real = 0.649, D_loss_fake = 0.622, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 44/46, D_loss_real = 0.586, D_loss_fake = 0.653, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 45/46, D_loss_real = 0.567, D_loss_fake = 0.649, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 487, Iteration 46/46, D_loss_real = 0.563, D_loss_fake = 0.661, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 1/46, D_loss_real = 0.634, D_loss_fake = 0.666, G_loss =
0.814
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 2/46, D_loss_real = 0.653, D_loss_fake = 0.636, G_loss =
0.825
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 3/46, D_loss_real = 0.697, D_loss_fake = 0.629, G_loss =
```

```
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 4/46, D_loss_real = 0.666, D_loss_fake = 0.641, G_loss =
0.790
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 5/46, D_loss_real = 0.641, D_loss_fake = 0.643, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 6/46, D_loss_real = 0.618, D_loss_fake = 0.642, G_loss =
0.862
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 7/46, D_loss_real = 0.625, D_loss_fake = 0.613, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 8/46, D_loss_real = 0.652, D_loss_fake = 0.653, G_loss =
0.870
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 9/46, D_loss_real = 0.606, D_loss_fake = 0.638, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 10/46, D_loss_real = 0.618, D_loss_fake = 0.624, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 11/46, D_loss_real = 0.585, D_loss_fake = 0.615, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 12/46, D_loss_real = 0.659, D_loss_fake = 0.672, G_loss =
0.885
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 13/46, D_loss_real = 0.669, D_loss_fake = 0.584, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 14/46, D_loss_real = 0.652, D_loss_fake = 0.558, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 15/46, D_loss_real = 0.662, D_loss_fake = 0.626, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 16/46, D_loss_real = 0.611, D_loss_fake = 0.583, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 17/46, D_loss_real = 0.616, D_loss_fake = 0.673, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 18/46, D_loss_real = 0.619, D_loss_fake = 0.635, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 19/46, D_loss_real = 0.648, D_loss_fake = 0.635, G_loss =
```

```
0.877
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.623, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 21/46, D_loss_real = 0.652, D_loss_fake = 0.611, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 22/46, D_loss_real = 0.636, D_loss_fake = 0.642, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 23/46, D_loss_real = 0.644, D_loss_fake = 0.684, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 24/46, D_loss_real = 0.639, D_loss_fake = 0.625, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 25/46, D_loss_real = 0.752, D_loss_fake = 0.602, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.640, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 27/46, D_loss_real = 0.726, D_loss_fake = 0.604, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 28/46, D_loss_real = 0.635, D_loss_fake = 0.655, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 29/46, D_loss_real = 0.627, D_loss_fake = 0.646, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 30/46, D_loss_real = 0.612, D_loss_fake = 0.681, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.620, G_loss =
0.864
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 32/46, D_loss_real = 0.666, D_loss_fake = 0.614, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 33/46, D_loss_real = 0.711, D_loss_fake = 0.594, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 34/46, D_loss_real = 0.588, D_loss_fake = 0.660, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 35/46, D_loss_real = 0.608, D_loss_fake = 0.712, G_loss =
```

```
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 36/46, D_loss_real = 0.628, D_loss_fake = 0.652, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 37/46, D_loss_real = 0.636, D_loss_fake = 0.635, G_loss =
0.876
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 38/46, D_loss_real = 0.661, D_loss_fake = 0.610, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 39/46, D_loss_real = 0.717, D_loss_fake = 0.631, G_loss =
0.893
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.588, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 41/46, D_loss_real = 0.594, D_loss_fake = 0.646, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 42/46, D_loss_real = 0.624, D_loss_fake = 0.654, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 43/46, D_loss_real = 0.666, D_loss_fake = 0.621, G_loss =
0.868
2/2 [==============================] - 0s 4ms/step
Epoch 488, Iteration 44/46, D_loss_real = 0.658, D_loss_fake = 0.605, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 45/46, D_loss_real = 0.648, D_loss_fake = 0.624, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 488, Iteration 46/46, D_loss_real = 0.650, D_loss_fake = 0.624, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 1/46, D_loss_real = 0.643, D_loss_fake = 0.622, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 2/46, D_loss_real = 0.652, D_loss_fake = 0.651, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 3/46, D_loss_real = 0.610, D_loss_fake = 0.646, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 4/46, D_loss_real = 0.676, D_loss_fake = 0.577, G_loss =
0.825
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 5/46, D_loss_real = 0.606, D_loss_fake = 0.625, G_loss =
```

```
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 6/46, D_loss_real = 0.617, D_loss_fake = 0.625, G_loss =
0.869
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 7/46, D_loss_real = 0.641, D_loss_fake = 0.635, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 8/46, D_loss_real = 0.605, D_loss_fake = 0.670, G_loss =
0.816
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 9/46, D_loss_real = 0.600, D_loss_fake = 0.636, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 10/46, D_loss_real = 0.566, D_loss_fake = 0.641, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 11/46, D_loss_real = 0.658, D_loss_fake = 0.655, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 12/46, D_loss_real = 0.664, D_loss_fake = 0.698, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 13/46, D_loss_real = 0.606, D_loss_fake = 0.671, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.609, G_loss =
0.883
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 15/46, D_loss_real = 0.669, D_loss_fake = 0.619, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 16/46, D_loss_real = 0.680, D_loss_fake = 0.570, G_loss =
0.822
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 17/46, D_loss_real = 0.652, D_loss_fake = 0.649, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 18/46, D_loss_real = 0.673, D_loss_fake = 0.590, G_loss =
0.867
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 19/46, D_loss_real = 0.589, D_loss_fake = 0.646, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 20/46, D_loss_real = 0.629, D_loss_fake = 0.641, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 21/46, D_loss_real = 0.657, D_loss_fake = 0.615, G_loss =
```

```
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 22/46, D_loss_real = 0.683, D_loss_fake = 0.632, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 23/46, D_loss_real = 0.622, D_loss_fake = 0.673, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 24/46, D_loss_real = 0.677, D_loss_fake = 0.571, G_loss =
0.848
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 25/46, D_loss_real = 0.615, D_loss_fake = 0.583, G_loss =
0.862
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 26/46, D_loss_real = 0.603, D_loss_fake = 0.618, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 27/46, D_loss_real = 0.614, D_loss_fake = 0.616, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 28/46, D_loss_real = 0.622, D_loss_fake = 0.621, G_loss =
0.802
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 29/46, D_loss_real = 0.597, D_loss_fake = 0.641, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 30/46, D_loss_real = 0.600, D_loss_fake = 0.650, G_loss =
0.816
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 31/46, D_loss_real = 0.629, D_loss_fake = 0.652, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 32/46, D_loss_real = 0.655, D_loss_fake = 0.655, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 33/46, D_loss_real = 0.677, D_loss_fake = 0.590, G_loss =
0.832
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 34/46, D_loss_real = 0.608, D_loss_fake = 0.596, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 35/46, D_loss_real = 0.598, D_loss_fake = 0.609, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 36/46, D_loss_real = 0.620, D_loss_fake = 0.596, G_loss =
0.838
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 37/46, D_loss_real = 0.677, D_loss_fake = 0.627, G_loss =
```

```
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 38/46, D_loss_real = 0.645, D_loss_fake = 0.612, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 39/46, D_loss_real = 0.650, D_loss_fake = 0.598, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 40/46, D_loss_real = 0.572, D_loss_fake = 0.666, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 41/46, D_loss_real = 0.612, D_loss_fake = 0.628, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 42/46, D_loss_real = 0.549, D_loss_fake = 0.624, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 43/46, D_loss_real = 0.659, D_loss_fake = 0.750, G_loss =
0.793
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 44/46, D_loss_real = 0.660, D_loss_fake = 0.606, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 489, Iteration 45/46, D_loss_real = 0.675, D_loss_fake = 0.714, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 489, Iteration 46/46, D_loss_real = 0.653, D_loss_fake = 0.644, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 1/46, D_loss_real = 0.602, D_loss_fake = 0.636, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 2/46, D_loss_real = 0.652, D_loss_fake = 0.571, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 3/46, D_loss_real = 0.722, D_loss_fake = 0.648, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 4/46, D_loss_real = 0.602, D_loss_fake = 0.653, G_loss =
0.895
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 5/46, D_loss_real = 0.650, D_loss_fake = 0.604, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 6/46, D_loss_real = 0.654, D_loss_fake = 0.613, G_loss =
0.893
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.586, G_loss =
```

```
0.895
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 8/46, D_loss_real = 0.702, D_loss_fake = 0.653, G_loss =
0.880
2/2 [==============================] - 0s 6ms/step
Epoch 490, Iteration 9/46, D_loss_real = 0.710, D_loss_fake = 0.575, G_loss =
0.856
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 10/46, D_loss_real = 0.671, D_loss_fake = 0.613, G_loss =
0.798
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 11/46, D_loss_real = 0.635, D_loss_fake = 0.631, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 12/46, D_loss_real = 0.632, D_loss_fake = 0.615, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 13/46, D_loss_real = 0.607, D_loss_fake = 0.690, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 14/46, D_loss_real = 0.720, D_loss_fake = 0.650, G_loss =
0.879
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 15/46, D_loss_real = 0.674, D_loss_fake = 0.607, G_loss =
0.907
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 16/46, D_loss_real = 0.608, D_loss_fake = 0.631, G_loss =
0.908
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 17/46, D_loss_real = 0.623, D_loss_fake = 0.607, G_loss =
0.882
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 18/46, D_loss_real = 0.635, D_loss_fake = 0.545, G_loss =
0.904
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 19/46, D_loss_real = 0.598, D_loss_fake = 0.636, G_loss =
0.874
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 20/46, D_loss_real = 0.622, D_loss_fake = 0.579, G_loss =
0.866
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 21/46, D_loss_real = 0.639, D_loss_fake = 0.670, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 22/46, D_loss_real = 0.601, D_loss_fake = 0.604, G_loss =
0.847
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 23/46, D_loss_real = 0.652, D_loss_fake = 0.673, G_loss =
```

```
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 24/46, D_loss_real = 0.675, D_loss_fake = 0.613, G_loss =
0.888
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 25/46, D_loss_real = 0.622, D_loss_fake = 0.601, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 26/46, D_loss_real = 0.610, D_loss_fake = 0.652, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 27/46, D_loss_real = 0.622, D_loss_fake = 0.702, G_loss =
0.894
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 28/46, D_loss_real = 0.671, D_loss_fake = 0.569, G_loss =
0.888
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 29/46, D_loss_real = 0.616, D_loss_fake = 0.561, G_loss =
0.880
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 30/46, D_loss_real = 0.659, D_loss_fake = 0.637, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 31/46, D_loss_real = 0.679, D_loss_fake = 0.589, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 32/46, D_loss_real = 0.628, D_loss_fake = 0.663, G_loss =
0.842
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 33/46, D_loss_real = 0.643, D_loss_fake = 0.625, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 34/46, D_loss_real = 0.608, D_loss_fake = 0.608, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 35/46, D_loss_real = 0.643, D_loss_fake = 0.673, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 36/46, D_loss_real = 0.634, D_loss_fake = 0.619, G_loss =
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 37/46, D_loss_real = 0.651, D_loss_fake = 0.614, G_loss =
0.778
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 38/46, D_loss_real = 0.612, D_loss_fake = 0.618, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 39/46, D_loss_real = 0.629, D_loss_fake = 0.644, G_loss =
```

```
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 40/46, D_loss_real = 0.606, D_loss_fake = 0.660, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 490, Iteration 41/46, D_loss_real = 0.631, D_loss_fake = 0.641, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 42/46, D_loss_real = 0.669, D_loss_fake = 0.592, G_loss =
0.882
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.612, G_loss =
0.859
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 44/46, D_loss_real = 0.662, D_loss_fake = 0.624, G_loss =
0.882
2/2 [==============================] - 0s 5ms/step
Epoch 490, Iteration 45/46, D_loss_real = 0.658, D_loss_fake = 0.592, G_loss =
0.887
2/2 [==============================] - 0s 4ms/step
Epoch 490, Iteration 46/46, D_loss_real = 0.677, D_loss_fake = 0.700, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.567, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 2/46, D_loss_real = 0.680, D_loss_fake = 0.592, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 3/46, D_loss_real = 0.650, D_loss_fake = 0.600, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 4/46, D_loss_real = 0.637, D_loss_fake = 0.627, G_loss =
0.820
2/2 [==============================] - 0s 4ms/step
Epoch 491, Iteration 5/46, D_loss_real = 0.619, D_loss_fake = 0.647, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 6/46, D_loss_real = 0.706, D_loss_fake = 0.588, G_loss =
0.801
2/2 [==============================] - 0s 4ms/step
Epoch 491, Iteration 7/46, D_loss_real = 0.640, D_loss_fake = 0.575, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 491, Iteration 8/46, D_loss_real = 0.607, D_loss_fake = 0.687, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 491, Iteration 9/46, D_loss_real = 0.635, D_loss_fake = 0.631, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 10/46, D_loss_real = 0.635, D_loss_fake = 0.625, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 11/46, D_loss_real = 0.662, D_loss_fake = 0.599, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 12/46, D_loss_real = 0.657, D_loss_fake = 0.589, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 13/46, D_loss_real = 0.686, D_loss_fake = 0.614, G_loss =
0.842
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 14/46, D_loss_real = 0.601, D_loss_fake = 0.652, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 15/46, D_loss_real = 0.662, D_loss_fake = 0.615, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 491, Iteration 16/46, D_loss_real = 0.635, D_loss_fake = 0.663, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 17/46, D_loss_real = 0.634, D_loss_fake = 0.635, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 18/46, D_loss_real = 0.654, D_loss_fake = 0.654, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 19/46, D_loss_real = 0.606, D_loss_fake = 0.670, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 20/46, D_loss_real = 0.622, D_loss_fake = 0.704, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 21/46, D_loss_real = 0.611, D_loss_fake = 0.616, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 22/46, D_loss_real = 0.726, D_loss_fake = 0.618, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 23/46, D_loss_real = 0.629, D_loss_fake = 0.598, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 491, Iteration 24/46, D_loss_real = 0.658, D_loss_fake = 0.612, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
```

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Epoch 491, Iteration 25/46, D_loss_real = 0.622, D_loss_fake = 0.628, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 26/46, D_loss_real = 0.678, D_loss_fake = 0.616, G_loss =
0.892
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 27/46, D_loss_real = 0.646, D_loss_fake = 0.571, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 28/46, D_loss_real = 0.640, D_loss_fake = 0.615, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 29/46, D_loss_real = 0.646, D_loss_fake = 0.575, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 491, Iteration 30/46, D_loss_real = 0.651, D_loss_fake = 0.644, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 31/46, D_loss_real = 0.518, D_loss_fake = 0.662, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 32/46, D_loss_real = 0.600, D_loss_fake = 0.653, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 33/46, D_loss_real = 0.683, D_loss_fake = 0.591, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 34/46, D_loss_real = 0.666, D_loss_fake = 0.606, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 35/46, D_loss_real = 0.580, D_loss_fake = 0.663, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 36/46, D_loss_real = 0.602, D_loss_fake = 0.607, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 491, Iteration 37/46, D_loss_real = 0.627, D_loss_fake = 0.608, G_loss =
0.882
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 38/46, D_loss_real = 0.654, D_loss_fake = 0.590, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 39/46, D_loss_real = 0.631, D_loss_fake = 0.601, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 40/46, D_loss_real = 0.543, D_loss_fake = 0.659, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
```

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Epoch 491, Iteration 41/46, D_loss_real = 0.646, D_loss_fake = 0.647, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 42/46, D_loss_real = 0.620, D_loss_fake = 0.641, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 43/46, D_loss_real = 0.600, D_loss_fake = 0.676, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 44/46, D_loss_real = 0.646, D_loss_fake = 0.672, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 45/46, D_loss_real = 0.616, D_loss_fake = 0.613, G_loss =
0.840
2/2 [==============================] - 0s 3ms/step
Epoch 491, Iteration 46/46, D_loss_real = 0.639, D_loss_fake = 0.638, G_loss =
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 1/46, D_loss_real = 0.623, D_loss_fake = 0.579, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 2/46, D_loss_real = 0.575, D_loss_fake = 0.704, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 3/46, D_loss_real = 0.589, D_loss_fake = 0.694, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 4/46, D_loss_real = 0.674, D_loss_fake = 0.688, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 5/46, D_loss_real = 0.677, D_loss_fake = 0.580, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 6/46, D_loss_real = 0.688, D_loss_fake = 0.575, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 7/46, D_loss_real = 0.663, D_loss_fake = 0.627, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 8/46, D_loss_real = 0.634, D_loss_fake = 0.624, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 9/46, D_loss_real = 0.612, D_loss_fake = 0.615, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 10/46, D_loss_real = 0.660, D_loss_fake = 0.657, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 492, Iteration 11/46, D_loss_real = 0.607, D_loss_fake = 0.608, G_loss =
0.876
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 12/46, D_loss_real = 0.634, D_loss_fake = 0.602, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 13/46, D_loss_real = 0.623, D_loss_fake = 0.597, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 14/46, D_loss_real = 0.637, D_loss_fake = 0.643, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 15/46, D_loss_real = 0.626, D_loss_fake = 0.633, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 16/46, D_loss_real = 0.646, D_loss_fake = 0.605, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 17/46, D_loss_real = 0.636, D_loss_fake = 0.632, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 18/46, D_loss_real = 0.590, D_loss_fake = 0.607, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 19/46, D_loss_real = 0.651, D_loss_fake = 0.620, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 20/46, D_loss_real = 0.679, D_loss_fake = 0.604, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 21/46, D_loss_real = 0.604, D_loss_fake = 0.650, G_loss =
0.873
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 22/46, D_loss_real = 0.651, D_loss_fake = 0.591, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 23/46, D_loss_real = 0.642, D_loss_fake = 0.622, G_loss =
0.872
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 24/46, D_loss_real = 0.657, D_loss_fake = 0.606, G_loss =
0.883
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 25/46, D_loss_real = 0.657, D_loss_fake = 0.675, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 26/46, D_loss_real = 0.632, D_loss_fake = 0.624, G_loss =
0.919
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 492, Iteration 27/46, D_loss_real = 0.670, D_loss_fake = 0.579, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 28/46, D_loss_real = 0.631, D_loss_fake = 0.596, G_loss =
0.895
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 29/46, D_loss_real = 0.613, D_loss_fake = 0.605, G_loss =
0.850
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 30/46, D_loss_real = 0.639, D_loss_fake = 0.579, G_loss =
0.873
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 31/46, D_loss_real = 0.647, D_loss_fake = 0.656, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 32/46, D_loss_real = 0.565, D_loss_fake = 0.636, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 33/46, D_loss_real = 0.621, D_loss_fake = 0.673, G_loss =
0.883
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 34/46, D_loss_real = 0.623, D_loss_fake = 0.597, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 35/46, D_loss_real = 0.632, D_loss_fake = 0.608, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 36/46, D_loss_real = 0.612, D_loss_fake = 0.583, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 37/46, D_loss_real = 0.629, D_loss_fake = 0.597, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 38/46, D_loss_real = 0.610, D_loss_fake = 0.832, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 39/46, D_loss_real = 0.600, D_loss_fake = 0.616, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 40/46, D_loss_real = 0.701, D_loss_fake = 0.692, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 492, Iteration 41/46, D_loss_real = 0.678, D_loss_fake = 0.638, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 42/46, D_loss_real = 0.645, D_loss_fake = 0.632, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 492, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.626, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 44/46, D_loss_real = 0.645, D_loss_fake = 0.615, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 45/46, D_loss_real = 0.638, D_loss_fake = 0.636, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 492, Iteration 46/46, D_loss_real = 0.610, D_loss_fake = 0.636, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.583, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 2/46, D_loss_real = 0.573, D_loss_fake = 0.690, G_loss =
0.901
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 3/46, D_loss_real = 0.634, D_loss_fake = 0.592, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.636, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.625, G_loss =
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 6/46, D_loss_real = 0.708, D_loss_fake = 0.622, G_loss =
0.890
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 7/46, D_loss_real = 0.655, D_loss_fake = 0.573, G_loss =
0.903
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 8/46, D_loss_real = 0.602, D_loss_fake = 0.627, G_loss =
0.891
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 9/46, D_loss_real = 0.656, D_loss_fake = 0.605, G_loss =
0.883
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 10/46, D_loss_real = 0.666, D_loss_fake = 0.585, G_loss =
0.843
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 11/46, D_loss_real = 0.653, D_loss_fake = 0.613, G_loss =
0.858
2/2 [==============================] - 0s 4ms/step
Epoch 493, Iteration 12/46, D_loss_real = 0.653, D_loss_fake = 0.589, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 493, Iteration 13/46, D_loss_real = 0.618, D_loss_fake = 0.596, G_loss =
0.835
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.681, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 15/46, D_loss_real = 0.578, D_loss_fake = 0.643, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 16/46, D_loss_real = 0.588, D_loss_fake = 0.672, G_loss =
0.844
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 17/46, D_loss_real = 0.604, D_loss_fake = 0.661, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 18/46, D_loss_real = 0.538, D_loss_fake = 0.624, G_loss =
0.779
2/2 [==============================] - 0s 4ms/step
Epoch 493, Iteration 19/46, D_loss_real = 0.659, D_loss_fake = 0.653, G_loss =
0.872
2/2 [==============================] - 0s 4ms/step
Epoch 493, Iteration 20/46, D_loss_real = 0.585, D_loss_fake = 0.658, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 493, Iteration 21/46, D_loss_real = 0.630, D_loss_fake = 0.681, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 22/46, D_loss_real = 0.562, D_loss_fake = 0.676, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 23/46, D_loss_real = 0.645, D_loss_fake = 0.600, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 24/46, D_loss_real = 0.638, D_loss_fake = 0.611, G_loss =
0.884
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 25/46, D_loss_real = 0.661, D_loss_fake = 0.629, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 26/46, D_loss_real = 0.709, D_loss_fake = 0.605, G_loss =
0.892
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 27/46, D_loss_real = 0.683, D_loss_fake = 0.648, G_loss =
0.914
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 28/46, D_loss_real = 0.738, D_loss_fake = 0.600, G_loss =
0.908
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 493, Iteration 29/46, D_loss_real = 0.643, D_loss_fake = 0.624, G_loss =
0.911
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.682, G_loss =
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 31/46, D_loss_real = 0.719, D_loss_fake = 0.611, G_loss =
0.894
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.558, G_loss =
0.888
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 33/46, D_loss_real = 0.712, D_loss_fake = 0.571, G_loss =
0.924
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 34/46, D_loss_real = 0.651, D_loss_fake = 0.569, G_loss =
0.896
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 35/46, D_loss_real = 0.679, D_loss_fake = 0.569, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 36/46, D_loss_real = 0.693, D_loss_fake = 0.577, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 37/46, D_loss_real = 0.677, D_loss_fake = 0.622, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 38/46, D_loss_real = 0.588, D_loss_fake = 0.637, G_loss =
0.833
2/2 [==============================] - 0s 4ms/step
Epoch 493, Iteration 39/46, D_loss_real = 0.665, D_loss_fake = 0.619, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 40/46, D_loss_real = 0.586, D_loss_fake = 0.673, G_loss =
0.797
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 41/46, D_loss_real = 0.620, D_loss_fake = 0.632, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 42/46, D_loss_real = 0.644, D_loss_fake = 0.660, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 493, Iteration 43/46, D_loss_real = 0.603, D_loss_fake = 0.670, G_loss =
0.783
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 44/46, D_loss_real = 0.657, D_loss_fake = 0.587, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 493, Iteration 45/46, D_loss_real = 0.571, D_loss_fake = 0.647, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 493, Iteration 46/46, D_loss_real = 0.619, D_loss_fake = 0.633, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 1/46, D_loss_real = 0.591, D_loss_fake = 0.672, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 2/46, D_loss_real = 0.615, D_loss_fake = 0.683, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 3/46, D_loss_real = 0.606, D_loss_fake = 0.634, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 4/46, D_loss_real = 0.650, D_loss_fake = 0.610, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 5/46, D_loss_real = 0.621, D_loss_fake = 0.643, G_loss =
0.885
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.611, G_loss =
0.882
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 7/46, D_loss_real = 0.667, D_loss_fake = 0.611, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 8/46, D_loss_real = 0.639, D_loss_fake = 0.644, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 9/46, D_loss_real = 0.682, D_loss_fake = 0.631, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 10/46, D_loss_real = 0.639, D_loss_fake = 0.582, G_loss =
0.872
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 11/46, D_loss_real = 0.596, D_loss_fake = 0.618, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 12/46, D_loss_real = 0.603, D_loss_fake = 0.567, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 13/46, D_loss_real = 0.612, D_loss_fake = 0.676, G_loss =
0.819
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 14/46, D_loss_real = 0.598, D_loss_fake = 0.613, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 494, Iteration 15/46, D_loss_real = 0.626, D_loss_fake = 0.631, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 16/46, D_loss_real = 0.664, D_loss_fake = 0.602, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 17/46, D_loss_real = 0.580, D_loss_fake = 0.658, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 18/46, D_loss_real = 0.634, D_loss_fake = 0.625, G_loss =
0.830
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 19/46, D_loss_real = 0.634, D_loss_fake = 0.604, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 20/46, D_loss_real = 0.598, D_loss_fake = 0.579, G_loss =
0.817
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 21/46, D_loss_real = 0.637, D_loss_fake = 0.657, G_loss =
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 22/46, D_loss_real = 0.670, D_loss_fake = 0.584, G_loss =
0.886
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 23/46, D_loss_real = 0.628, D_loss_fake = 0.552, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 24/46, D_loss_real = 0.654, D_loss_fake = 0.638, G_loss =
0.850
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 25/46, D_loss_real = 0.663, D_loss_fake = 0.657, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 26/46, D_loss_real = 0.693, D_loss_fake = 0.613, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 27/46, D_loss_real = 0.669, D_loss_fake = 0.619, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 28/46, D_loss_real = 0.740, D_loss_fake = 0.626, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 29/46, D_loss_real = 0.651, D_loss_fake = 0.617, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 30/46, D_loss_real = 0.604, D_loss_fake = 0.637, G_loss =
0.874
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 494, Iteration 31/46, D_loss_real = 0.639, D_loss_fake = 0.614, G_loss =
0.873
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 32/46, D_loss_real = 0.664, D_loss_fake = 0.609, G_loss =
0.870
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 33/46, D_loss_real = 0.642, D_loss_fake = 0.566, G_loss =
0.874
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 34/46, D_loss_real = 0.666, D_loss_fake = 0.626, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 35/46, D_loss_real = 0.648, D_loss_fake = 0.601, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 36/46, D_loss_real = 0.675, D_loss_fake = 0.614, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.620, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 38/46, D_loss_real = 0.605, D_loss_fake = 0.605, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 39/46, D_loss_real = 0.601, D_loss_fake = 0.626, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 40/46, D_loss_real = 0.569, D_loss_fake = 0.702, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 41/46, D_loss_real = 0.625, D_loss_fake = 0.617, G_loss =
0.868
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 42/46, D_loss_real = 0.604, D_loss_fake = 0.610, G_loss =
0.874
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 43/46, D_loss_real = 0.623, D_loss_fake = 0.572, G_loss =
0.906
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 44/46, D_loss_real = 0.689, D_loss_fake = 0.608, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 494, Iteration 45/46, D_loss_real = 0.641, D_loss_fake = 0.660, G_loss =
0.860
2/2 [==============================] - 0s 3ms/step
Epoch 494, Iteration 46/46, D_loss_real = 0.667, D_loss_fake = 0.654, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 495, Iteration 1/46, D_loss_real = 0.625, D_loss_fake = 0.614, G_loss =
0.820
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 2/46, D_loss_real = 0.635, D_loss_fake = 0.620, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.645, G_loss =
0.837
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 4/46, D_loss_real = 0.659, D_loss_fake = 0.609, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 5/46, D_loss_real = 0.627, D_loss_fake = 0.600, G_loss =
0.871
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 6/46, D_loss_real = 0.656, D_loss_fake = 0.622, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 7/46, D_loss_real = 0.650, D_loss_fake = 0.609, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 8/46, D_loss_real = 0.523, D_loss_fake = 0.647, G_loss =
0.798
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 9/46, D_loss_real = 0.601, D_loss_fake = 0.622, G_loss =
0.785
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 10/46, D_loss_real = 0.629, D_loss_fake = 0.620, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 11/46, D_loss_real = 0.670, D_loss_fake = 0.658, G_loss =
0.822
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 12/46, D_loss_real = 0.642, D_loss_fake = 0.598, G_loss =
0.791
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 13/46, D_loss_real = 0.660, D_loss_fake = 0.636, G_loss =
0.830
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 14/46, D_loss_real = 0.613, D_loss_fake = 0.607, G_loss =
0.802
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 15/46, D_loss_real = 0.674, D_loss_fake = 0.667, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 16/46, D_loss_real = 0.588, D_loss_fake = 0.611, G_loss =
0.809
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 495, Iteration 17/46, D_loss_real = 0.616, D_loss_fake = 0.696, G_loss =
0.851
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 18/46, D_loss_real = 0.635, D_loss_fake = 0.653, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 19/46, D_loss_real = 0.628, D_loss_fake = 0.613, G_loss =
0.883
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 20/46, D_loss_real = 0.672, D_loss_fake = 0.598, G_loss =
0.896
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.616, G_loss =
0.812
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 22/46, D_loss_real = 0.661, D_loss_fake = 0.619, G_loss =
0.853
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 23/46, D_loss_real = 0.689, D_loss_fake = 0.617, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 24/46, D_loss_real = 0.577, D_loss_fake = 0.669, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 25/46, D_loss_real = 0.619, D_loss_fake = 0.612, G_loss =
0.783
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 26/46, D_loss_real = 0.647, D_loss_fake = 0.713, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 27/46, D_loss_real = 0.638, D_loss_fake = 0.585, G_loss =
0.821
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 28/46, D_loss_real = 0.640, D_loss_fake = 0.657, G_loss =
0.829
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 29/46, D_loss_real = 0.700, D_loss_fake = 0.661, G_loss =
0.814
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 30/46, D_loss_real = 0.655, D_loss_fake = 0.654, G_loss =
0.847
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 31/46, D_loss_real = 0.632, D_loss_fake = 0.623, G_loss =
0.856
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 32/46, D_loss_real = 0.622, D_loss_fake = 0.639, G_loss =
0.792
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 495, Iteration 33/46, D_loss_real = 0.614, D_loss_fake = 0.643, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 34/46, D_loss_real = 0.612, D_loss_fake = 0.730, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 35/46, D_loss_real = 0.670, D_loss_fake = 0.634, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 36/46, D_loss_real = 0.609, D_loss_fake = 0.637, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 37/46, D_loss_real = 0.582, D_loss_fake = 0.638, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 38/46, D_loss_real = 0.626, D_loss_fake = 0.614, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 39/46, D_loss_real = 0.654, D_loss_fake = 0.581, G_loss =
0.863
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 40/46, D_loss_real = 0.601, D_loss_fake = 0.581, G_loss =
0.857
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 41/46, D_loss_real = 0.618, D_loss_fake = 0.718, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 42/46, D_loss_real = 0.663, D_loss_fake = 0.604, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 43/46, D_loss_real = 0.598, D_loss_fake = 0.647, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 44/46, D_loss_real = 0.617, D_loss_fake = 0.644, G_loss =
0.859
2/2 [==============================] - 0s 4ms/step
Epoch 495, Iteration 45/46, D_loss_real = 0.640, D_loss_fake = 0.591, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 495, Iteration 46/46, D_loss_real = 0.593, D_loss_fake = 0.637, G_loss =
0.804
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 1/46, D_loss_real = 0.599, D_loss_fake = 0.688, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 2/46, D_loss_real = 0.692, D_loss_fake = 0.590, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 496, Iteration 3/46, D_loss_real = 0.680, D_loss_fake = 0.583, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 4/46, D_loss_real = 0.668, D_loss_fake = 0.625, G_loss =
0.899
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 5/46, D_loss_real = 0.703, D_loss_fake = 0.627, G_loss =
0.881
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 6/46, D_loss_real = 0.687, D_loss_fake = 0.647, G_loss =
0.886
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.625, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 8/46, D_loss_real = 0.677, D_loss_fake = 0.585, G_loss =
0.852
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 9/46, D_loss_real = 0.715, D_loss_fake = 0.609, G_loss =
0.806
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 10/46, D_loss_real = 0.653, D_loss_fake = 0.649, G_loss =
0.872
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 11/46, D_loss_real = 0.629, D_loss_fake = 0.636, G_loss =
0.807
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 12/46, D_loss_real = 0.635, D_loss_fake = 0.584, G_loss =
0.841
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 13/46, D_loss_real = 0.587, D_loss_fake = 0.649, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 14/46, D_loss_real = 0.590, D_loss_fake = 0.610, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 15/46, D_loss_real = 0.637, D_loss_fake = 0.603, G_loss =
0.870
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 16/46, D_loss_real = 0.651, D_loss_fake = 0.618, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 17/46, D_loss_real = 0.623, D_loss_fake = 0.578, G_loss =
0.871
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 18/46, D_loss_real = 0.619, D_loss_fake = 0.653, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 496, Iteration 19/46, D_loss_real = 0.643, D_loss_fake = 0.693, G_loss =
0.881
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 20/46, D_loss_real = 0.623, D_loss_fake = 0.660, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 21/46, D_loss_real = 0.628, D_loss_fake = 0.562, G_loss =
0.868
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 22/46, D_loss_real = 0.626, D_loss_fake = 0.624, G_loss =
0.840
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 23/46, D_loss_real = 0.636, D_loss_fake = 0.610, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 24/46, D_loss_real = 0.649, D_loss_fake = 0.653, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 25/46, D_loss_real = 0.650, D_loss_fake = 0.639, G_loss =
0.837
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 26/46, D_loss_real = 0.643, D_loss_fake = 0.624, G_loss =
0.885
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 27/46, D_loss_real = 0.555, D_loss_fake = 0.605, G_loss =
0.839
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 28/46, D_loss_real = 0.660, D_loss_fake = 0.636, G_loss =
0.868
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 29/46, D_loss_real = 0.607, D_loss_fake = 0.649, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 30/46, D_loss_real = 0.648, D_loss_fake = 0.613, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 31/46, D_loss_real = 0.589, D_loss_fake = 0.635, G_loss =
0.861
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 32/46, D_loss_real = 0.629, D_loss_fake = 0.564, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 33/46, D_loss_real = 0.646, D_loss_fake = 0.632, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 34/46, D_loss_real = 0.639, D_loss_fake = 0.578, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 496, Iteration 35/46, D_loss_real = 0.693, D_loss_fake = 0.590, G_loss =
0.898
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 36/46, D_loss_real = 0.637, D_loss_fake = 0.671, G_loss =
0.888
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 37/46, D_loss_real = 0.656, D_loss_fake = 0.590, G_loss =
0.897
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 38/46, D_loss_real = 0.623, D_loss_fake = 0.588, G_loss =
0.914
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 39/46, D_loss_real = 0.644, D_loss_fake = 0.625, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 40/46, D_loss_real = 0.626, D_loss_fake = 0.608, G_loss =
0.898
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 41/46, D_loss_real = 0.587, D_loss_fake = 0.606, G_loss =
0.867
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 42/46, D_loss_real = 0.639, D_loss_fake = 0.708, G_loss =
0.899
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 43/46, D_loss_real = 0.668, D_loss_fake = 0.587, G_loss =
0.861
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 44/46, D_loss_real = 0.709, D_loss_fake = 0.617, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 496, Iteration 45/46, D_loss_real = 0.658, D_loss_fake = 0.621, G_loss =
0.877
2/2 [==============================] - 0s 4ms/step
Epoch 496, Iteration 46/46, D_loss_real = 0.588, D_loss_fake = 0.606, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 1/46, D_loss_real = 0.551, D_loss_fake = 0.627, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 2/46, D_loss_real = 0.580, D_loss_fake = 0.639, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 3/46, D_loss_real = 0.710, D_loss_fake = 0.620, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 4/46, D_loss_real = 0.557, D_loss_fake = 0.707, G_loss =
0.819
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 497, Iteration 5/46, D_loss_real = 0.634, D_loss_fake = 0.645, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.603, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 7/46, D_loss_real = 0.630, D_loss_fake = 0.634, G_loss =
0.904
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 8/46, D_loss_real = 0.623, D_loss_fake = 0.613, G_loss =
0.866
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 9/46, D_loss_real = 0.667, D_loss_fake = 0.642, G_loss =
0.864
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 10/46, D_loss_real = 0.622, D_loss_fake = 0.597, G_loss =
0.876
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 11/46, D_loss_real = 0.609, D_loss_fake = 0.669, G_loss =
0.858
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 12/46, D_loss_real = 0.658, D_loss_fake = 0.652, G_loss =
0.892
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 13/46, D_loss_real = 0.669, D_loss_fake = 0.551, G_loss =
0.895
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 14/46, D_loss_real = 0.703, D_loss_fake = 0.701, G_loss =
0.878
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 15/46, D_loss_real = 0.683, D_loss_fake = 0.641, G_loss =
0.872
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 16/46, D_loss_real = 0.704, D_loss_fake = 0.608, G_loss =
0.881
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 17/46, D_loss_real = 0.626, D_loss_fake = 0.585, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 18/46, D_loss_real = 0.676, D_loss_fake = 0.708, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 19/46, D_loss_real = 0.629, D_loss_fake = 0.583, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 20/46, D_loss_real = 0.641, D_loss_fake = 0.563, G_loss =
0.820
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 497, Iteration 21/46, D_loss_real = 0.675, D_loss_fake = 0.605, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 22/46, D_loss_real = 0.589, D_loss_fake = 0.619, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 23/46, D_loss_real = 0.614, D_loss_fake = 0.628, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 24/46, D_loss_real = 0.690, D_loss_fake = 0.671, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 25/46, D_loss_real = 0.691, D_loss_fake = 0.630, G_loss =
0.827
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.603, G_loss =
0.845
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 27/46, D_loss_real = 0.630, D_loss_fake = 0.665, G_loss =
0.910
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 28/46, D_loss_real = 0.646, D_loss_fake = 0.583, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 29/46, D_loss_real = 0.651, D_loss_fake = 0.596, G_loss =
0.877
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 30/46, D_loss_real = 0.592, D_loss_fake = 0.636, G_loss =
0.893
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 31/46, D_loss_real = 0.656, D_loss_fake = 0.590, G_loss =
0.825
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 32/46, D_loss_real = 0.693, D_loss_fake = 0.620, G_loss =
0.879
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.630, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 34/46, D_loss_real = 0.643, D_loss_fake = 0.656, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 35/46, D_loss_real = 0.714, D_loss_fake = 0.574, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 36/46, D_loss_real = 0.621, D_loss_fake = 0.574, G_loss =
0.869
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 497, Iteration 37/46, D_loss_real = 0.671, D_loss_fake = 0.686, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 38/46, D_loss_real = 0.628, D_loss_fake = 0.573, G_loss =
0.852
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 39/46, D_loss_real = 0.710, D_loss_fake = 0.623, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 40/46, D_loss_real = 0.704, D_loss_fake = 0.591, G_loss =
0.866
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 41/46, D_loss_real = 0.651, D_loss_fake = 0.634, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 42/46, D_loss_real = 0.613, D_loss_fake = 0.623, G_loss =
0.812
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 43/46, D_loss_real = 0.642, D_loss_fake = 0.709, G_loss =
0.792
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 44/46, D_loss_real = 0.652, D_loss_fake = 0.599, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 497, Iteration 45/46, D_loss_real = 0.695, D_loss_fake = 0.658, G_loss =
0.806
2/2 [==============================] - 0s 4ms/step
Epoch 497, Iteration 46/46, D_loss_real = 0.692, D_loss_fake = 0.571, G_loss =
0.838
2/2 [==============================] - 0s 5ms/step
Epoch 498, Iteration 1/46, D_loss_real = 0.659, D_loss_fake = 0.627, G_loss =
0.795
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 2/46, D_loss_real = 0.597, D_loss_fake = 0.680, G_loss =
0.813
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 3/46, D_loss_real = 0.607, D_loss_fake = 0.711, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 4/46, D_loss_real = 0.642, D_loss_fake = 0.616, G_loss =
0.839
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 5/46, D_loss_real = 0.731, D_loss_fake = 0.617, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 6/46, D_loss_real = 0.656, D_loss_fake = 0.681, G_loss =
0.815
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 498, Iteration 7/46, D_loss_real = 0.587, D_loss_fake = 0.662, G_loss =
0.789
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 8/46, D_loss_real = 0.662, D_loss_fake = 0.699, G_loss =
0.823
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 9/46, D_loss_real = 0.620, D_loss_fake = 0.681, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 10/46, D_loss_real = 0.624, D_loss_fake = 0.639, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 11/46, D_loss_real = 0.574, D_loss_fake = 0.590, G_loss =
0.872
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 12/46, D_loss_real = 0.619, D_loss_fake = 0.644, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 13/46, D_loss_real = 0.620, D_loss_fake = 0.612, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 14/46, D_loss_real = 0.671, D_loss_fake = 0.648, G_loss =
0.829
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 15/46, D_loss_real = 0.583, D_loss_fake = 0.639, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 16/46, D_loss_real = 0.668, D_loss_fake = 0.653, G_loss =
0.821
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 17/46, D_loss_real = 0.608, D_loss_fake = 0.632, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 18/46, D_loss_real = 0.584, D_loss_fake = 0.606, G_loss =
0.854
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 19/46, D_loss_real = 0.652, D_loss_fake = 0.646, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 20/46, D_loss_real = 0.556, D_loss_fake = 0.716, G_loss =
0.887
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 21/46, D_loss_real = 0.659, D_loss_fake = 0.664, G_loss =
0.899
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 22/46, D_loss_real = 0.738, D_loss_fake = 0.588, G_loss =
0.834
2/2 [==============================] - 0s 3ms/step
```

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Epoch 498, Iteration 23/46, D_loss_real = 0.658, D_loss_fake = 0.577, G_loss =
0.885
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 24/46, D_loss_real = 0.643, D_loss_fake = 0.581, G_loss =
0.856
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.624, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 26/46, D_loss_real = 0.621, D_loss_fake = 0.626, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 27/46, D_loss_real = 0.588, D_loss_fake = 0.670, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 28/46, D_loss_real = 0.670, D_loss_fake = 0.651, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 29/46, D_loss_real = 0.688, D_loss_fake = 0.631, G_loss =
0.823
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 30/46, D_loss_real = 0.603, D_loss_fake = 0.643, G_loss =
0.853
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.654, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 32/46, D_loss_real = 0.633, D_loss_fake = 0.642, G_loss =
0.771
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 33/46, D_loss_real = 0.648, D_loss_fake = 0.588, G_loss =
0.863
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 34/46, D_loss_real = 0.647, D_loss_fake = 0.632, G_loss =
0.795
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 35/46, D_loss_real = 0.607, D_loss_fake = 0.691, G_loss =
0.850
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.626, G_loss =
0.824
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 37/46, D_loss_real = 0.593, D_loss_fake = 0.631, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 38/46, D_loss_real = 0.652, D_loss_fake = 0.622, G_loss =
0.833
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 498, Iteration 39/46, D_loss_real = 0.607, D_loss_fake = 0.623, G_loss =
0.879
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.640, G_loss =
0.889
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 41/46, D_loss_real = 0.598, D_loss_fake = 0.561, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 42/46, D_loss_real = 0.652, D_loss_fake = 0.635, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 498, Iteration 43/46, D_loss_real = 0.591, D_loss_fake = 0.691, G_loss =
0.838
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 44/46, D_loss_real = 0.741, D_loss_fake = 0.668, G_loss =
0.849
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 45/46, D_loss_real = 0.671, D_loss_fake = 0.597, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 498, Iteration 46/46, D_loss_real = 0.708, D_loss_fake = 0.640, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 1/46, D_loss_real = 0.660, D_loss_fake = 0.587, G_loss =
0.862
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 2/46, D_loss_real = 0.679, D_loss_fake = 0.595, G_loss =
0.890
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 3/46, D_loss_real = 0.704, D_loss_fake = 0.661, G_loss =
0.911
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 4/46, D_loss_real = 0.698, D_loss_fake = 0.560, G_loss =
0.921
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 5/46, D_loss_real = 0.672, D_loss_fake = 0.590, G_loss =
0.927
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 6/46, D_loss_real = 0.642, D_loss_fake = 0.587, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 7/46, D_loss_real = 0.636, D_loss_fake = 0.573, G_loss =
0.847
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 8/46, D_loss_real = 0.622, D_loss_fake = 0.617, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 499, Iteration 9/46, D_loss_real = 0.634, D_loss_fake = 0.629, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 10/46, D_loss_real = 0.647, D_loss_fake = 0.631, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 11/46, D_loss_real = 0.626, D_loss_fake = 0.656, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 12/46, D_loss_real = 0.670, D_loss_fake = 0.652, G_loss =
0.857
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 13/46, D_loss_real = 0.666, D_loss_fake = 0.631, G_loss =
0.810
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 14/46, D_loss_real = 0.632, D_loss_fake = 0.611, G_loss =
0.861
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 15/46, D_loss_real = 0.656, D_loss_fake = 0.603, G_loss =
0.859
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 16/46, D_loss_real = 0.662, D_loss_fake = 0.591, G_loss =
0.855
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 17/46, D_loss_real = 0.640, D_loss_fake = 0.644, G_loss =
0.818
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 18/46, D_loss_real = 0.618, D_loss_fake = 0.622, G_loss =
0.821
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 19/46, D_loss_real = 0.637, D_loss_fake = 0.630, G_loss =
0.791
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 20/46, D_loss_real = 0.598, D_loss_fake = 0.659, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 21/46, D_loss_real = 0.628, D_loss_fake = 0.648, G_loss =
0.802
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 22/46, D_loss_real = 0.616, D_loss_fake = 0.626, G_loss =
0.793
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 23/46, D_loss_real = 0.657, D_loss_fake = 0.685, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 24/46, D_loss_real = 0.652, D_loss_fake = 0.626, G_loss =
0.836
2/2 [==============================] - 0s 3ms/step
```

```
Epoch 499, Iteration 25/46, D_loss_real = 0.643, D_loss_fake = 0.659, G_loss =
0.813
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 26/46, D_loss_real = 0.647, D_loss_fake = 0.587, G_loss =
0.811
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 27/46, D_loss_real = 0.547, D_loss_fake = 0.611, G_loss =
0.828
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 28/46, D_loss_real = 0.579, D_loss_fake = 0.622, G_loss =
0.855
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 29/46, D_loss_real = 0.608, D_loss_fake = 0.654, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 30/46, D_loss_real = 0.640, D_loss_fake = 0.644, G_loss =
0.863
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 31/46, D_loss_real = 0.660, D_loss_fake = 0.576, G_loss =
0.875
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 32/46, D_loss_real = 0.652, D_loss_fake = 0.610, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 33/46, D_loss_real = 0.654, D_loss_fake = 0.666, G_loss =
0.808
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 34/46, D_loss_real = 0.676, D_loss_fake = 0.621, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 35/46, D_loss_real = 0.663, D_loss_fake = 0.658, G_loss =
0.799
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 36/46, D_loss_real = 0.645, D_loss_fake = 0.626, G_loss =
0.804
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 37/46, D_loss_real = 0.679, D_loss_fake = 0.807, G_loss =
0.889
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 38/46, D_loss_real = 0.649, D_loss_fake = 0.579, G_loss =
0.906
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 39/46, D_loss_real = 0.671, D_loss_fake = 0.597, G_loss =
0.883
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 40/46, D_loss_real = 0.658, D_loss_fake = 0.572, G_loss =
0.880
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 499, Iteration 41/46, D_loss_real = 0.633, D_loss_fake = 0.596, G_loss =
0.824
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 42/46, D_loss_real = 0.628, D_loss_fake = 0.628, G_loss =
0.831
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 43/46, D_loss_real = 0.604, D_loss_fake = 0.694, G_loss =
0.781
2/2 [==============================] - 0s 3ms/step
Epoch 499, Iteration 44/46, D_loss_real = 0.646, D_loss_fake = 0.682, G_loss =
0.784
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 45/46, D_loss_real = 0.569, D_loss_fake = 0.723, G_loss =
0.815
2/2 [==============================] - 0s 4ms/step
Epoch 499, Iteration 46/46, D_loss_real = 0.639, D_loss_fake = 0.697, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 1/46, D_loss_real = 0.641, D_loss_fake = 0.602, G_loss =
0.845
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 2/46, D_loss_real = 0.626, D_loss_fake = 0.662, G_loss =
0.864
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 3/46, D_loss_real = 0.620, D_loss_fake = 0.573, G_loss =
0.921
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 4/46, D_loss_real = 0.670, D_loss_fake = 0.607, G_loss =
0.843
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 5/46, D_loss_real = 0.663, D_loss_fake = 0.605, G_loss =
0.846
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 6/46, D_loss_real = 0.664, D_loss_fake = 0.644, G_loss =
0.895
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 7/46, D_loss_real = 0.630, D_loss_fake = 0.684, G_loss =
0.876
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 8/46, D_loss_real = 0.625, D_loss_fake = 0.587, G_loss =
0.866
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 9/46, D_loss_real = 0.662, D_loss_fake = 0.594, G_loss =
0.923
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 10/46, D_loss_real = 0.706, D_loss_fake = 0.580, G_loss =
0.850
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 500, Iteration 11/46, D_loss_real = 0.639, D_loss_fake = 0.592, G_loss =
0.826
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 12/46, D_loss_real = 0.618, D_loss_fake = 0.619, G_loss =
0.832
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 13/46, D_loss_real = 0.640, D_loss_fake = 0.617, G_loss =
0.805
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 14/46, D_loss_real = 0.627, D_loss_fake = 0.644, G_loss =
0.851
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 15/46, D_loss_real = 0.624, D_loss_fake = 0.648, G_loss =
0.816
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 16/46, D_loss_real = 0.616, D_loss_fake = 0.622, G_loss =
0.828
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 17/46, D_loss_real = 0.581, D_loss_fake = 0.624, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 18/46, D_loss_real = 0.607, D_loss_fake = 0.623, G_loss =
0.827
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 19/46, D_loss_real = 0.571, D_loss_fake = 0.650, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 20/46, D_loss_real = 0.608, D_loss_fake = 0.687, G_loss =
0.808
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 21/46, D_loss_real = 0.615, D_loss_fake = 0.654, G_loss =
0.860
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 22/46, D_loss_real = 0.698, D_loss_fake = 0.628, G_loss =
0.831
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 23/46, D_loss_real = 0.727, D_loss_fake = 0.617, G_loss =
0.875
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 24/46, D_loss_real = 0.645, D_loss_fake = 0.603, G_loss =
0.854
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 25/46, D_loss_real = 0.644, D_loss_fake = 0.624, G_loss =
0.880
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 26/46, D_loss_real = 0.625, D_loss_fake = 0.662, G_loss =
0.849
2/2 [==============================] - 0s 4ms/step
```

```
Epoch 500, Iteration 27/46, D_loss_real = 0.655, D_loss_fake = 0.575, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 28/46, D_loss_real = 0.613, D_loss_fake = 0.595, G_loss =
0.889
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 29/46, D_loss_real = 0.654, D_loss_fake = 0.597, G_loss =
0.848
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 30/46, D_loss_real = 0.640, D_loss_fake = 0.616, G_loss =
0.836
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 31/46, D_loss_real = 0.607, D_loss_fake = 0.671, G_loss =
0.865
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.650, G_loss =
0.791
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 33/46, D_loss_real = 0.640, D_loss_fake = 0.636, G_loss =
0.811
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 34/46, D_loss_real = 0.624, D_loss_fake = 0.615, G_loss =
0.885
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 35/46, D_loss_real = 0.647, D_loss_fake = 0.604, G_loss =
0.861
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 36/46, D_loss_real = 0.613, D_loss_fake = 0.633, G_loss =
0.850
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 37/46, D_loss_real = 0.677, D_loss_fake = 0.657, G_loss =
0.826
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 38/46, D_loss_real = 0.589, D_loss_fake = 0.591, G_loss =
0.867
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.634, G_loss =
0.834
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.611, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 41/46, D_loss_real = 0.697, D_loss_fake = 0.605, G_loss =
0.846
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 42/46, D_loss_real = 0.643, D_loss_fake = 0.645, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
```

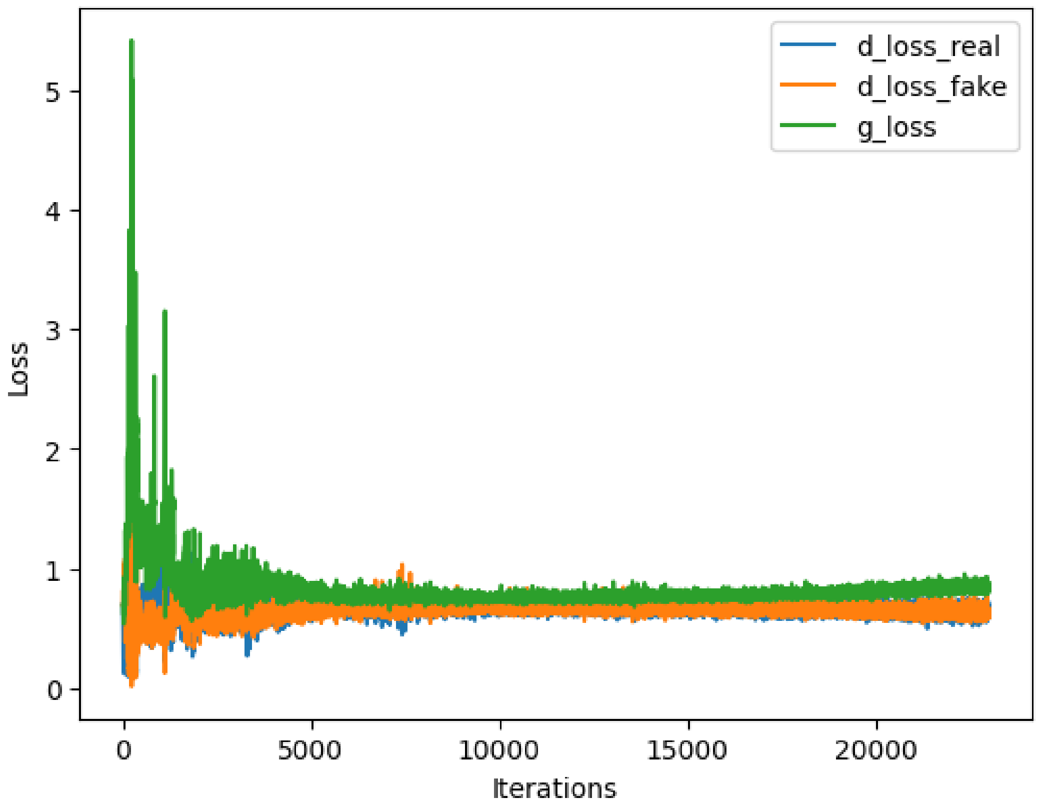
```
Epoch 500, Iteration 43/46, D_loss_real = 0.670, D_loss_fake = 0.644, G_loss =
0.835
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 44/46, D_loss_real = 0.637, D_loss_fake = 0.642, G_loss =
0.817
2/2 [==============================] - 0s 3ms/step
Epoch 500, Iteration 45/46, D_loss_real = 0.649, D_loss_fake = 0.645, G_loss =
0.841
2/2 [==============================] - 0s 4ms/step
Epoch 500, Iteration 46/46, D_loss_real = 0.640, D_loss_fake = 0.599, G_loss =
0.844
2/2 [==============================] - 0s 4ms/step
```

[13]:

```
### Plot the training loss of the Generator and the Discriminator over␣
↪iterations

# Initialize values of x-axis (iterations)
gx = np.linspace(0, len(g_losses), len(g_losses))
dx_real = np.linspace(0, len(d_losses_real), len(d_losses_real))
dx_fake = np.linspace(0, len(d_losses_fake), len(d_losses_fake))

# Plot the training loss of G and D
plt.plot(dx_real, d_losses_real)
plt.plot(dx_fake, d_losses_fake)
plt.plot(gx, g_losses)
plt.ylabel('Loss')
plt.xlabel('Iterations')
plt.legend(['d_loss_real', 'd_loss_fake', 'g_loss'], loc='upper right')
plt.show()
```



10 Save and load the models

[14]:
```
# Save the Generator, Discriminator and GAN model
g_model.save('g_model.sav')
d_model.save('d_model.sav')
gan_model.save('gan_model.sav')
```

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metrics` will be empty until you train or evaluate the model.

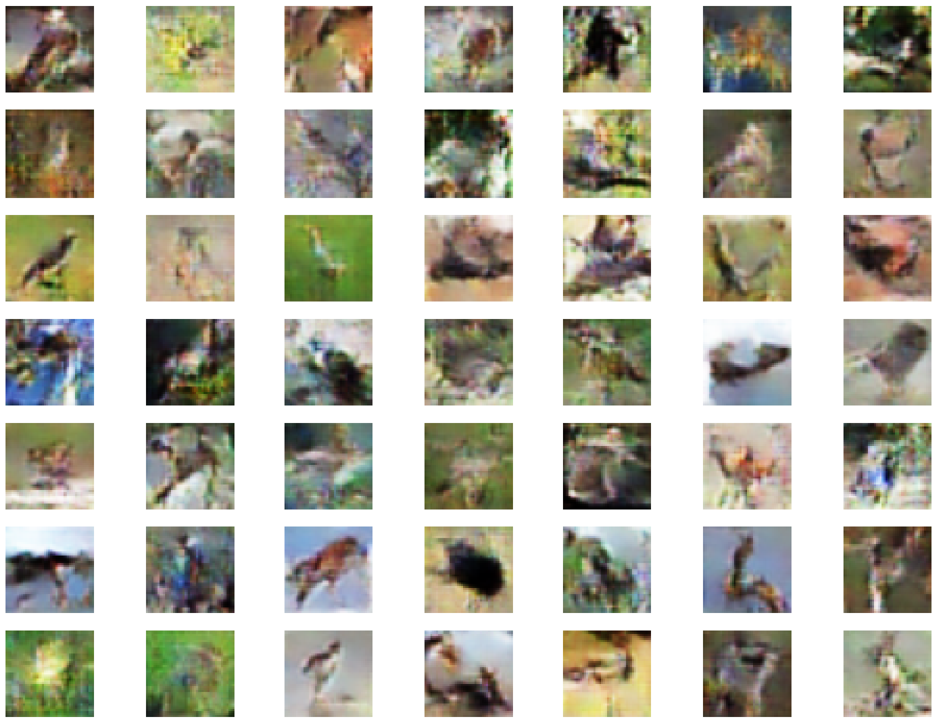
[15]:
```
# Load the Generator from saved model
g_model = keras.models.load_model("g_model.sav")

# Generate 49 fake images by passing 256-dimensional random vectors through the␣
 ↪Generator
X_fake, _ = generate_fake_samples(g_model, 256, 49)
```

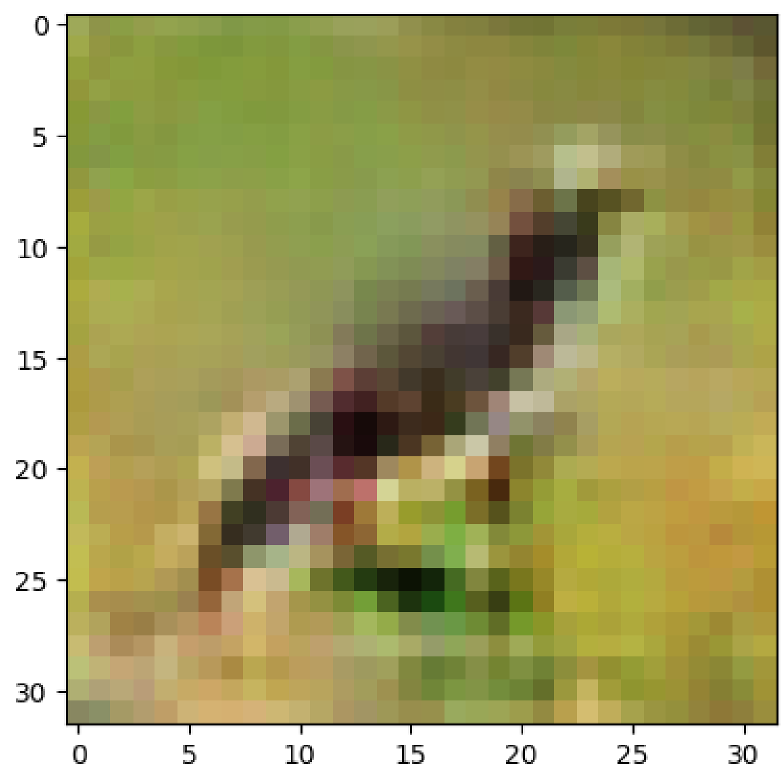
WARNING:tensorflow:No training configuration found in save file, so the model was *not* compiled. Compile it manually.

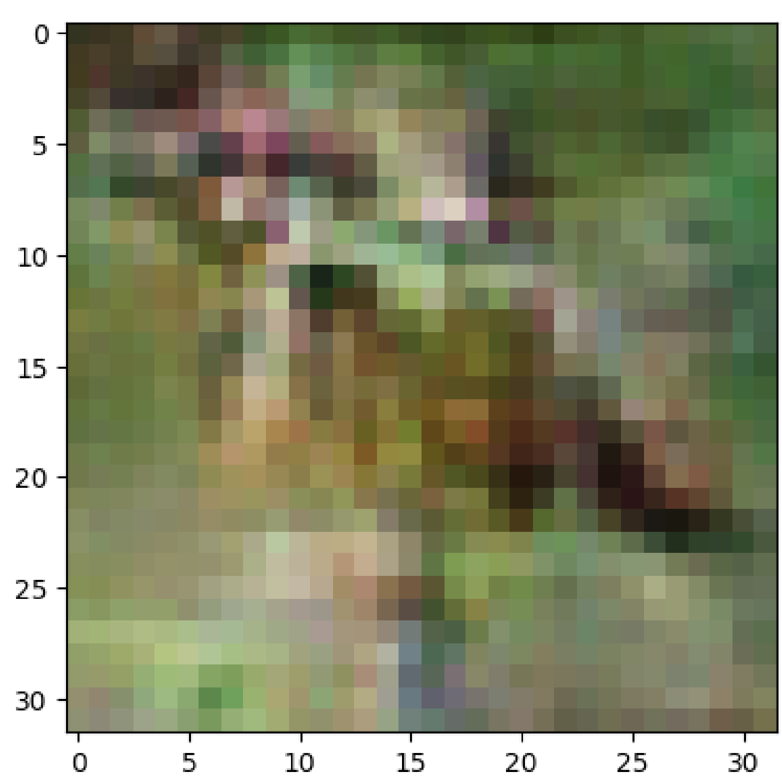
2/2 [==============================] - 0s 4ms/step

[16]:
```
# Visualize all generated images
for i in range(49):
  plt.subplot(7, 7, 1 + i)
  plt.axis('off')
  plt.imshow(((X_fake[i] * 127.5) + 127.5).astype(np.uint8))  # Convert pixel␣
↪values from range [-1,1] into range [0,255]
plt.show()
```



[27]:
```
# Visualize some individual generated images
imgplot = plt.imshow(((X_fake[14] * 127.5) + 127.5).astype(np.uint8))
plt.show()
imgplot = plt.imshow(((X_fake[25] * 127.5) + 127.5).astype(np.uint8))
plt.show()
```


0
5
10
15
20
25
30
0
5
10
15
20
25
30


0
5
10
15
20
25
30
0
5
10
15
20
25
30

[18]:
```
# Compress output folders for downloading
!zip -r /content/d_model.zip /content/d_model.sav
!zip -r /content/g_model.zip /content/g_model.sav
!zip -r /content/gan_model.zip /content/gan_model.sav
```

```
  adding: content/d_model.sav/ (stored 0%)
  adding: content/d_model.sav/keras_metadata.pb (deflated 92%)
  adding: content/d_model.sav/variables/ (stored 0%)
  adding: content/d_model.sav/variables/variables.index (deflated 67%)
  adding: content/d_model.sav/variables/variables.data-00000-of-00001 (deflated
8%)
  adding: content/d_model.sav/fingerprint.pb (stored 0%)
  adding: content/d_model.sav/saved_model.pb (deflated 88%)
  adding: content/d_model.sav/assets/ (stored 0%)
  adding: content/g_model.sav/ (stored 0%)
  adding: content/g_model.sav/keras_metadata.pb (deflated 92%)
  adding: content/g_model.sav/variables/ (stored 0%)
  adding: content/g_model.sav/variables/variables.index (deflated 52%)
  adding: content/g_model.sav/variables/variables.data-00000-of-00001 (deflated
7%)
  adding: content/g_model.sav/fingerprint.pb (stored 0%)
  adding: content/g_model.sav/saved_model.pb (deflated 89%)
  adding: content/g_model.sav/assets/ (stored 0%)
  adding: content/gan_model.sav/ (stored 0%)
  adding: content/gan_model.sav/keras_metadata.pb (deflated 95%)
  adding: content/gan_model.sav/variables/ (stored 0%)
  adding: content/gan_model.sav/variables/variables.index (deflated 71%)
  adding: content/gan_model.sav/variables/variables.data-00000-of-00001
(deflated 9%)
  adding: content/gan_model.sav/fingerprint.pb (stored 0%)
  adding: content/gan_model.sav/saved_model.pb (deflated 89%)
  adding: content/gan_model.sav/assets/ (stored 0%)
```

[19]:
```
# Download the Generator, Discriminator and GAN model
files.download("/content/g_model.zip")
files.download("/content/d_model.zip")
files.download("/content/gan_model.zip")
```

```
<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>
```

```
<IPython.core.display.Javascript object>
```